

16. *Assertion (A) :* The image of an object placed between f and $2f$ from a convex lens can be seen on a screen. If the screen is removed, image is not formed.

Reason (R) : Rays from a given point on the object placed between f and $2f$, after passing through a convex lens, do not converge on a point in space.

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16. *Assertion (A) :* The image of an object placed between f and $2f$ from a convex lens can be seen on a screen. If the screen is removed, image is not formed.

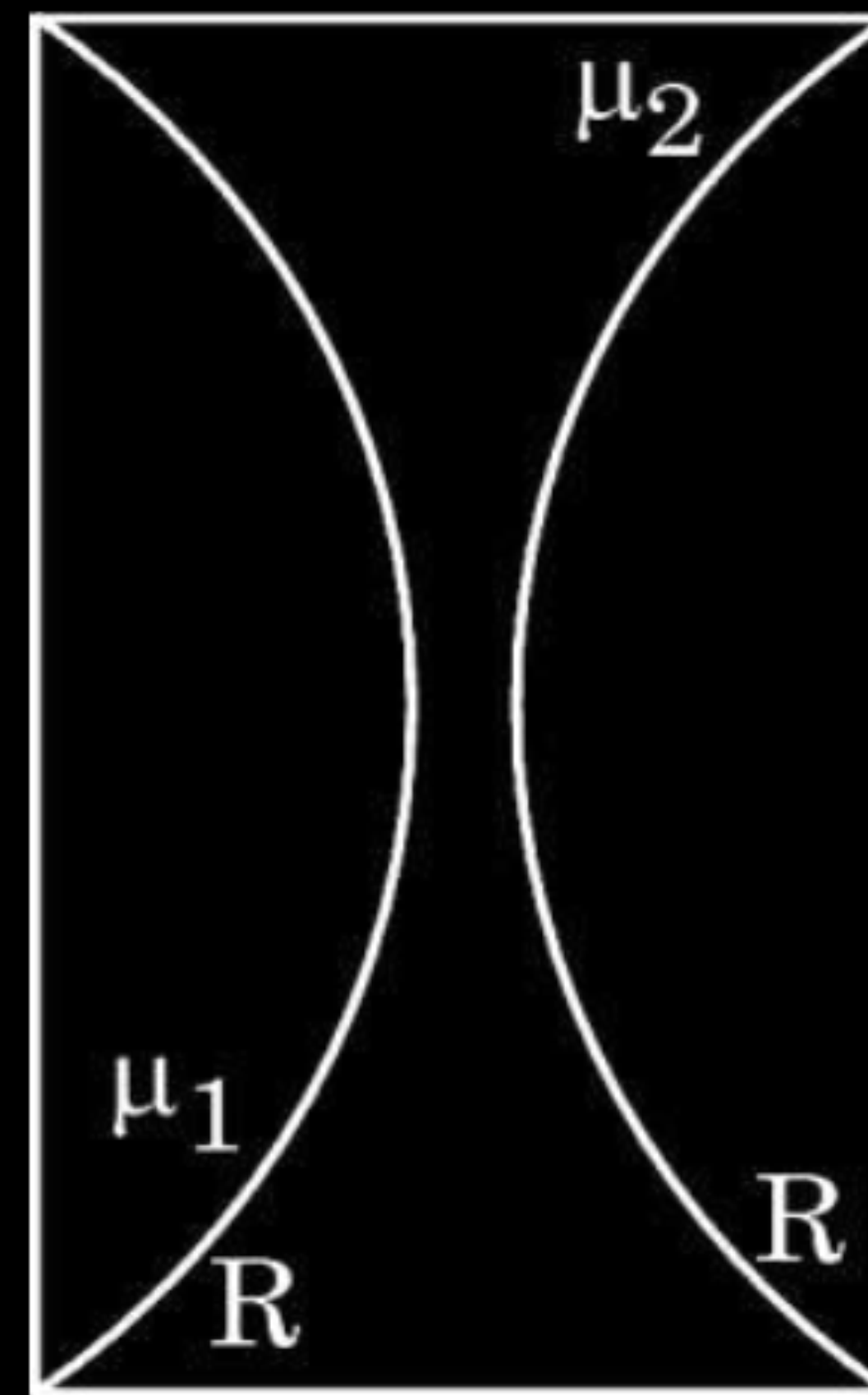
Reason (R) : Rays from a given point on the object placed between f and $2f$, after passing through a convex lens, do not converge on a point in space.

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(D) Both Assertion (A) and Reason (R) are false.

- 19.** (a) A plano-convex lens of refractive index μ_1 is placed coaxially in contact with a biconcave lens of refractive index μ_2 as shown in the figure. All curved faces are of radius of curvature R each. Obtain the expression for the focal length of the combined lens.

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$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

For plano convex lens

$$\frac{1}{f_1} = (\mu_1 - 1) \left(\frac{1}{\infty} - \frac{1}{-R} \right)$$

$$\frac{1}{f_1} = \frac{(\mu_1 - 1)}{R}$$

For concave lens

$$\frac{1}{f_2} = (\mu_2 - 1) \left(\frac{1}{-R} - \frac{1}{R} \right)$$

$$\frac{1}{f_2} = -2 \frac{(\mu_2 - 1)}{R}$$

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$f = \frac{R}{\mu_1 - 2\mu_2 + 1}$$

$$\frac{1}{2}$$

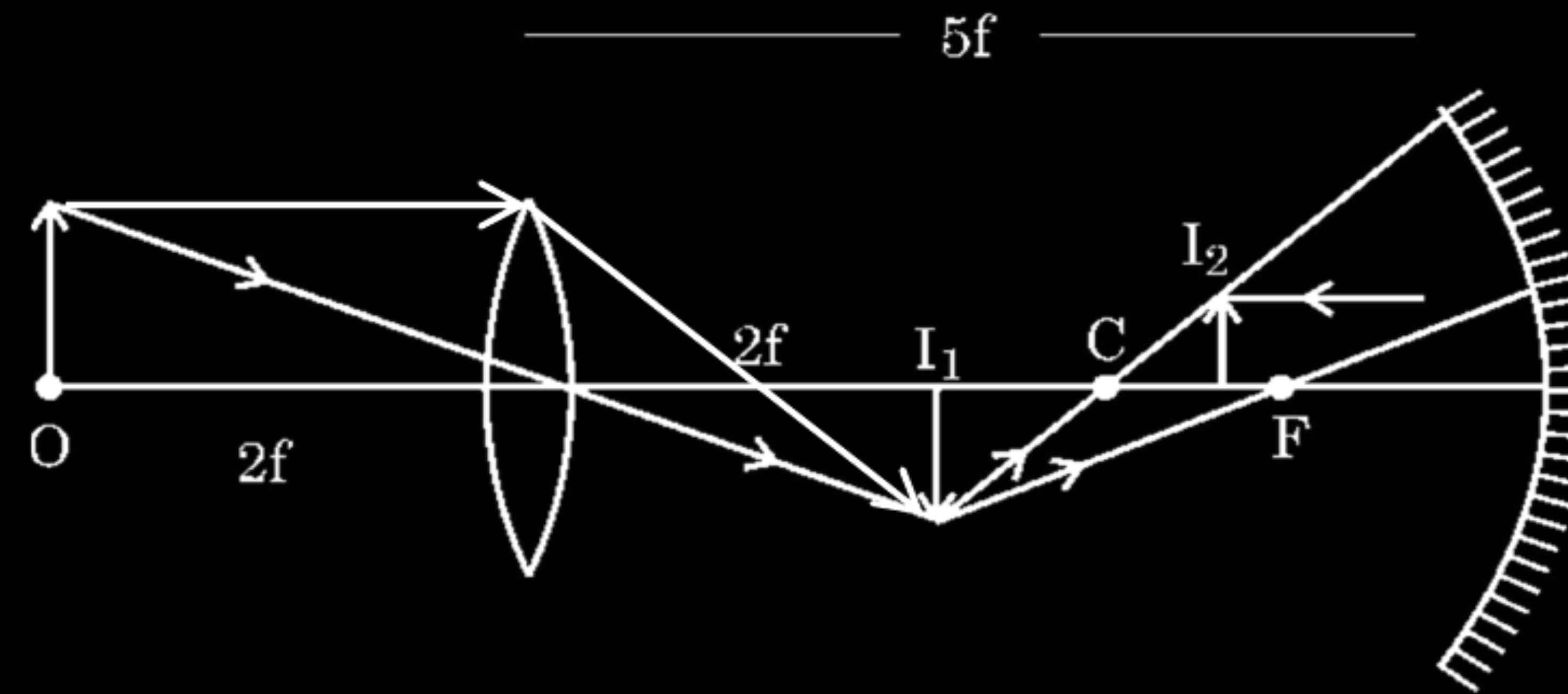
$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

- (b) A convex lens is kept coaxially on the left side of a concave mirror at a distance of $5f$ from it where f is focal length of each of them. An object is kept at a distance of $2f$ on the left side of the convex lens. Draw the ray diagram showing the formation of the image by the combination. Find the distance of the final image from the mirror. 2

- (b) A convex lens is kept coaxially on the left side of a concave mirror at a distance of $5f$ from it where f is focal length of each of them. An object is kept at a distance of $2f$ on the left side of the convex lens. Draw the ray diagram showing the formation of the image by the combination. Find the distance of the final image from the mirror. 2



$$u = -3f$$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{-1}{f} = \frac{-1}{3f} + \frac{1}{v}$$

$$v = -\frac{3}{2}f$$

1

 $\frac{1}{2}$ $\frac{1}{2}$

- (b) A convex lens of focal length 10 cm, a concave lens of focal length 15 cm and a third lens of unknown focal length are placed coaxially in contact. If the focal length of the combination is +12 cm, find the nature and focal length of the third lens, if all lenses are thin. Will the answer change if the lenses were thick ?

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- (b) A convex lens of focal length 10 cm, a concave lens of focal length 15 cm and a third lens of unknown focal length are placed coaxially in contact. If the focal length of the combination is +12 cm, find the nature and focal length of the third lens, if all lenses are thin. Will the answer change if the lenses were thick ?

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2

$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3}$	1/2
$\frac{1}{12} = \frac{1}{10} - \frac{1}{15} + \frac{1}{f_3}$	
$\frac{1}{f_3} = \frac{5-6+4}{60}$	
$f_3 = 20 \text{ cm}$	1/2
Nature: Convex	1/2
Yes	1/2

30. When light travels from an optically denser medium to an optically rarer medium, at the interface it is partly reflected back into the same medium and partly refracted to the second medium. The angle of incidence corresponding to an angle of refraction 90° is called the critical angle (i_c) for the given pair of media. This angle is related to the refractive index of medium 1 with respect to medium 2.

Refraction of light through a prism involves refraction at two plane interfaces. A relation for the refractive index of the material of the prism can be obtained in terms of the refracting angle of the prism and the angle of minimum deviation. For a thin prism, this relation reduces to a simple equation.

Laws of refraction are also valid for refraction of light at a spherical interface. When an object is placed in front of a spherical surface separating two media, its image is formed. A relation between object and image distance, in terms of refractive indices of two media and the radius of curvature of the spherical surface can be obtained. Using this relation for two surfaces of a lens, 'lens maker formula' is obtained.

(i) An object is placed in front of a convex spherical glass surface ($n = 1.5$ and radius of curvature R) at a distance of $4R$ from it. As the object is moved slowly close to the surface, the image formed is : *1*

- (A) always real
- (B) always virtual
- (C) first real and then virtual
- (D) first virtual and then real

(i) An object is placed in front of a convex spherical glass surface ($n = 1.5$ and radius of curvature R) at a distance of $4R$ from it. As the object is moved slowly close to the surface, the image formed is : *1*

- (A) always real
- (B) always virtual
- (C) first real and then virtual
- (D) first virtual and then real

(C) First real and then virtual

(ii) A double-convex lens, made of glass of refractive index 1.5, has focal length 10 cm. The radius of curvature of its each face, is : *1*

(A) 10 cm

(B) 15 cm

(C) 20 cm

(D) 40 cm

(ii) A double-convex lens, made of glass of refractive index 1.5, has focal length 10 cm. The radius of curvature of its each face, is : *1*

(A) 10 cm

(B) 15 cm

(C) 20 cm

(D) 40 cm

(A) 10cm

- (iii) A small bulb is placed at the bottom of a tank containing a transparent liquid (refractive index n) to a depth H . The radius of the circular area of the surface of liquid, through which light from the bulb can emerge out, is R . Then $\left(\frac{R}{H}\right)$ is : *1*

(A) $\frac{1}{\sqrt{n^2 - 1}}$

(B) $\sqrt{n^2 - 1}$

(C) $\frac{1}{\sqrt{n^2 + 1}}$

(D) $\sqrt{n^2 + 1}$

- (iii) A small bulb is placed at the bottom of a tank containing a transparent liquid (refractive index n) to a depth H . The radius of the circular area of the surface of liquid, through which light from the bulb can emerge out, is R . Then $\left(\frac{R}{H}\right)$ is : *1*

(A) $\frac{1}{\sqrt{n^2 - 1}}$

(B) $\sqrt{n^2 - 1}$

(C) $\frac{1}{\sqrt{n^2 + 1}}$

(D) $\sqrt{n^2 + 1}$

(A) $1/\sqrt{n^2 - 1}$

- (iv) (a) A parallel beam of light is incident on a face of a prism with refracting angle 60° . The angle of minimum deviation is found to be 30° . The refractive index of the material of the prism is close to :

- (A) 1.3 (B) 1.4
(C) 1.5 (D) 1.6

(B) 1.4

(b) The angle of minimum deviation for a ray of light incident on a thin prism, made of crown glass ($n = 1.52$) is D_m . If the prism was made of dense flint glass ($n = 1.62$) instead of crown glass, the angle of minimum deviation will :

1

- | | |
|---------------------|---------------------|
| (A) decrease by 4% | (B) increase by 4% |
| (C) decrease by 19% | (D) increase by 19% |

(b) The angle of minimum deviation for a ray of light incident on a thin prism, made of crown glass ($n = 1.52$) is D_m . If the prism was made of dense flint glass ($n = 1.62$) instead of crown glass, the angle of minimum deviation will :

1

- | | |
|---------------------|---------------------|
| (A) decrease by 4% | (B) increase by 4% |
| (C) decrease by 19% | (D) increase by 19% |

(D) increase by 19%

- (b) A point source of light in air is kept at a distance of 12 cm in front of a convex spherical surface of glass of refractive index 1.5 and radius of curvature 30 cm. Find the nature and position of the image formed.

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2

$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$	$\frac{1}{2}$
$\frac{1.5}{v} - \frac{1}{(-12)} = \frac{1.5 - 1}{30}$	$\frac{1}{2}$
$v = -22.5 \text{ cm}$	$\frac{1}{2}$
Image is virtual and erect.	$\frac{1}{2}$

8. Two thin lenses of focal lengths $+40$ cm and -20 cm and placed coaxially in contact. An object is placed at infinity, in front of the combination. The image formed by the combination will lie – **1**
- (A) at a distance of 60 cm, in front of the combination.
 - (B) at a distance of 40 cm, in front of the combination.
 - (C) at a distance of 20 cm, behind the combination.
 - (D) at infinity

8. Two thin lenses of focal lengths $+40$ cm and -20 cm and placed coaxially in contact. An object is placed at infinity, in front of the combination. The image formed by the combination will lie – **1**
- (A) at a distance of 60 cm, in front of the combination.
 - (B) at a distance of 40 cm, in front of the combination.
 - (C) at a distance of 20 cm, behind the combination.
 - (D) at infinity

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(B) at a distance of 40 cm in front of the combination

- (b) An object is placed 1.50 m in front of a lens of power 4.0 D.
Find the nature and position of the image formed. **2**

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- (b) An object is placed 1.50 m in front of a lens of power 4.0 D.
Find the nature and position of the image formed.

2

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$$f = \frac{1}{P} = \frac{1}{4} = 0.25 \text{ m} = 25 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{25} = \frac{1}{v} - \left(\frac{1}{-150} \right)$$

On solving, $v = 30 \text{ cm}$

The image formed is real and inverted.

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- (b) (A) A ray of light is incident at an angle of incidence, $\frac{A}{2}$ on face AB of a prism ABC of angle of prism A. The ray emerges normally from the opposite face AC. If the refractive index of the material of the prism be n , find the relation between A and n .
- (B) The focal length of a convex lens of refractive index 1.5 is 20 cm. Find the focal length of this lens when immersed in a liquid of refractive index 1.25. 5

(A) As refracted ray emerges normally from opposite face AC of the prism, So $r_2 = 0$	$\frac{1}{2}$
$r_1 + r_2 = A$	$\frac{1}{2}$
$r_1 = A$	
$n = \frac{\sin i_1}{\sin r_1} = \frac{\sin A/2}{\sin A}$	$\frac{1}{2}$
$n = \frac{\sin A/2}{2 \sin A/2 \cos A/2}$	$\frac{1}{2}$
$n = \frac{1}{2 \cos A/2}$	$\frac{1}{2}$

(B) From lens maker formula	
$\frac{1}{f} = ({}^1n_2 - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$	$\frac{1}{2}$
$\frac{1}{f_{air}} = (1.5 - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$	
$\frac{1}{f_{air}} = 0.5 \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$	$\frac{1}{2}$
$\frac{1}{f_{liquid}} = \left(\frac{1.5}{1.25} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$	
$\frac{1}{f_{liquid}} = 0.2 \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$	$\frac{1}{2}$
$\frac{f_l}{f_{air}} = \frac{5}{2}$	$\frac{1}{2}$
$f_l = \frac{5}{2} \times f_{air} = \frac{5}{2} \times 20 = 50 \text{ cm}$	$\frac{1}{2}$

- 20.** (a) A point object is placed in air at a distance $R/3$ in front of a convex surface of radius of curvature R , separating air from a medium of refractive index n (< 4). Find the nature and position of the image formed.

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- 20.** (a) A point object is placed in air at a distance $R/3$ in front of a convex surface of radius of curvature R , separating air from a medium of refractive index n (< 4). Find the nature and position of the image formed.

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2

For refraction at convex surface

$$\frac{n_1}{-u} + \frac{n_2}{v} = \frac{n_2 - n_1}{R}$$

$$\frac{n}{v} = \frac{[n-1-3]}{R}$$

$$v = \frac{nR}{n-4}$$

For all values of $n < 4$, the value of v is negative and greater than R . Therefore the nature of image is virtual and is formed in front of convex surface.

$\frac{1}{2}$

$\frac{1}{2}$

1

(b) (i)

- (ii) A thin planoconcave lens with its curved face of radius of curvature R is made of glass of refractive index n_1 . It is placed coaxially in contact with a thin equiconvex lens of same radius of curvature of refractive index n_2 . Obtain the power of the combination lens.

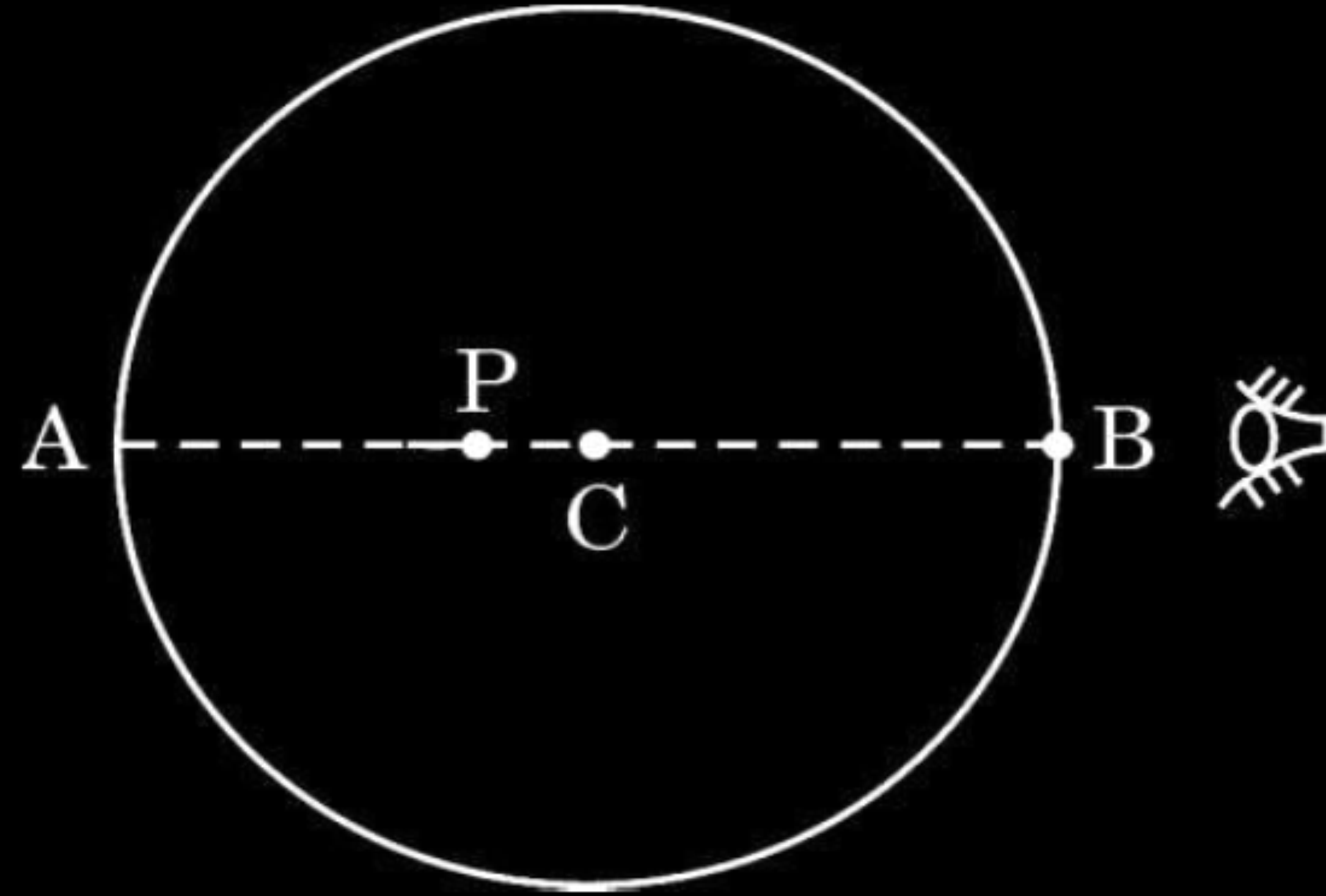
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(ii)	Power of plano concave lens	=	$P_1 = \frac{-(n_1-1)}{R}$	$\frac{1}{2}$
	Power of convex lens	=	$P_2 = (n_2-1) \left(\frac{2}{R} \right)$	$\frac{1}{2}$
			$P = P_1 + P_2$	$\frac{1}{2}$
			$= \frac{(2n_2 - n_1 - 1)}{R}$	$\frac{1}{2}$

- 26.** An air bubble is trapped at point P ($CP = 1.75$ cm) in a spherical glass ball ($n = 1.5$) of radius 7 cm as shown in the figure. Find the nature and position of the image when viewed from side B. Show the image formation by drawing a ray diagram.

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26. An air bubble is trapped at point P (CP = 1.75 cm) in a spherical glass ball (n = 1.5) of radius 7 cm as shown in the figure. Find the nature and position of the image when viewed from side B. Show the image formation by drawing a ray diagram.

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3

For light going from denser to rarer medium

$$\frac{n_2}{-u} + \frac{n_1}{v} = \frac{n_1 - n_2}{R}$$

$$\frac{1.5}{8.75} + \frac{1}{v} = \frac{(1 - 1.5)}{-7}$$

$$\frac{1}{v} = \frac{0.5}{7} - \frac{1.5}{8.75}$$

$$= \frac{-1}{10}$$

$$v = -10 \text{ cm}$$

Nature of image is virtual

<u>1</u>
$\frac{1}{2}$
$\frac{1}{2}$
$\frac{1}{2}$
$\frac{1}{2}$

- 19.** The two surfaces of a biconvex lens are of radius of curvature ' R ' each. Obtain the condition under which its focal length ' f ' be equal to ' R '. If one of the two surfaces of this lens is made plane, what will be the new focal length of the lens ?

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19. The two surfaces of a biconvex lens are of radius of curvature 'R' each. Obtain the condition under which its focal length 'f' be equal to 'R'. If one of the two surfaces of this lens is made plane, what will be the new focal length of the lens ?

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2

$$\frac{1}{f} = (n-1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

For $f = R, R_1 = R, R_2 = -R$

$$\frac{1}{R} = (n-1) \left(\frac{1}{R} + \frac{1}{R} \right)$$

$$n = \frac{3}{2}$$

For convex lens

$$\frac{1}{f} = (n-1) \left(\frac{2}{R} \right)$$

For plano Convex lens $R_1 = R, R_2 = \infty$

$$\frac{1}{f_p} = (n-1) \left(\frac{1}{R} \right)$$

$$f_p = 2f$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- 31.** (a) (i) An object is placed 30 cm from a thin convex lens of focal length 10 cm. The lens forms a sharp image on a screen. If a thin concave lens is placed in contact with the convex lens, the sharp image on the screen is formed when the screen is moved by 45 cm from its initial position. Calculate the focal length of the concave lens.

For real image form by Convex lens

$$\frac{1}{f_1} = \frac{1}{v_1} - \frac{1}{u_1}$$

$$\frac{1}{10} = \frac{1}{v_1} - \frac{1}{(-30)}$$

$$v_1 = 15 \text{ cm}$$

For Combination of lenses, let the focal length of combination of lens is f_3

$$\frac{1}{f_3} = \frac{1}{v_3} - \frac{1}{u_3}$$

$$\frac{1}{f_3} = \frac{1}{(15+45)} + \frac{1}{30}$$

$$f_3 = 20 \text{ cm}$$

Let the focal length of concave lens is f_2

$$\frac{1}{f_3} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$\frac{1}{f_2} = \frac{1}{20} - \frac{1}{10}$$

$$f_2 = -20 \text{ cm}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

- 19.** Find the focal length of plano-convex lens of refractive index 1.5 and radius of curvature 10 cm when it is immersed in a liquid of refractive index 1.25.

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$$\frac{1}{f} = \left(\frac{n_2}{n_1} - 1 \right) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

For plano convex lens

$$R_1 = R \text{ and } R_2 = \infty$$

$$\frac{1}{f} = \left(\frac{n_2}{n_1} - 1 \right) \frac{1}{R}$$

$$= \left(\frac{1.5}{1.25} - 1 \right) \times \frac{1}{10}$$

$$\frac{1}{f} = \frac{1}{50}$$

$$\therefore f = 50 \text{ cm}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- 19.** An equiconvex lens is made of glass of refractive index 1.55. If the focal length of the lens is 15.0 cm, calculate the radius of curvature of its surfaces.

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2

$\frac{1}{f} = (n_{21} - 1) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$	$\frac{1}{2}$
$R_1 = R \text{ and } R_2 = -R$	$\frac{1}{2}$
$\frac{1}{15} = (1.55 - 1) \left[\frac{2}{R} \right]$	$\frac{1}{2}$
$R = 16.5 \text{ cm}$	$\frac{1}{2}$

9. The plane face of a planoconvex lens is silvered. The refractive index of material and radius of curvature of the curved surface of the lens are n and R respectively. This lens will behave as a concave mirror of focal length **1**

(A) $\frac{R}{n}$

(B) $\frac{R}{(n - 1)}$

(C) nR

(D) $\frac{R}{2(n - 1)}$

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9. The plane face of a planoconvex lens is silvered. The refractive index of material and radius of curvature of the curved surface of the lens are n and R respectively. This lens will behave as a concave mirror of focal length **1**

(A) $\frac{R}{n}$

(B) $\frac{R}{(n - 1)}$

(C) nR

(D) $\frac{R}{2(n - 1)}$

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(D) $\frac{R}{2(n-1)}$

19. A light beam converges at a point O. In the path of this beam, a concave lens of focal length 15 cm is placed at a distance of 10 cm before point O. The beam now converges at a point O'. Find the magnitude and the direction of shift OO'.

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19. A light beam converges at a point O. In the path of this beam, a concave lens of focal length 15 cm is placed at a distance of 10 cm before point O. The beam now converges at a point O'. Find the magnitude and the direction of shift OO'.

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2

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = -\frac{1}{15} + \frac{1}{10}$$

On solving: -

$$v = 30 \text{ cm}$$

$$\text{Magnitude of } OO' = 30 - 10 = 20 \text{ cm}$$

Direction is away from the lens.

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

19. Two convex lenses A and B, each of focal length 10.0 cm, are mounted on an optical bench at 50.0 cm and 70.0 cm respectively. An object is mounted at 20.0 cm. Find the nature and position of the final image formed by the combination.

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19. Two convex lenses A and B, each of focal length 10.0 cm, are mounted on an optical bench at 50.0 cm and 70.0 cm respectively. An object is mounted at 20.0 cm. Find the nature and position of the final image formed by the combination.

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2

For the first lens: -

$$\frac{1}{v_1} - \frac{1}{u_1} = \frac{1}{f_1}$$

$$\frac{1}{v_1} + \frac{1}{30} = \frac{1}{10}$$

$$v_1 = 15 \text{ cm}$$

For the second lens: -

$$u_2 = -20 - (-15) = -5 \text{ cm}$$

$$\frac{1}{v_2} - \left(-\frac{1}{5} \right) = \frac{1}{10}$$

On solving: -

$$v_2 = -10 \text{ cm}$$

Image is virtual.

$\frac{1}{2}$

$\frac{1}{2}$

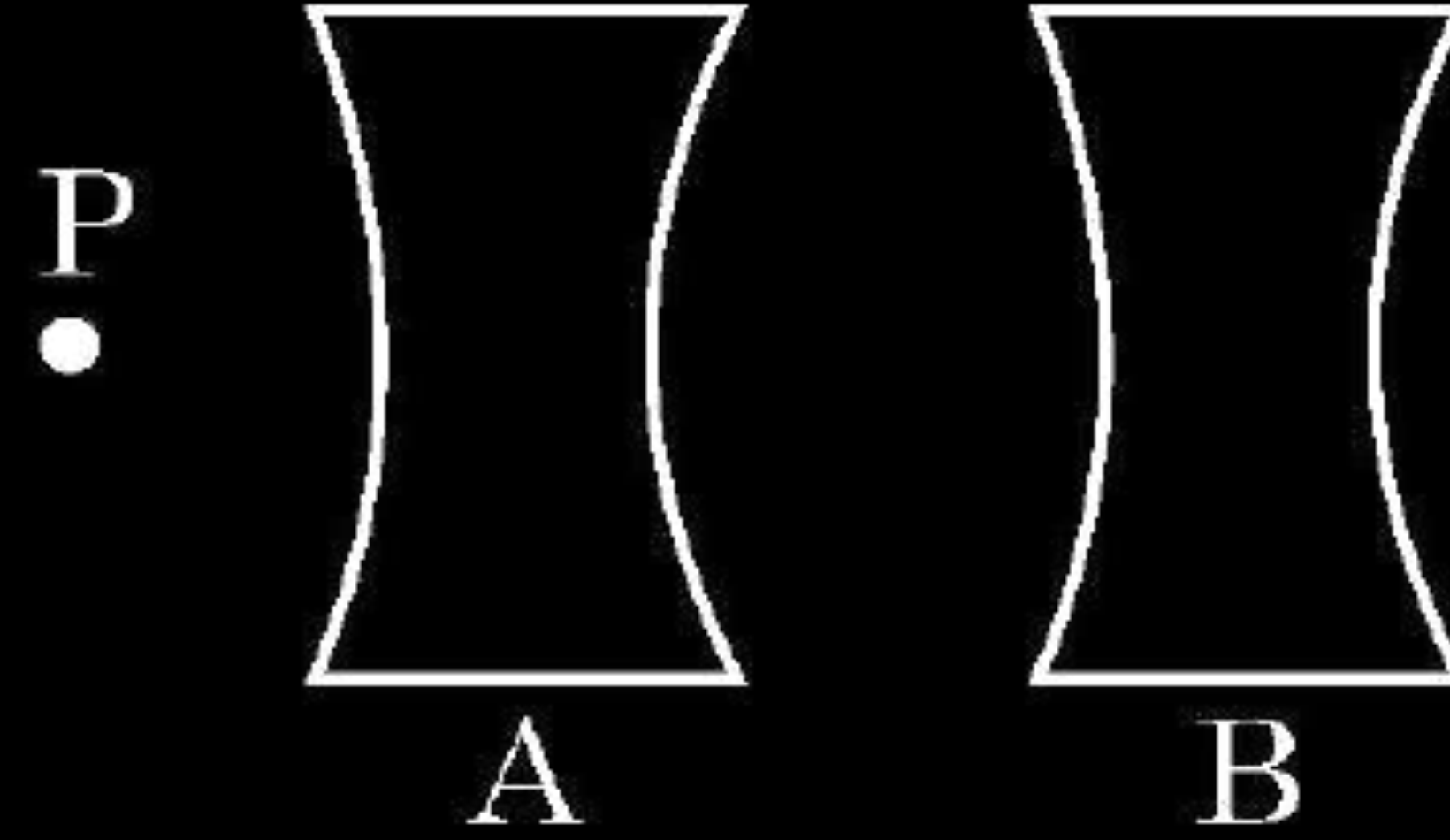
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19. Two concave lenses A and B, each of focal length 8.0 cm are arranged coaxially 16 cm apart as shown in figure. An object P is placed at a distance of 4.0 cm from A. Find the position and nature of the final image formed.

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2



For the first lens: -

$$\frac{1}{v_1} - \frac{1}{u_1} = \frac{1}{f_1}$$

$$\frac{1}{v_1} + \frac{1}{4} = -\frac{1}{8}$$

$$v_1 = -\frac{8}{3}\text{cm}$$

For the second lens: -

$$u_2 = -16 - \frac{8}{3} = -\frac{56}{3}\text{cm}$$

$$\frac{1}{v_2} - \left(-\frac{3}{56}\right) = -\frac{1}{8}$$

$$v_2 = -5.6\text{cm}$$

Image is virtual.

1/2

1/2

1/2

1/2

19. A double convex lens of glass has both faces of the same radius of curvature 17 cm. Find its focal length if it is immersed in water. The refractive indices of glass and water are 1.5 and 1.33 respectively. **2**

19. A double convex lens of glass has both faces of the same radius of curvature 17 cm. Find its focal length if it is immersed in water. The refractive indices of glass and water are 1.5 and 1.33 respectively. **2**

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$$\frac{1}{f} = \left(\frac{n_g}{n_w} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

For double convex lens $R_1=R$ and $R_2=-R$

$$\frac{1}{f} = \left(\frac{1.5}{1.33} - 1 \right) \left(\frac{2}{R} \right)$$

$$= \left(\frac{1.5 - 1.33}{1.33} \right) \left(\frac{2}{17} \right)$$

$$f = 66.5 \text{ cm}$$

$\frac{1}{2}$

$\frac{1}{2}$

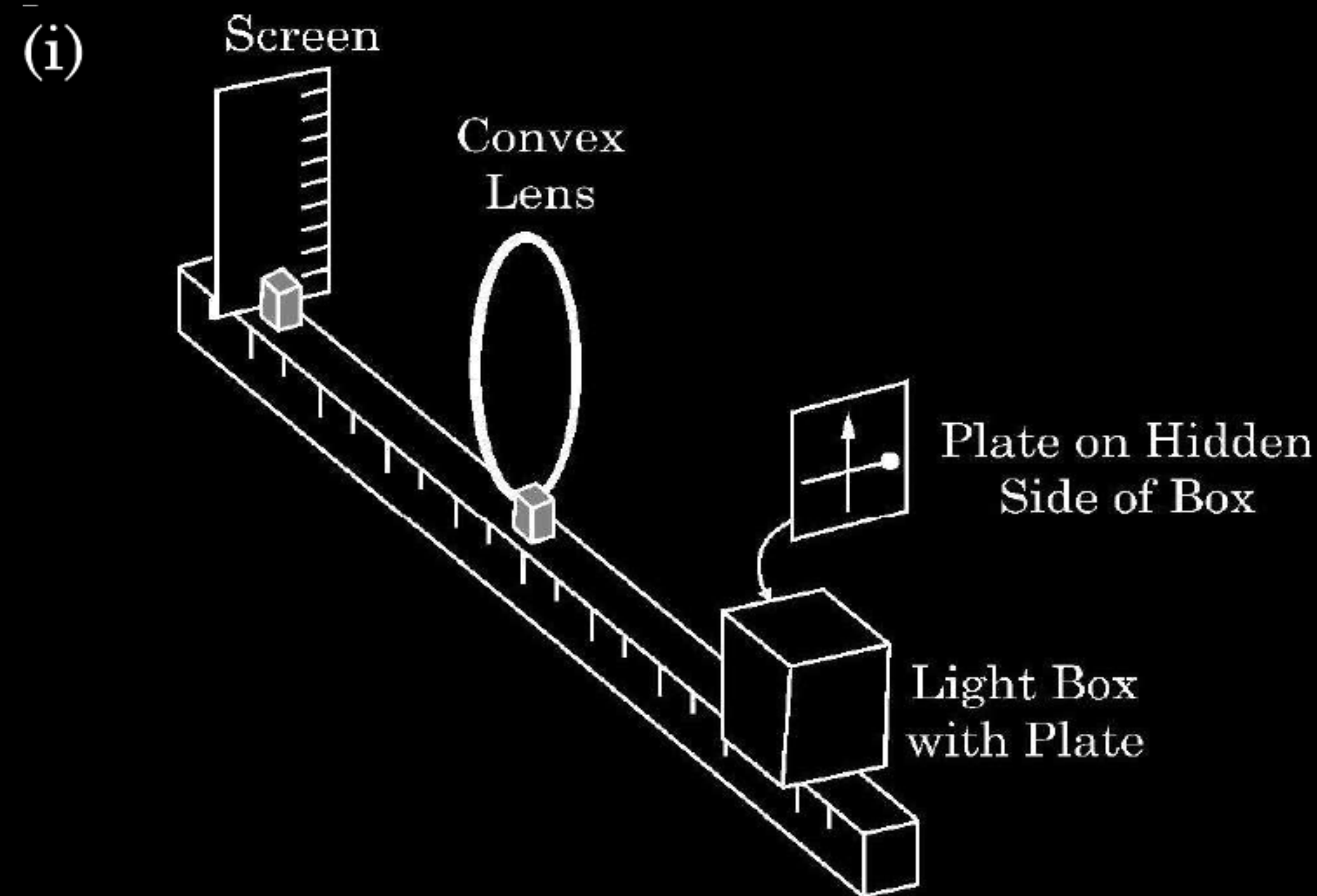
$\frac{1}{2}$

$\frac{1}{2}$

29. A thin lens is a transparent optical medium bounded by two surfaces, at least one of which should be spherical. Applying the formula for image formation by a single spherical surface successively at the two surfaces of a lens, one can obtain the 'lens maker formula' and then the 'lens formula'. A lens has two foci – called 'first focal point' and 'second focal point' of the lens, one on each side.

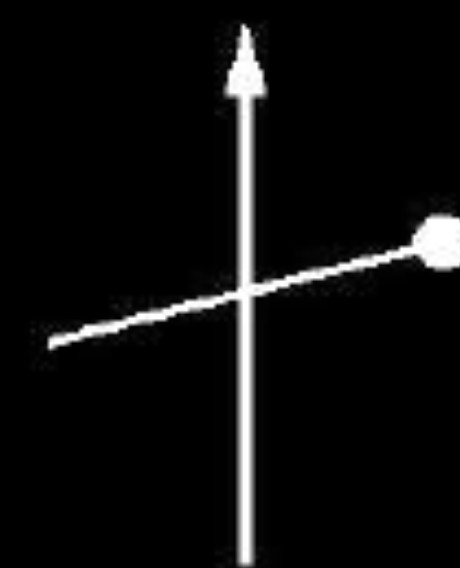
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$4 \times 1 = 4$

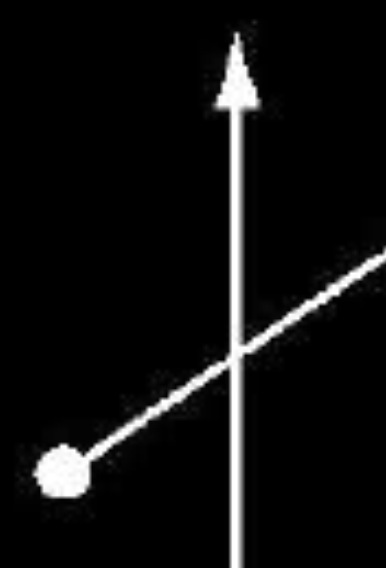


Consider the arrangement shown in figure. A black vertical arrow and a horizontal thick line with a ball are painted on a glass plate. It serves as the object. When the plate is illuminated, its real image is formed on the screen.

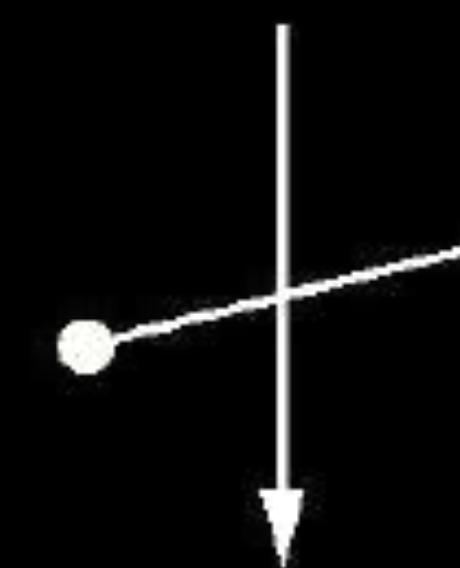
Which of the following correctly represents the image formed on the screen ?



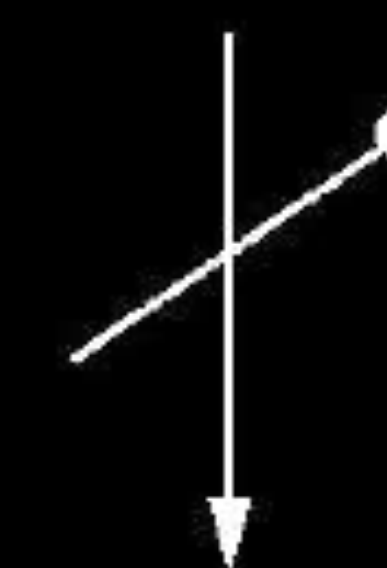
(A)



(B)



(C)

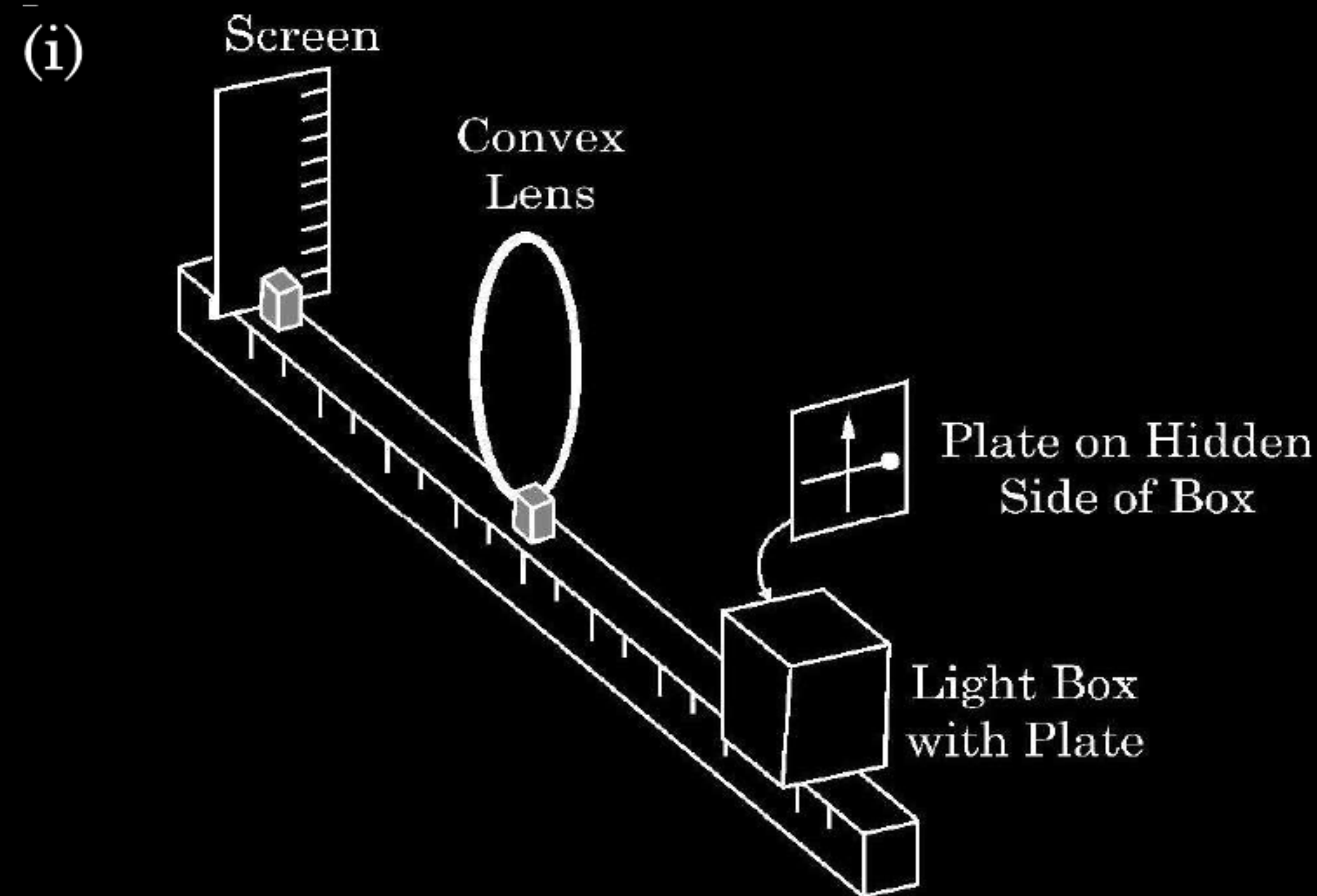


(D)

29. A thin lens is a transparent optical medium bounded by two surfaces, at least one of which should be spherical. Applying the formula for image formation by a single spherical surface successively at the two surfaces of a lens, one can obtain the 'lens maker formula' and then the 'lens formula'. A lens has two foci – called 'first focal point' and 'second focal point' of the lens, one on each side.

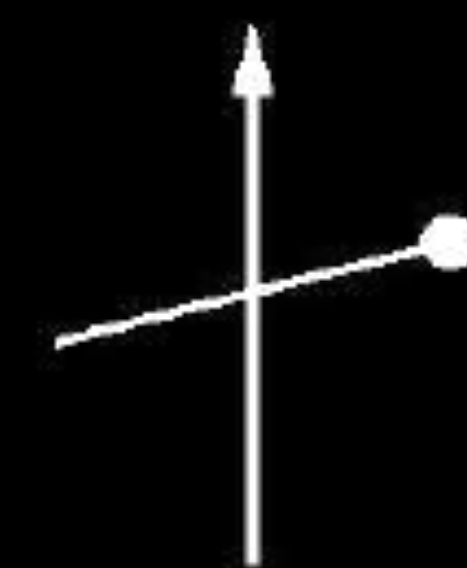
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$4 \times 1 = 4$

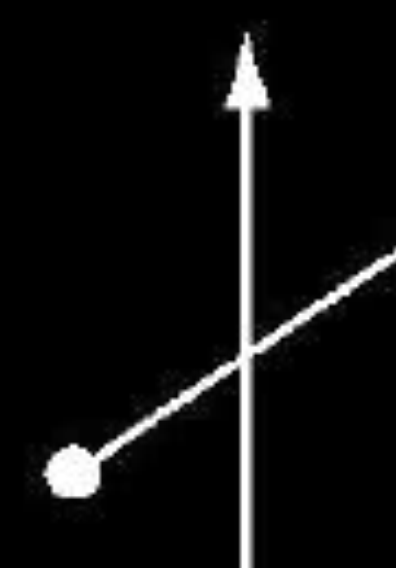


Consider the arrangement shown in figure. A black vertical arrow and a horizontal thick line with a ball are painted on a glass plate. It serves as the object. When the plate is illuminated, its real image is formed on the screen.

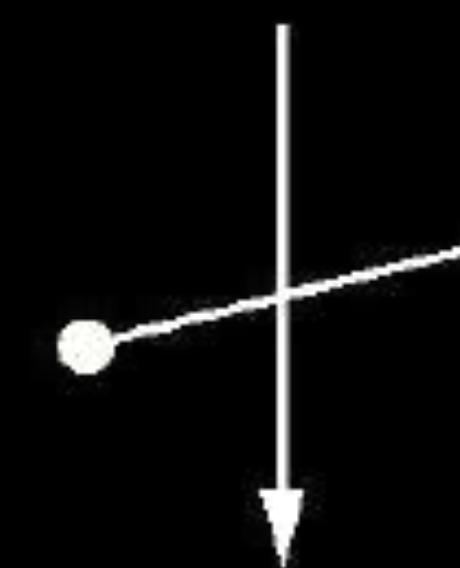
Which of the following correctly represents the image formed on the screen ?



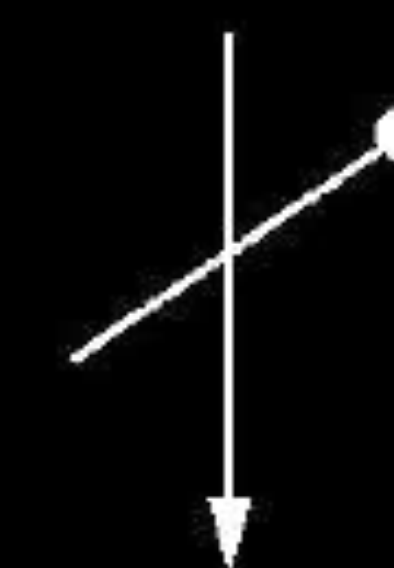
(A)



(B)

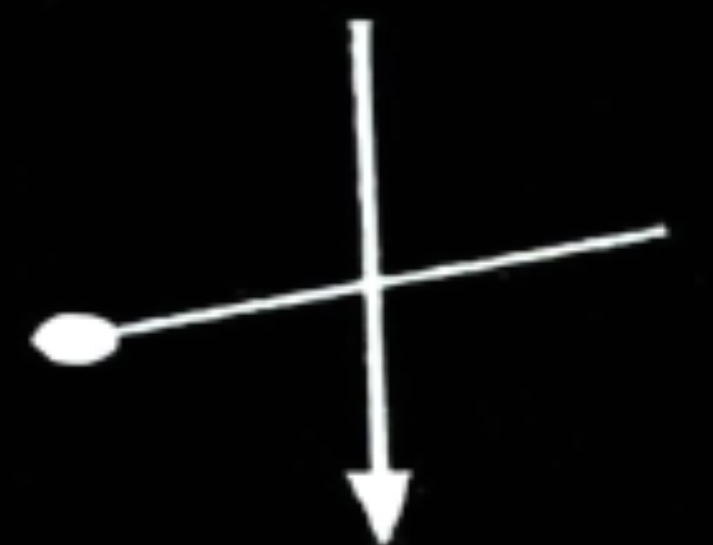


(C)



(D)

i) (C)



- (ii) Which of the following statements is incorrect ?
- (A) For a convex mirror magnification is always negative.
 - (B) For all virtual images formed by a mirror magnification is positive.
 - (C) For a concave lens magnification is always positive.
 - (D) For real and inverted images, magnification is always negative.

- (ii) Which of the following statements is incorrect ?
- (A) For a convex mirror magnification is always negative.
 - (B) For all virtual images formed by a mirror magnification is positive.
 - (C) For a concave lens magnification is always positive.
 - (D) For real and inverted images, magnification is always negative.

(A) For a convex mirror magnification is always negative

(iii) A convex lens of focal length 'f' is cut into two equal parts perpendicular to the principal axis. The focal length of each part will be :

(A) f

(B) $2f$

(C) $\frac{f}{2}$

(D) $\frac{f}{4}$

(iii) A convex lens of focal length 'f' is cut into two equal parts perpendicular to the principal axis. The focal length of each part will be :

(A) f

(B) $2f$

(C) $\frac{f}{2}$

(D) $\frac{f}{4}$

(B) $2f$

(iii) If an object in case (i) above is 20 cm from the lens and the screen is 50 cm away from the object, the focal length of the lens used is

(A) 10 cm

(B) 12 cm

(C) 16 cm

(D) 20 cm

(iii) If an object in case (i) above is 20 cm from the lens and the screen is 50 cm away from the object, the focal length of the lens used is

(A) 10 cm

(B) 12 cm

(C) 16 cm

(D) 20 cm

(B) 12 cm

(iv) The distance of an object from first focal point of a biconvex lens is X_1 and distance of the image from second focal point is X_2 . The focal length of the lens is

(A) $X_1 X_2$

(B) $\sqrt{X_1 + X_2}$

(C) $\sqrt{X_1 X_2}$

(D) $\sqrt{\frac{X_2}{X_1}}$

(iv) The distance of an object from first focal point of a biconvex lens is X_1 and distance of the image from second focal point is X_2 . The focal length of the lens is

(A) $X_1 X_2$

(B) $\sqrt{X_1 + X_2}$

(C) $\sqrt{X_1 X_2}$

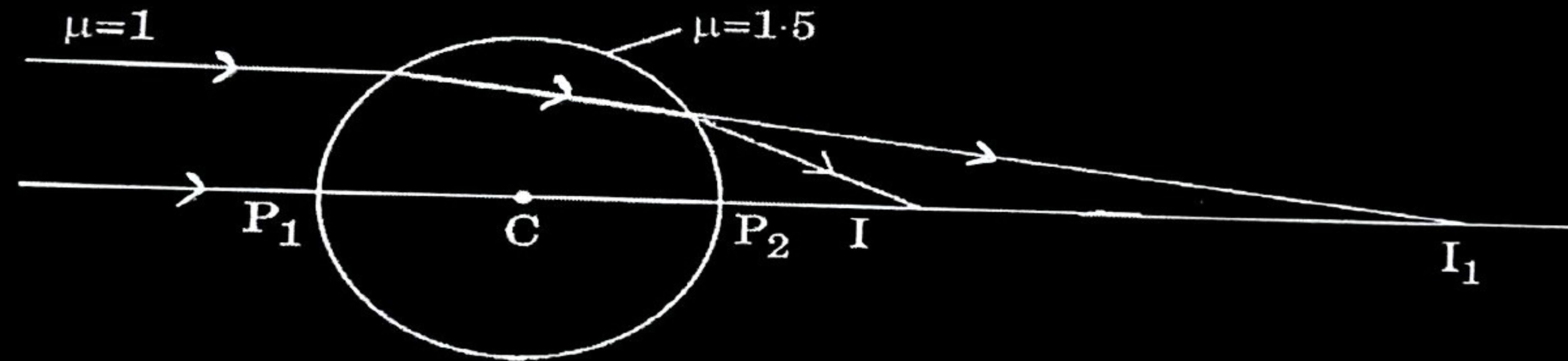
(D) $\sqrt{\frac{X_2}{X_1}}$

(C) $\sqrt{X_1 X_2}$

- (ii) A beam of light coming from a distant source is refracted by a spherical glass ball (refractive index 1.5) of radius 15 cm. Draw the ray diagram and obtain the position of the final image formed.

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ii)



From 1st surface , Refraction is from rarer to denser medium and object is at ∞

$$n_1 = 1, \quad n_2 = 1.5, \quad R = 15 \text{ cm}, \quad u = \infty$$

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

$$\frac{1.5}{v} - \frac{1}{\infty} = \frac{1.5 - 1}{15}$$

$$v = 45 \text{ cm}$$

From 2nd surface , Refraction is from denser to rarer medium and object is at 15 cm

$$n_1 = 1, \quad n_2 = 1.5, \quad R = -15 \text{ cm}, \quad u = 15 \text{ cm}$$

$$\frac{n_1}{v} - \frac{n_2}{u} = \frac{n_1 - n_2}{R}$$

$$\frac{1}{v} - \frac{1.5}{15} = \frac{1 - 1.5}{-15}$$

$$v = 7.5 \text{ cm}$$

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

19. A spherical convex surface of radius of curvature R separates glass (refractive index 1.5) from air. Light from a point source placed in air at distance $R/2$ from the surface falls on it. Find the position and nature of the image formed.

19. A spherical convex surface of radius of curvature R separates glass (refractive index 1.5) from air. Light from a point source placed in air at distance $R/2$ from the surface falls on it. Find the position and nature of the image formed.

2

Refraction from rarer to denser medium

$$\frac{n_1}{-u} + \frac{n_2}{v} = \frac{n_2 - n_1}{R}$$

$$u = -\frac{R}{2}, n_1 = 1, n_2 = 1.5$$

$$\frac{2}{R} + \frac{1.5}{v} = \frac{1.5 - 1}{R}$$

$$\frac{1.5}{v} = \frac{0.5}{R} - \frac{2}{R}$$

$$\frac{1.5}{v} = -\frac{1.5}{R}$$

$$v = -R$$

The image is virtual in air at distance R .

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

6. A point object is placed in air at a distance of $4R$ on the principal axis of a convex spherical surface of radius of curvature R separating two mediums, air and glass. As the object is moved towards the surface, the image formed is :

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- (A) always real
- (B) always virtual
- (C) first virtual and then real
- (D) first real and then virtual

6. A point object is placed in air at a distance of $4R$ on the principal axis of a convex spherical surface of radius of curvature R separating two mediums, air and glass. As the object is moved towards the surface, the image formed is :

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- (A) always real
- (B) always virtual
- (C) first virtual and then real
- (D) first real and then virtual

(D) first real and then virtual.

13. *Assertion (A) :* Although the surfaces of a goggle lens are curved, it does not have any power.

Reason (R) : In case of goggles, both the curved surfaces are curved on the same side and have equal radii of curvature.

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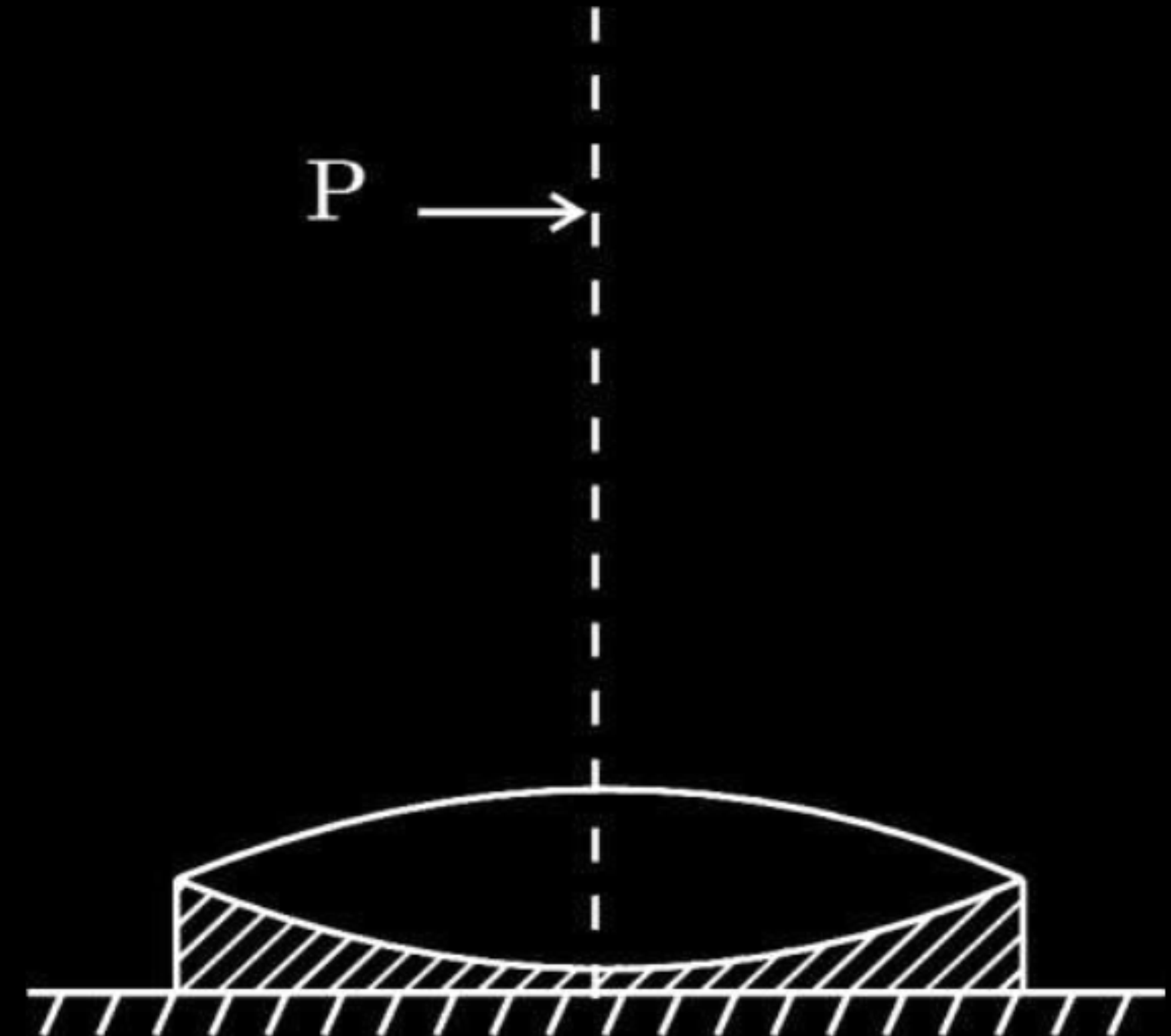
13. *Assertion (A) :* Although the surfaces of a goggle lens are curved, it does not have any power.

Reason (R) : In case of goggles, both the curved surfaces are curved on the same side and have equal radii of curvature.

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(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion(A).

- (ii) The figure below shows an equiconvex lens (of refractive index 1.50) in contact with a liquid layer on top of a plane mirror. A small needle with its tip on the principal axis is moved along the axis until its inverted image is found at the position of the needle. The distance of the needle from the lens is measured to be 45.0 cm. When the liquid is removed and the experiment is repeated, the new distance is 30.0 cm. Find the refractive index of the liquid. CBSE 2024 C



(ii) $\frac{1}{f} = (\mu - 1) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$	$\frac{1}{2}$
Focal length of convex lens, $f_1 = 30 \text{ cm}$	
$\frac{1}{30} = (1.5 - 1) \left[\frac{1}{R} - \frac{1}{(-R)} \right]$	$\frac{1}{2}$
$R = 30 \text{ cm}$	$\frac{1}{2}$
focal length of combination, $f = 45 \text{ cm}$	
focal length of plane concave lens of liquid.	
$\frac{1}{f_2} = \frac{1}{f} - \frac{1}{f_1}$	$\frac{1}{2}$
$\frac{1}{f_2} = \frac{1}{45} - \frac{1}{30}$	
$f_2 = -90 \text{ cm}$	
Using lens maker's formula	$\frac{1}{2}$
$\frac{1}{-90} = (\mu_l - 1) \left[\frac{1}{-30} - \frac{1}{\infty} \right]$	
$\mu_l = \frac{4}{3}$	$\frac{1}{2}$

26. A glass biconvex lens of focal length 0.6 m and refractive index 1.5 is immersed in a medium of refractive index 2.0. Find the new power of the lens and the nature of lens.

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26. A glass biconvex lens of focal length 0.6 m and refractive index 1.5 is immersed in a medium of refractive index 2.0. Find the new power of the lens and the nature of lens.

CBSE 2024 C

3

According to Lens- Makers formula:

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$\frac{1}{2}$

For biconvex lens

$$\frac{1}{0.6} = (1.5 - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \text{-----(1)}$$

$\frac{1}{2}$

When immersed in medium:-

$$\frac{1}{f} = \left(\frac{1.5}{2} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \text{-----(2)}$$

$\frac{1}{2}$

On solving equation (1) and (2):-

$$\frac{1}{f} = -\frac{5}{6}m$$

$$P = -\frac{5}{6}D$$

1

Nature of lens will be concave or diverging.

$\frac{1}{2}$

33. (a) (i) Write the relation between u , v and R for a convex spherical surface when refraction takes place from optically rarer to denser medium. Using it, derive 'lens maker formula' for a double convex lens.
- (ii) Two thin lenses of power $+ 25 \text{ D}$ and $- 5 \text{ D}$ are kept in contact with each other.
- (1) Calculate the focal length of the combination.
 - (2) An object of 4 cm height is placed at a distance of 25 cm from this combination. What is the position and size of the image formed ?

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- (i) For a convex spherical surface, for light going from optically rarer to optically denser medium-

$$\frac{n_1}{-u} + \frac{n_2}{v} = \frac{n_2 - n_1}{R}$$

1/2

For the refraction through the first convex surface (light going from rarer to denser medium) , the relation will be

$$\frac{n_1}{-u} + \frac{n_2}{v_1} = \frac{n_2 - n_1}{R_1} \text{ ----- (1)}$$

1/2

For the refraction through the second convex surface (light going from denser to rarer medium) , the relation will be

$$\frac{-n_2}{v_1} + \frac{n_1}{v} = \frac{n_1 - n_2}{R_2} \text{ ----- (2)}$$

1/2

Adding equation (1) and (2)

$$\frac{n_1}{v} - \frac{n_1}{u} = (n_2 - n_1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \text{ ----- (3)}$$

From lens formula

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

Using equation (3) and (4)

$$\frac{1}{f} = (n_{21} - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

1/2

(ii)

$$(1) P_1 = +25D \Rightarrow f_1 = 4 \text{ cm}$$

$$P_2 = -5D \Rightarrow f_2 = -20 \text{ cm}$$

$$f = \frac{f_1 f_2}{f_1 + f_2} = \frac{4 \times (-20)}{-16}$$

$$f = 5 \text{ cm}$$

(2) From lens formula

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{5-1}{25}$$

on solving

$$v = \frac{25}{4} \text{ cm}$$

$$\text{Magnification } m = \frac{I}{O} = \frac{v}{u}$$

$$I = \frac{v}{u} \times O = \frac{25}{4 \times (-25)} \times 4$$

$$I = -1 \text{ cm}$$

1

$\frac{1}{2}$

1

$\frac{1}{2}$

12. A point object is kept 60 cm in front of a spherical convex surface ($n = 1.5$, radius of curvature 40 cm). The image formed is
- (A) real, at a distance 1.8 m from the surface.
 - (B) virtual, at a distance 1.8 m from the surface.
 - (C) real, at a distance 3.6 m from the surface.
 - (D) virtual, at a distance 3.6 m from the surface.

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12. A point object is kept 60 cm in front of a spherical convex surface ($n = 1.5$, radius of curvature 40 cm). The image formed is
- (A) real, at a distance 1.8 m from the surface.
 - (B) virtual, at a distance 1.8 m from the surface.
 - (C) real, at a distance 3.6 m from the surface.
 - (D) virtual, at a distance 3.6 m from the surface.

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(D) virtual, at a distance of 3.6 m from the surface.

19. A thin converging lens of focal length 10 cm is placed coaxially in contact with a thin diverging lens of focal length 15 cm. How will the combination behave ? Justify your answer.

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It will behave like a converging lens.

Power of converging lens is more than the power of diverging lens. Hence the combination will behave like a converging lens.

Alternatively

$$P = P_1 + P_2$$
$$= \frac{100}{10} + \frac{100}{-15}$$

$$P = \frac{10}{3} \text{ D}$$

Alternatively

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$\frac{1}{f} = \frac{1}{10} - \frac{1}{15}$$

$$\frac{1}{f} = \frac{1}{30}$$

$$f = 30 \text{ cm}$$

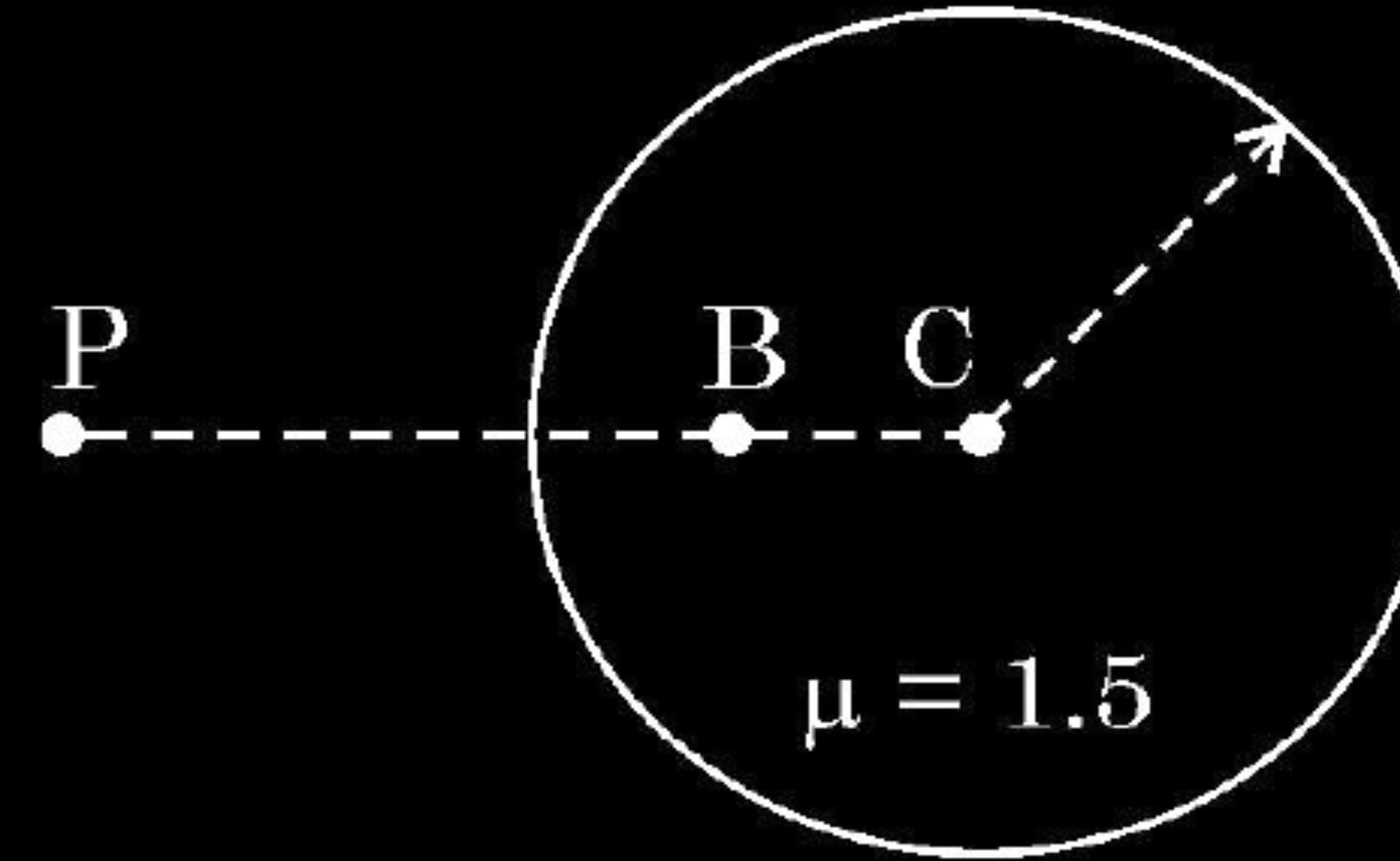
1

1

17. (a) An air bubble is trapped at point B ($CB = 20$ cm) in a glass sphere of radius 40 cm and refractive index 1.5 as shown in figure. Find the nature and position of the image of the bubble as seen by an observer at point P.

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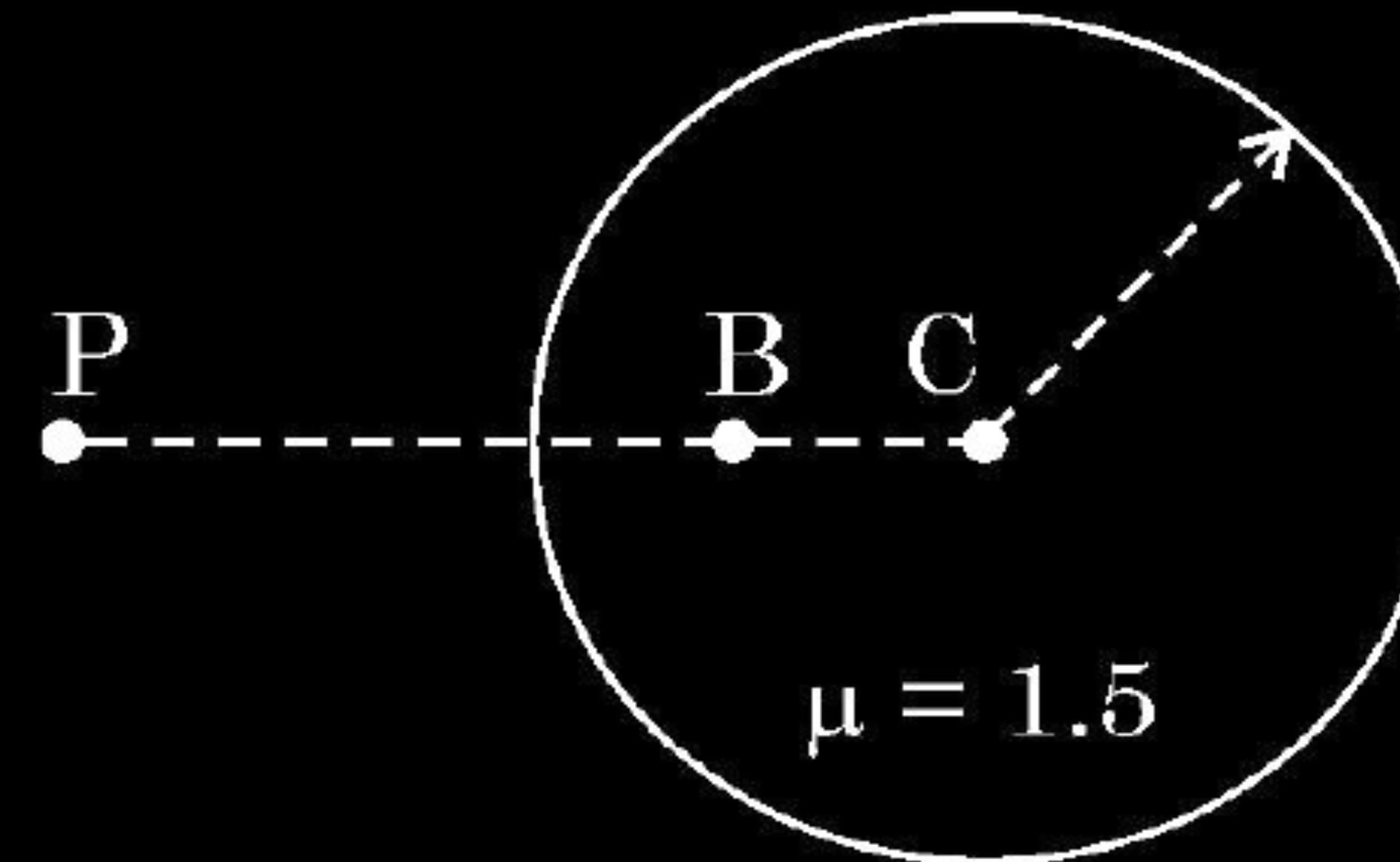
2



17. (a) An air bubble is trapped at point B (CB = 20 cm) in a glass sphere of radius 40 cm and refractive index 1.5 as shown in figure. Find the nature and position of the image of the bubble as seen by an observer at point P.

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2



Using refraction formula at spherical surface from denser to rarer medium

n_1 = refractive index of rarer medium

n_2 = refractive index of denser medium

$$\frac{n_1}{v} - \frac{n_2}{u} = \frac{n_1 - n_2}{R}$$

$$u = -20 \text{ cm}, R = -40 \text{ cm}, n_1 = 1, n_2 = 1.5$$

$$\frac{1}{v} - \frac{1.5}{(-20)} = \frac{1 - 1.5}{(-40)}$$

$$v = -16 \text{ cm}$$

Nature of image is virtual.

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

30. A lens is a transparent optical medium bounded by two surfaces; at least one of which should be spherical. Applying the formula of image formation by a single spherical surface successively at the two surfaces of a thin lens, a formula known as lens maker's formula and hence the basic lens formula can be obtained. The focal length (or power) of a lens depends on the radii of its surfaces and the refractive index of its material with respect to the surrounding medium. The refractive index of a material depends on the wavelength of light used. Combination of lenses helps us to obtain diverging or converging lenses of desired power and magnification.

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$4 \times 1 = 4$

- (i) A thin converging lens of focal length 20 cm and a thin diverging lens of focal length 15 cm are placed coaxially in contact. The power of the combination is

(A) $-\frac{5}{6} \text{ D}$

(B) $-\frac{5}{3} \text{ D}$

(C) $\frac{4}{3} \text{ D}$

(D) $\frac{3}{2} \text{ D}$

30. A lens is a transparent optical medium bounded by two surfaces; at least one of which should be spherical. Applying the formula of image formation by a single spherical surface successively at the two surfaces of a thin lens, a formula known as lens maker's formula and hence the basic lens formula can be obtained. The focal length (or power) of a lens depends on the radii of its surfaces and the refractive index of its material with respect to the surrounding medium. The refractive index of a material depends on the wavelength of light used. Combination of lenses helps us to obtain diverging or converging lenses of desired power and magnification.

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$4 \times 1 = 4$

- (i) A thin converging lens of focal length 20 cm and a thin diverging lens of focal length 15 cm are placed coaxially in contact. The power of the combination is

(A) $\frac{-5}{6} D$

(B) $\frac{-5}{3} D$

(C) $\frac{4}{3} D$

(D) $\frac{3}{2} D$

(B) $\frac{-5}{3} D$

- (ii) The radii of curvature of two surfaces of a convex lens are R and $2R$.
If the focal length of this lens is $\left(\frac{4}{3}\right)R$, the refractive index of the material of the lens is :

(A) $\frac{5}{3}$

(B) $\frac{4}{3}$

(C) $\frac{3}{2}$

(D) $\frac{7}{5}$

(ii) The radii of curvature of two surfaces of a convex lens are R and $2R$.
If the focal length of this lens is $\left(\frac{4}{3}\right)R$, the refractive index of the material of the lens is :

(A) $\frac{5}{3}$

(B) $\frac{4}{3}$

(C) $\frac{3}{2}$

(D) $\frac{7}{5}$

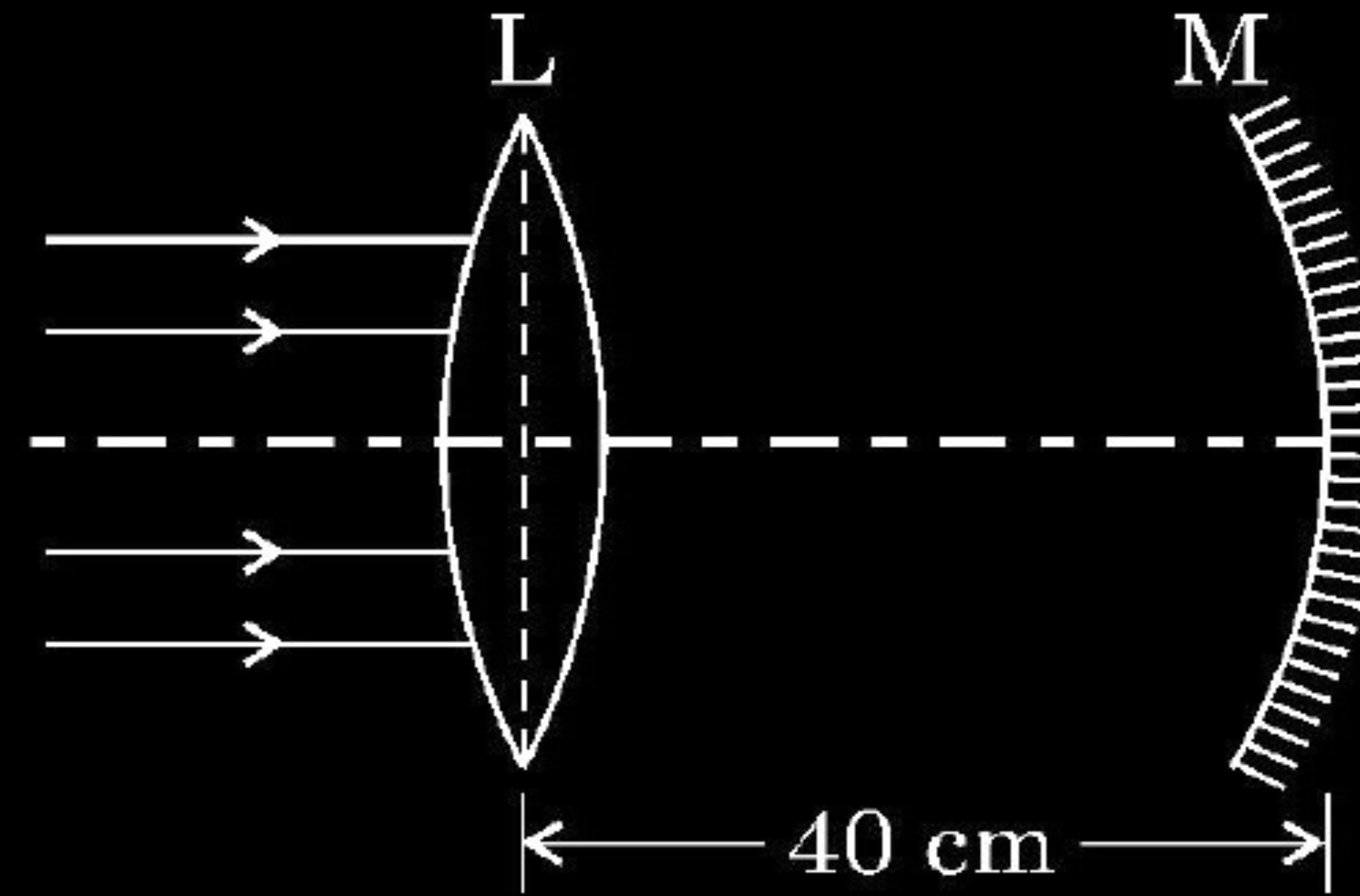
(C) $\frac{3}{2}$

- (iii) The focal length of an equiconvex lens
- (A) increases when the lens is dipped in water.
 - (B) increases when the wavelength of incident light decreases.
 - (C) increases with decrease in radius of curvature of its surface.
 - (D) decreases when the lens is cut into two identical parts along its principal axis.

- (iii) The focal length of an equiconvex lens
- (A) increases when the lens is dipped in water.
 - (B) increases when the wavelength of incident light decreases.
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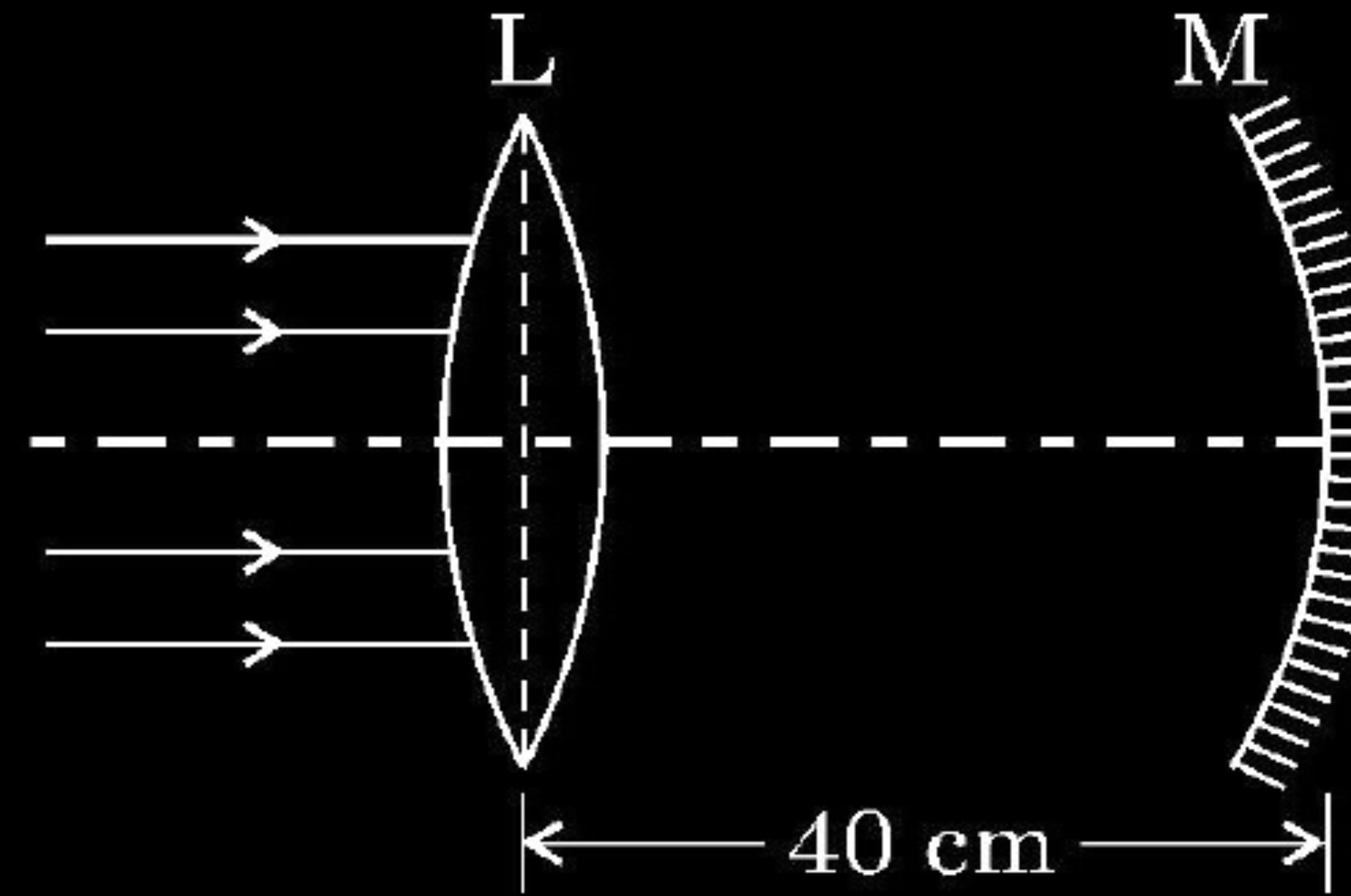
(A) increases when a lens is dipped in water.

- (iv) (a) A thin convex lens L of focal length 10 cm and a concave mirror M of focal length 15 cm are placed coaxially 40 cm apart as shown in figure. A beam of light coming parallel to the principal axis is incident on the lens. The final image will be formed at a distance of



- | | |
|-------------------------|--------------------------|
| (A) 10 cm, left of lens | (B) 10 cm, right of lens |
| (C) 20 cm, left of lens | (D) 20 cm, right of lens |

- (iv) (a) A thin convex lens L of focal length 10 cm and a concave mirror M of focal length 15 cm are placed coaxially 40 cm apart as shown in figure. A beam of light coming parallel to the principal axis is incident on the lens. The final image will be formed at a distance of



- | | |
|-------------------------|--------------------------|
| (A) 10 cm, left of lens | (B) 10 cm, right of lens |
| (C) 20 cm, left of lens | (D) 20 cm, right of lens |

(B) 10 cm , right from lens.

(iv) (b) A beam of light coming parallel to the principal axis of a convex lens L_1 of focal length 16 cm is incident on it. Another convex lens L_2 of focal length 12 cm is placed coaxially at a distance 40 cm from L_1 . The nature and distance of the final image from L_2 will be

(A) real, 24 cm

(B) virtual, 12 cm

(C) real, 32 cm

(D) virtual, 18 cm

(iv) (b) A beam of light coming parallel to the principal axis of a convex lens L_1 of focal length 16 cm is incident on it. Another convex lens L_2 of focal length 12 cm is placed coaxially at a distance 40 cm from L_1 . The nature and distance of the final image from L_2 will be

(A) real, 24 cm

(B) virtual, 12 cm

(C) real, 32 cm

(D) virtual, 18 cm

(A) real , 24 cm

16. *Assertion (A) :* A convex lens, when immersed in a liquid, disappears.

Reason (R) : The refractive indices of material of the lens and the liquid are equal.

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16. *Assertion (A) :* A convex lens, when immersed in a liquid, disappears.

Reason (R) : The refractive indices of material of the lens and the liquid are equal.

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(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

- 21.** A convex lens ($n = 1.52$) has a focal length of 15.0 cm in air. Find its focal length when it is immersed in liquid of refractive index 1.65 . What will be the nature of the lens ?

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2

- 21.** A convex lens ($n = 1.52$) has a focal length of 15.0 cm in air. Find its focal length when it is immersed in liquid of refractive index 1.65 . What will be the nature of the lens ?

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2

For convex lens in air	
$\frac{1}{f_a} = \left(\frac{n_g}{n_a} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$	
For convex lens in liquid.	
$\frac{1}{f_l} = \left(\frac{n_g}{n_l} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$	$\frac{1}{2}$
$\frac{f_l}{f_a} = \frac{\frac{1.52 - 1}{1.52 - 1.65}}{1.65}$	$\frac{1}{2}$
$= -6.6$	
$f_l = -6.6 f_a$	$\frac{1}{2}$
$= -99 \text{ cm}$	
Nature of the lens: Diverging/ behaves like a concave lens.	$\frac{1}{2}$

- (b) A plano-convex lens of focal length 16 cm is made of a material of refractive index 1.4. Calculate the radius of the curved surface of the lens.

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2

- (b) A plano-convex lens of focal length 16 cm is made of a material of refractive index 1.4. Calculate the radius of the curved surface of the lens.

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2

$$\begin{aligned}\frac{1}{f} &= (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \\ \frac{1}{16} &= (1.4 - 1) \left(\frac{1}{R} - \frac{1}{\infty} \right) \\ \frac{1}{16} &= 0.4 \times \frac{1}{R} \\ R &= 16 \times 0.4 \\ R &= 6.4 \text{ cm}\end{aligned}$$

1

1

- 29.** A lens is a transparent medium bounded by two surfaces, with one or both surfaces being spherical. The focal length of a lens is determined by the radii of curvature of its two surfaces and the refractive index of its medium with respect to that of the surrounding medium. The power of a lens is reciprocal of its focal length. If a number of lenses are kept in contact, the power of the combination is the algebraic sum of the powers of the individual lenses.

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- (i) A double-convex lens, with each face having same radius of curvature R , is made of glass of refractive index n . Its power is : *1*

(A) $\frac{2(n-1)}{R}$

(B) $\frac{(2n-1)}{R}$

(C) $\frac{(n-1)}{2R}$

(D) $\frac{(2n-1)}{2R}$

- 29.** A lens is a transparent medium bounded by two surfaces, with one or both surfaces being spherical. The focal length of a lens is determined by the radii of curvature of its two surfaces and the refractive index of its medium with respect to that of the surrounding medium. The power of a lens is reciprocal of its focal length. If a number of lenses are kept in contact, the power of the combination is the algebraic sum of the powers of the individual lenses.

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- (i) A double-convex lens, with each face having same radius of curvature R , is made of glass of refractive index n . Its power is : 1

(A) $\frac{2(n-1)}{R}$

(B) $\frac{(2n-1)}{R}$

(C) $\frac{(n-1)}{2R}$

(D) $\frac{(2n-1)}{2R}$

(A) $\frac{2(n-1)}{R}$

(ii) A double-convex lens of power P , with each face having same radius of curvature, is cut into two equal parts perpendicular to its principal axis. The power of one part of the lens will be :

1

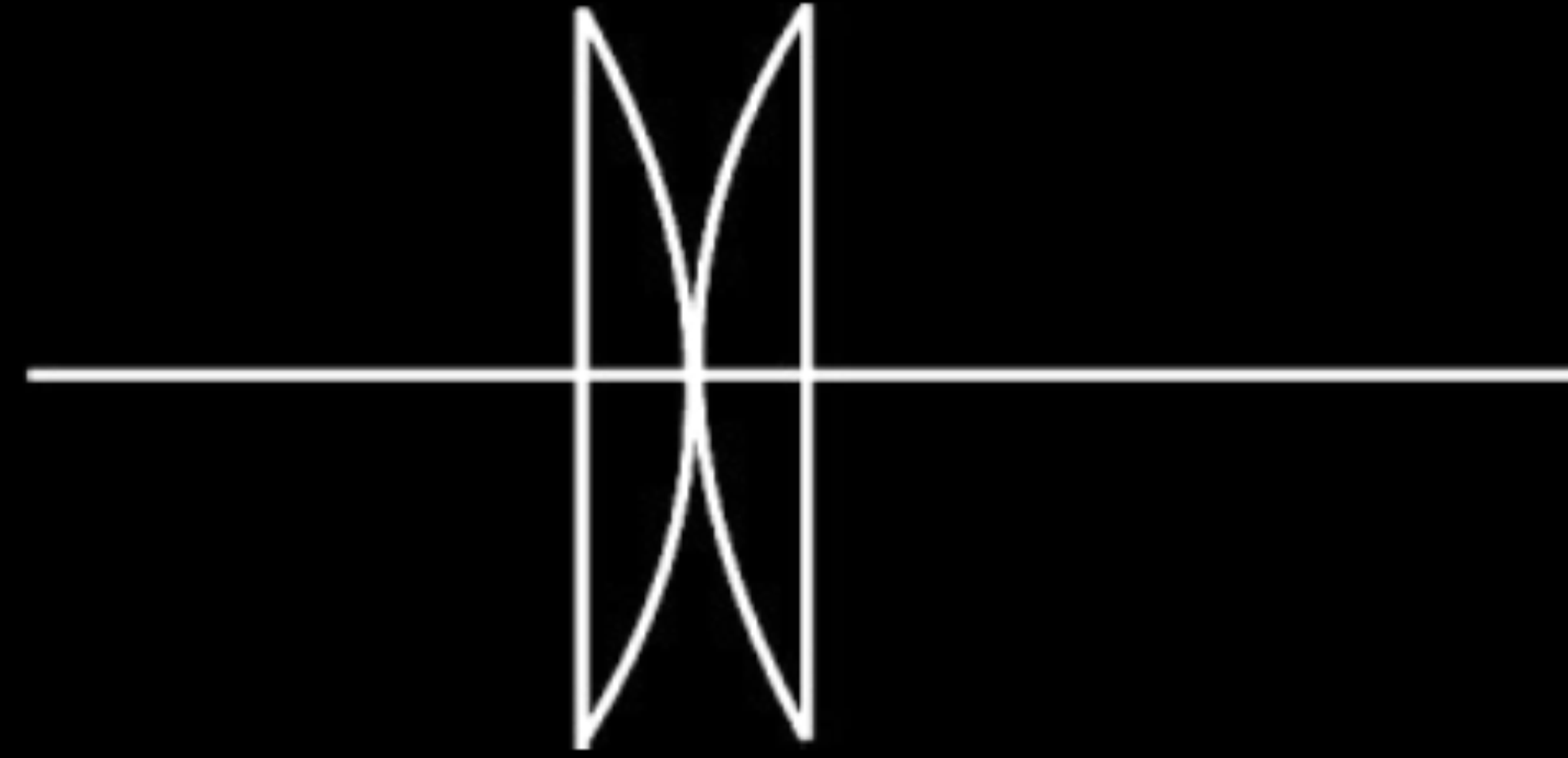
- (A) $2P$ (B) P (C) $4P$ (D) $\frac{P}{2}$

(ii) A double-convex lens of power P , with each face having same radius of curvature, is cut into two equal parts perpendicular to its principal axis. The power of one part of the lens will be : *1*

- (A) $2P$ (B) P (C) $4P$ (D) $\frac{P}{2}$

(D) $P/2$

- (iii) The above two parts are kept in contact with each other as shown in the figure. The power of the combination will be : 1



(A) $\frac{P}{2}$

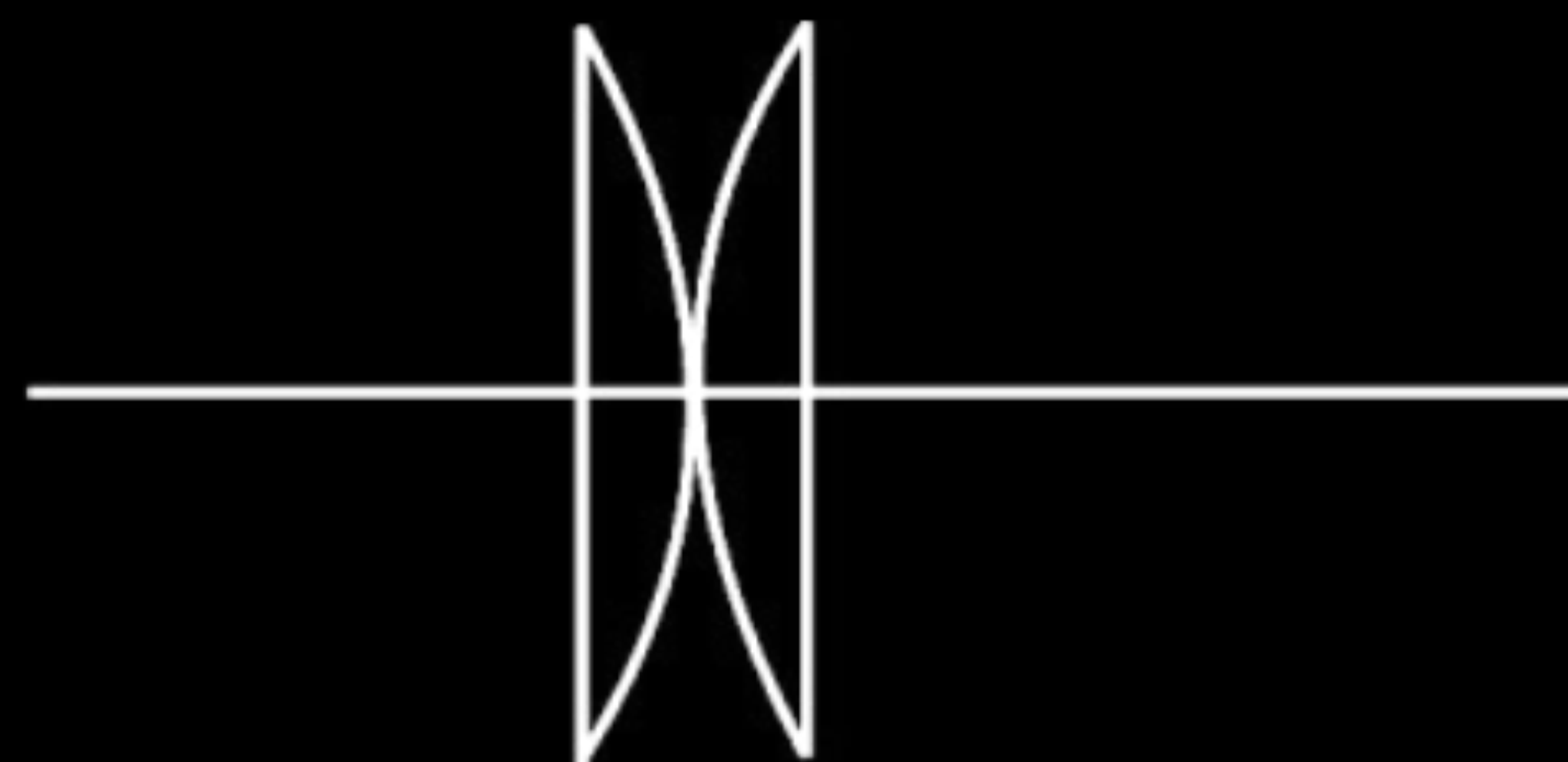
(B) P

(C) $2P$

(D) $\frac{P}{4}$

- (iii) The above two parts are kept in contact with each other as shown in the figure. The power of the combination will be :

1



(A) $\frac{P}{2}$

(B) P

(C) $2P$

(D) $\frac{P}{4}$

(B) P

- (iv) (a) A double-convex lens of power P , with each face having same radius of curvature, is cut along its principal axis. The two parts are arranged as shown in the figure. The power of the combination will be :

1



(A) Zero

(B) P

(C) $2P$

(D) $\frac{P}{2}$

- (iv) (a) A double-convex lens of power P , with each face having same radius of curvature, is cut along its principal axis. The two parts are arranged as shown in the figure. The power of the combination will be :

1



(A) Zero

(B) P

(C) $2P$

(D) $\frac{P}{2}$

(C) $2P$

(b) Two convex lenses of focal lengths 60 cm and 20 cm are held coaxially in contact with each other. The power of the combination is :

1

(A) 6.6 D

(B) 15 D

(C) $\frac{1}{15}$ D

(D) $\frac{1}{80}$ D

(b) Two convex lenses of focal lengths 60 cm and 20 cm are held coaxially in contact with each other. The power of the combination is :

1

(A) 6.6 D

(B) 15 D

(C) $\frac{1}{15}$ D

(D) $\frac{1}{80}$ D

(A) 6.6 D

- 18.** (a) Name the factors on which focal length of a lens depends.
How will a thin biconvex glass lens submerged in water
behave ? Give reason for your answer.

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18. (a) Name the factors on which focal length of a lens depends.
How will a thin biconvex glass lens submerged in water
behave ? Give reason for your answer.

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2

Two factors:

Radius of curvature

Refractive index of the material of the lens w.r.t. the surroundings

$\frac{1}{2} + \frac{1}{2}$

Nature of the lens will not change but the focal length will increase

$\frac{1}{2}$

Reason:

Focal length of a lens is inversely proportional to the refractive index of the material of lens w.r.t. the surroundings.

$\frac{1}{2}$

- 19.** The power of the combination of two thin convex lenses L_1 and L_2 , kept coaxially in contact, is +10 D. If the focal length of L_1 is four times that of L_2 , find their focal lengths.

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19. The power of the combination of two thin convex lenses L_1 and L_2 , kept coaxially in contact, is +10 D. If the focal length of L_1 is four times that of L_2 , find their focal lengths.

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2

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$f_1 = 4f_2 \quad (\text{given})$$

$$P = \frac{1}{f} = \frac{1}{4f_2} + \frac{1}{f_2}$$

$$10 = \frac{5}{4f_2}$$

$$f_2 = \frac{50}{4} \text{ cm}$$

$$\therefore f_1 = 50 \text{ cm}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- 22.** An equiconvex lens of focal length f is cut into two equal halves perpendicular to its principal axis. Find the focal length of each half of the lens.

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2

- 22.** An equiconvex lens of focal length f is cut into two equal halves perpendicular to its principal axis. Find the focal length of each half of the lens.

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2

$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

For equiconvex lens $R_1 = R$, $R_2 = -R$

$$\frac{1}{f} = (n-1)\left(\frac{1}{R} + \frac{1}{R}\right)$$

$$\frac{1}{f} = (n-1)\left(\frac{2}{R}\right) \quad \text{_____ (i)}$$

After cutting in two parts

$$R_1 = R \quad R_2 = \infty$$

$$\frac{1}{f'} = (n-1)\left(\frac{1}{R_1} + \frac{1}{\infty}\right)$$

$$\frac{1}{f'} = (n-1)\left(\frac{1}{R}\right) \quad \text{_____ (ii)}$$

From (i) and (ii)

$$f' = 2f$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

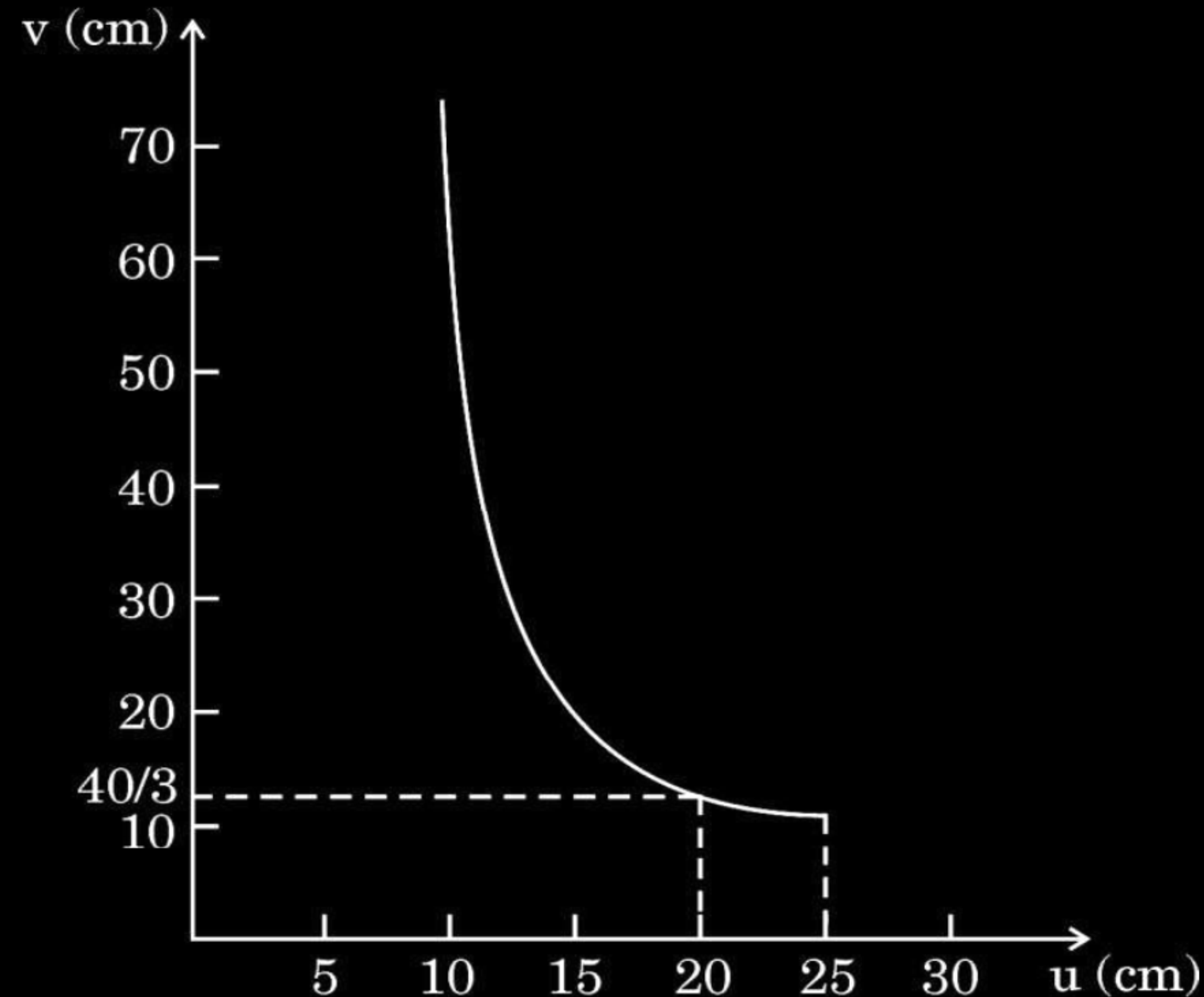
$\frac{1}{2}$

- (b) (i) Name the factors on which the focal length of a lens depends.
- (ii) A converging lens is immersed in a transparent medium. Explain the condition under which the lens (1) becomes invisible and (2) behaves as a diverging lens.
- (iii) Is it possible for two lenses in contact to have zero power ? Explain.

(i) Radius of curvature of two faces of lens Refractive index of lens with respect to surrounding	1
(ii) (1) If the refractive index of material of surrounding medium is equal to refractive index of lens	1
(2) If the refractive of surrounding medium is more than the refractive index of converging lens	1
(iii) yes	$\frac{1}{2}$
If one lens is converging and other is diverging, both having same power.	$\frac{1}{2}$

- (a) Two converging lenses of focal lengths f_1 and f_2 are placed coaxially in contact. Derive the expression for the focal length of the combination using the ray diagram for the formation of the image of an object by the combination.
- (b) The figure shows the variation of image distance v with object distance u for a lens.
- What is the nature of the lens ?
 - Using the graph, find the focal length of the lens.

5



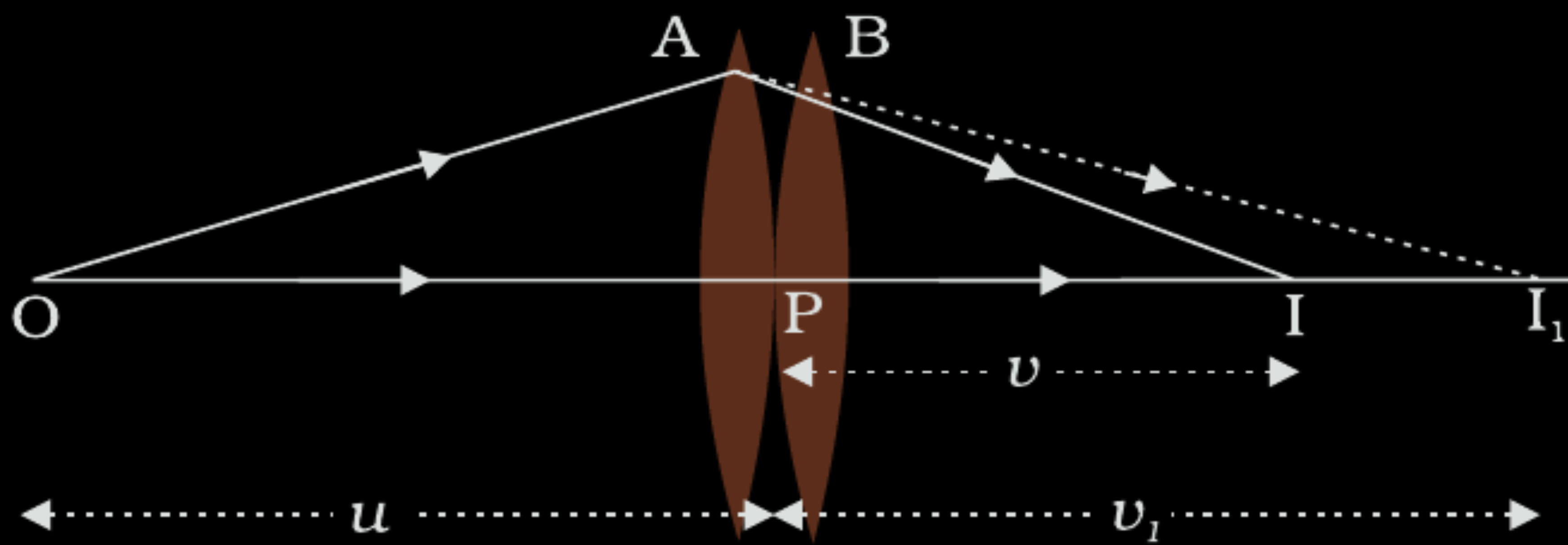


Image formed by the first lens A,

$$\frac{1}{v_1} - \frac{1}{u} = \frac{1}{f_1} \text{—————(1)}$$

Image formed by second lens B,

$$\frac{1}{v} - \frac{1}{v_1} = \frac{1}{f_2} \text{—————(2)}$$

Adding eq (1) and (2), we get,

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f_1} + \frac{1}{f_2} \text{—————(3)}$$

If the two lens system is regarded as equivalent to a single lens of focal length f, we have

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \text{—————(4)}$$

From equation (3) and (4) ,we get

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$

½

½

½

½

(b) (i) From graph we see that as the object moves away from the lens the image comes nearer. Therefore it is a convex lens

(ii) $u = -20cm$

$$v = \frac{40}{3} cm$$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$= \frac{3}{40} + \frac{1}{20}$$

$$f = 8cm$$

½+½

½

½

31.

- (ii) A biconvex lens of focal length 20 cm is immersed in water, whose refractive index is $\frac{4}{3}$. Find the change, if any, in the nature and the focal length of the lens. Refractive index of the material of convex lens is $\frac{3}{2}$.

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$$(ii) \frac{1}{f_a} = (n_g - 1) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

$$\frac{1}{f_a} = \left(\frac{n_g}{n_w} - 1 \right) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

$$\frac{f_w}{f_a} = \frac{(n_g - 1)}{\left(\frac{n_g}{n_w} - 1 \right)}$$

$$\frac{f_w}{20} = \frac{\left(\frac{3}{2} - 1 \right)}{\left(\frac{3}{2} \times \frac{3}{4} - 1 \right)} = \frac{\cancel{1/2}}{\cancel{1/8}} = 4$$

$$f_w = 80 \text{ cm}$$

$$\begin{aligned} \text{Changes in focal length } \Delta f &= 80 - 60 \\ &= 20 \text{ cm} \end{aligned}$$

Nature of the lense is converging because f_w is positive

1/2

1/2

1/2

1/2

1/2

1/2

(b)

- (ii) An equiconvex lens of radius of curvature R and made of glass of refractive index μ is cut into two identical plano-convex lenses. Find the focal length of the plano-convex lenses.

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(ii)	
$\frac{1}{f} = (n - 1) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$	$\frac{1}{2}$
$\frac{1}{f_1} = (n - 1) \left[\frac{2}{R} \right]$	$\frac{1}{2}$
$\frac{1}{f_2} = (n - 1) \left[\frac{1}{R} \right]$	$\frac{1}{2}$
$f_2 = 2 f_1$	

22. A convex lens is made of the material of refractive index n_1 . What will happen if it were immersed in a medium of refractive index n_2 in the following cases ?

2

(a) $n_2 > n_1$

(b) $n_2 = n_1$

22. A convex lens is made of the material of refractive index n_1 . What will happen if it were immersed in a medium of refractive index n_2 in the following cases ?

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2

(a) $n_2 > n_1$

(b) $n_2 = n_1$

Focal length of a lens in a medium

$$\frac{1}{f} = \left(\frac{n_1}{n_2} - 1\right) \left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

n_1 = refractive index of lens

n_2 = refractive index of medium

(a) $n_2 > n_1$

f becomes negative

The convex lens will behave as a concave lens/diverging lens

1

(b) $n_2 = n_1$

$$\frac{1}{f} = 0$$

$$f = \infty$$

Lens will be invisible in the medium / behave as a glass plate.

1

33.

- (ii) A thin converging lens of focal length 10 cm is placed coaxially in contact with a thin diverging lens of focal length 15 cm. Find the nature and focal length of the combined lens.

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(ii) Focal length of combination of two lenses

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$f_1 = +10 \text{ cm}$$

$$f_2 = -15 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{10} + \frac{1}{-15} = \frac{3 - 2}{30} = \frac{1}{30}$$

$$f = +30 \text{ cm}$$

Nature- Converging

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

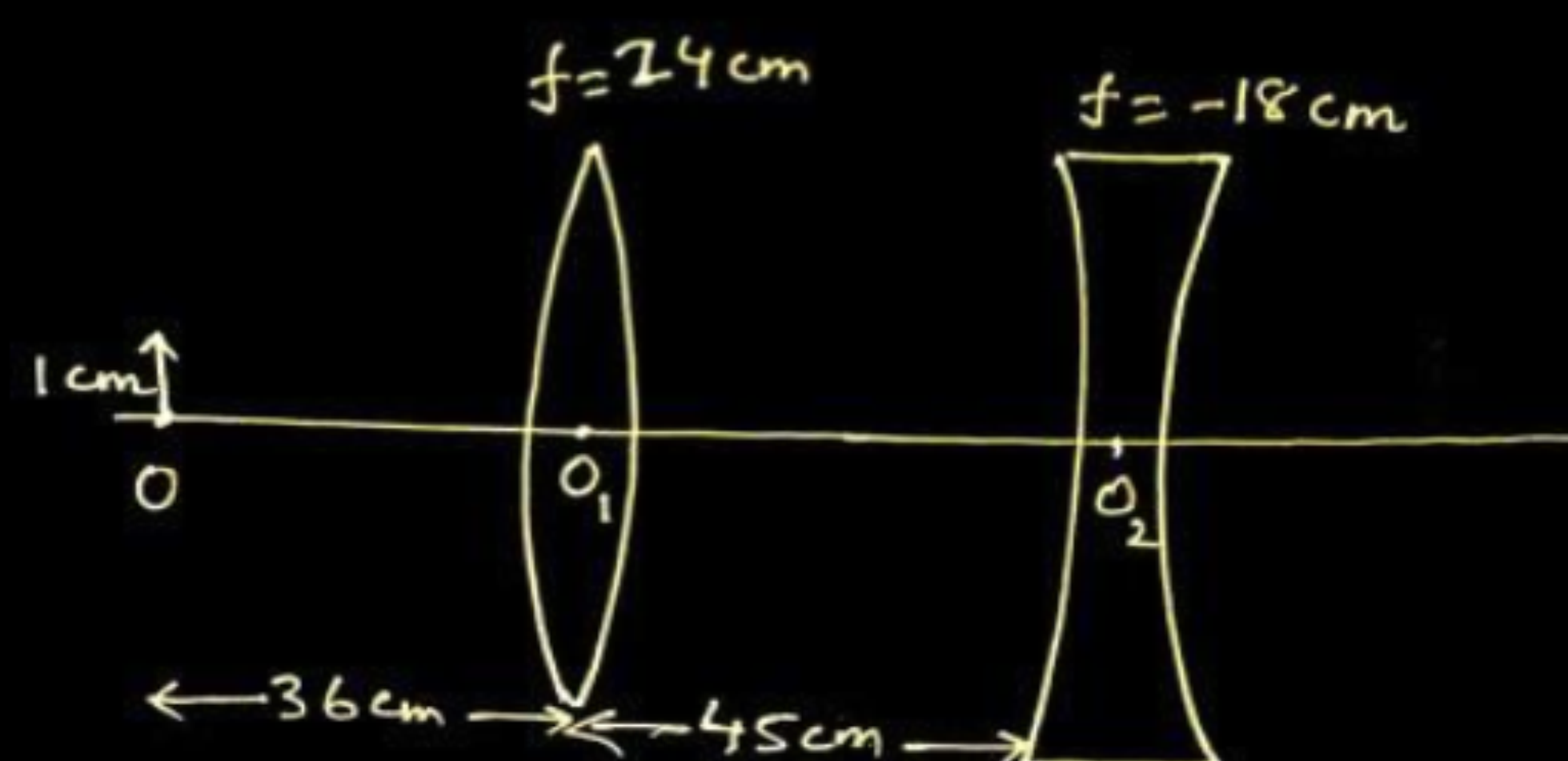
$\frac{1}{2}$

33. (a)

- (ii) Two thin lenses L_1 and L_2 , L_1 being a convex lens of focal length 24 cm and L_2 a concave lens of focal length 18 cm are placed coaxially at a separation of 45 cm. A 1 cm tall object is placed in front of the lens L_1 at a distance of 36 cm. Find the location and height of the image formed by the combination.

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(11)

For lens L_1 ,

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} - \frac{1}{-36} = \frac{1}{24}$$

$$\frac{1}{v} = \frac{1}{24} - \frac{1}{36}$$

$$\frac{1}{v} = \frac{3-2}{72} = \frac{1}{72}$$

$$v = 72 \text{ cm}$$

For lens L_2 :

$$\frac{1}{v'} - \frac{1}{u'} = \frac{1}{f'}$$

$$\frac{1}{v'} - \frac{1}{(72-45)} = \frac{1}{-18}$$

$$\frac{1}{v'} = \frac{1}{-18} + \frac{1}{27}$$

$$\frac{1}{v'} = \frac{-3+2}{54} = \frac{-1}{54}$$

$$v' = -54 \text{ cm}$$

Final distance $v'_1 = -54 - (-45)$

$$v'_1 = -9 \text{ cm (to the left of convex lens)}$$

Magnification $\frac{h_i}{h_o} = \frac{v'_1}{u}$

$$\frac{h_i}{1} = \frac{-9}{-36} \Rightarrow h_i = +\frac{1}{4} \text{ cm}$$

 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

- 35.** The lens maker's formula is useful to design lenses of desired focal lengths using surfaces of suitable radii of curvature. The focal length also depends on the refractive index of the material of the lens and the surrounding medium. The refractive index depends on the wavelength of the light used. The power of a lens is related to its focal length.

Answer the following questions based on the above :

- (a) How will the power of a lens be affected with an increase of wavelength of light ? 1
- (b) The radius of curvature of two surfaces of a convex lens is R each. For what value of μ of its material will its focal length become equal to R ? 1

(a) $P = \frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$ $\mu \propto \frac{1}{\lambda} \Rightarrow P \propto \frac{1}{\lambda}$ So P decreases as λ increases.	1
(b) $\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$ $\frac{1}{R} = (\mu - 1) \frac{2}{R} \quad (R_1 = R, R_2 = -R)$ $\Rightarrow \mu = \frac{3}{2}$	$\frac{1}{2}$

- (c) The focal length of a concave lens of $\mu = 1.5$ is 20 cm in air. It is completely immersed in water of $\mu = \frac{4}{3}$. Calculate its focal length in water. 2

OR

- (c) An object is placed in front of a lens which forms its erect image of magnification 3. The power of the lens is 5 D. Calculate the distance of the object and the image from the lens. 2

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(c) $\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$	$\frac{1}{2}$
$\frac{1}{20} = \left(\frac{3}{2} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$	$\frac{1}{2}$
When the lens is immersed in water	
$\frac{1}{f_{\text{unknown}}} = \left(\frac{9}{8} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$	$\frac{1}{2}$
$\frac{\frac{1}{20}}{\frac{1}{f}} = \frac{\cancel{\frac{1}{2}}}{\cancel{\frac{1}{8}}} = \Rightarrow f = 80 \text{ cm}$	$\frac{1}{2}$
OR	
$m = 3 = \frac{v}{u} \Rightarrow v = 3u$	$\frac{1}{2}$
$f = \frac{100}{+5} = +20 \text{ cm}$	$\frac{1}{2}$
$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$	
$\frac{1}{3u} - \frac{1}{u} = \frac{1}{20}$	
$\frac{1-3}{3u} = \frac{1}{20}$	
$u = -(40/3) \text{ cm}$	$\frac{1}{2}$
$v = -40 \text{ cm}$	$\frac{1}{2}$

31. (a)

- (ii) A 1.8 m tall person stands in front of a convex lens of focal length 1 m, at a distance of 5 m. Find the position and height of the image formed.

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(ii) For lens: $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$	$\frac{1}{2}$
$u = -5m; \quad f = +1m$	
$\frac{1}{v} - \frac{1}{-5} = \frac{1}{(+1)}$	
$\Rightarrow v = \frac{5}{4}m = 1.25m$	$\frac{1}{2}$
$m = \frac{I}{O} = \frac{v}{u} = \frac{\left(+\frac{5}{4}\right)}{(-5)}$	$\frac{1}{2}$
$I = (-0.25) \times (1.8)$	
$I = -0.45 \text{ m}$	$\frac{1}{2}$

(b)

- (ii) The radii of curvature of the two surfaces of a concave lens are 20 cm each. Find the refractive index of the material of the lens if its power is -5.0 D.

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(ii) Given; P = - 5D	
$f \text{ (in cm)} = \frac{100}{(-5)} = - 20 \text{ cm}$	$\frac{1}{2}$
Using Lens Maker's formula ; $\frac{1}{f} = (\mu - 1) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$	$\frac{1}{2}$
$\frac{1}{(-20)} = (\mu - 1) \left[\frac{1}{(-20)} - \frac{1}{(+20)} \right]$	
$\frac{1}{(-20)} = (\mu - 1) \left[-\frac{1}{10} \right]; \quad \mu - 1 = \frac{1}{2}$	$\frac{1}{2}$
$\Rightarrow \mu = \frac{3}{2} = 1.5$	$\frac{1}{2}$

- 19.** The radii of curvature of the faces of a double-convex lens are 20 cm and 30 cm. Its power is $\frac{25}{6}$ D. What is the refractive index of the glass of the lens ?

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- 19.** The radii of curvature of the faces of a double-convex lens are 20 cm and 30 cm. Its power is $\frac{25}{6}$ D. What is the refractive index of the glass of the lens ?

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2

$$R_1 = 20 \text{ cm}, R_2 = -30 \text{ cm} \quad P = \frac{25}{6} D = \frac{25}{600} \text{ cm}^{-1}$$

$$P = (n-1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\frac{25}{600} = (n-1) \left(\frac{1}{20} + \frac{1}{30} \right)$$

$$\frac{25}{600} = (n-1) \left(\frac{50}{20 \times 30} \right)$$

$$25 = (n-1)50$$

$$n-1 = \frac{1}{2}$$

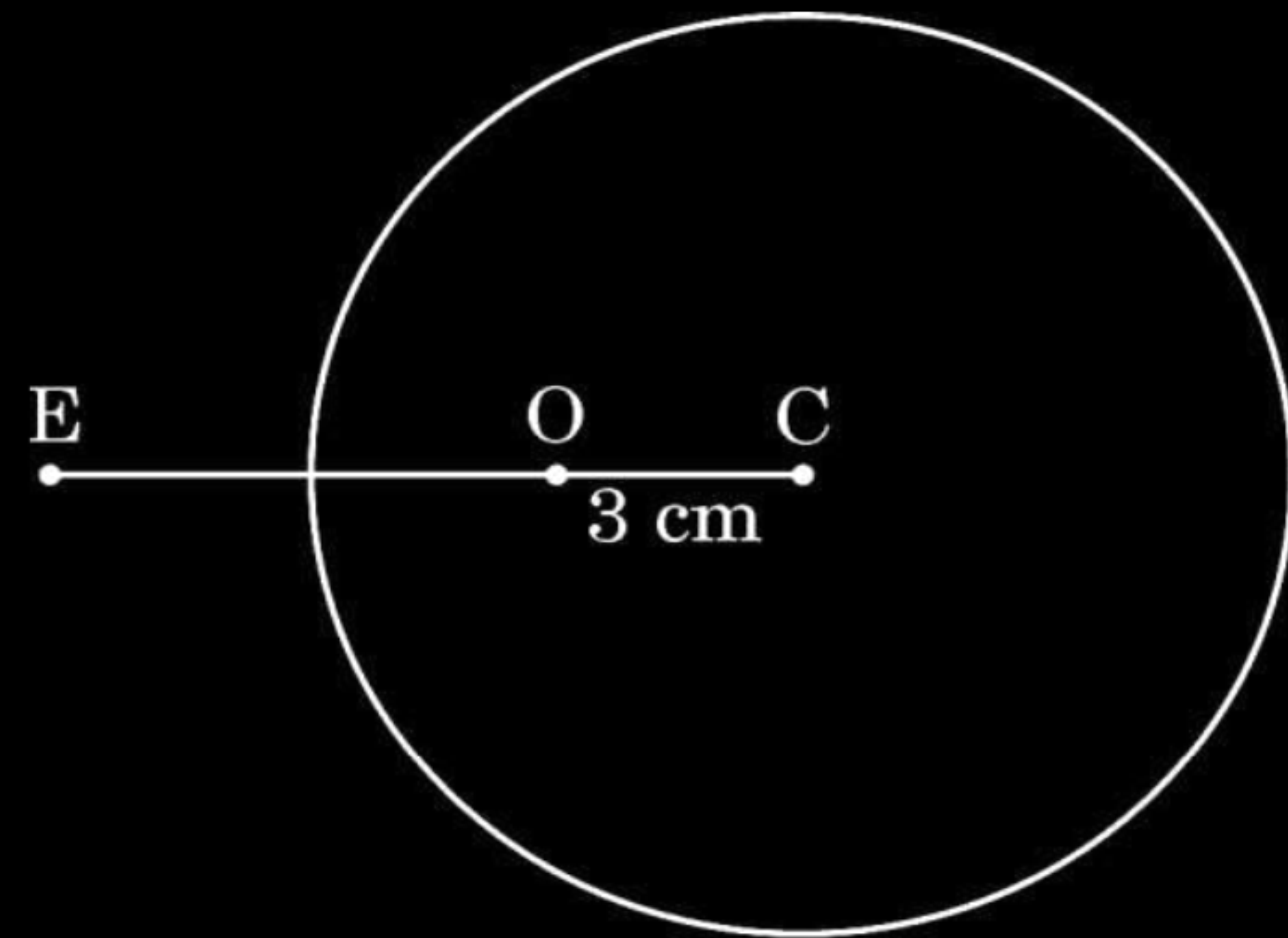
$$n = \frac{3}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$



- (b) (i) A spherical surface of radius of curvature R separates two media of refractive indices n_1 and n_2 . A point object is placed in front of the surface at distance u in medium of refractive index n_1 and its image is formed by the surface at distance v , in the medium of refractive index n_2 . Derive a relation between u and v .

- (ii) A solid glass sphere of radius 6.0 cm has a small air bubble trapped at a distance 3.0 cm from its centre C as shown in the figure. The refractive index of the material of the sphere is 1.5 . Find the apparent position of this bubble when seen through the surface of the sphere from an outside point E in air.

Assume that the aperture of the surface is small as compared to other distance involved, so that small angle approximation can be made.
 For small angles

for $\triangle NOC$, I is the exterior angle

$$\therefore I = \angle NOM + \angle NCM$$

$$i = \frac{MN}{OM} + \frac{MN}{MC} \tag{i}$$

$$\begin{aligned} \text{Similarly } r &= \angle NCM - \angle NIM \\ &= \frac{MN}{MC} - \frac{MN}{MI} \tag{ii} \end{aligned}$$

By Snell's law
 $n_1 \sin I = n_2 \sin r$
 for small angles
 $n_1 I = n_2 r$

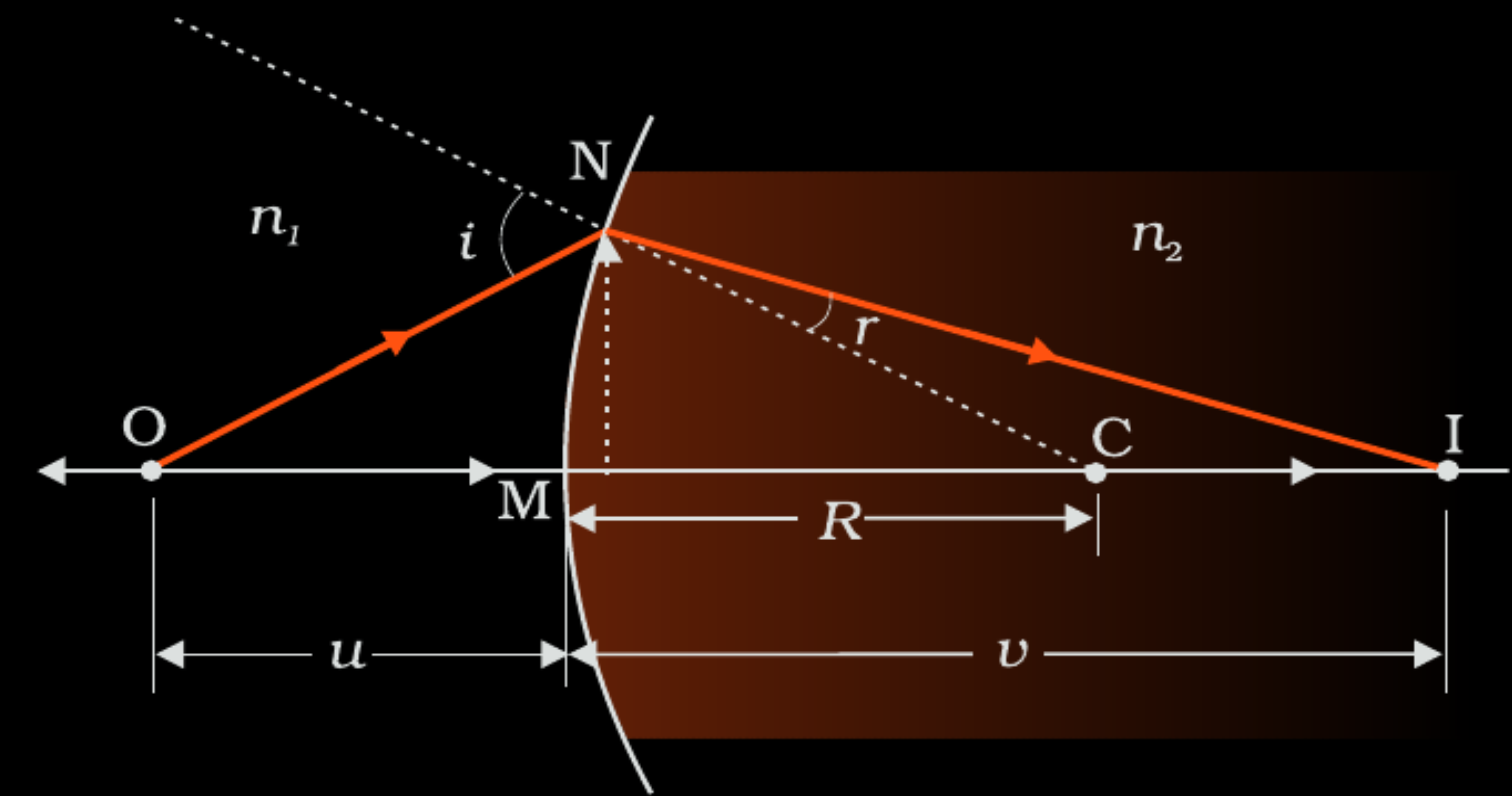
substituting I and r from (i) and (ii) we get

$$\frac{n_1}{OM} + \frac{n_2}{MI} = \frac{n_2 - n_1}{MC}$$

Applying Cartesian coordinates

$$OM = -u, MI = +v, MC = +R$$

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$



(ii) $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$	$\frac{1}{2}$
$R = -6 \text{ cm}, u = -3 \text{ cm}, n_1 = 1.5, n_2 = 1$	
$\frac{1}{v} + \frac{1.5}{3} = \frac{1 - 1.5}{-6}$	$\frac{1}{2}$
$\frac{1}{v} = \frac{0.5}{6} - \frac{1.5}{3}$	$\frac{1}{2}$
$\frac{1}{v} = \frac{0.5 - 3}{6}$	
$\frac{1}{v} = \frac{-2.5}{6}$	$\frac{1}{2}$
$v = -2.4 \text{ cm}$	
from the left surface inside the sphere	$\frac{1}{2}$

3. The minimum distance between an object and its real image formed by a convex lens of focal length f is :

(a) f

(b) $2f$

(c) $3f$

(d) $4f$

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3. The minimum distance between an object and its real image formed by a convex lens of focal length f is :

(a) f

(b) $2f$

(c) $3f$

(d) $4f$

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(d) $4f$

- 24.** A double-convex lens of power 4 D is manufactured from a glass of refractive index 1.5. What is the radius of curvature of each face of this lens if both faces have the same radius of curvature ?

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2

- 24.** A double-convex lens of power 4 D is manufactured from a glass of refractive index 1.5. What is the radius of curvature of each face of this lens if both faces have the same radius of curvature ?

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2

$P = 4D, \quad n = 1.5$	
$P = \frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$	$\frac{1}{2}$
$R_1 = R, \quad R_2 = -R$	
$P = \frac{1}{f} = (n - 1) \left(\frac{2}{R} \right)$	$\frac{1}{2}$
$4 = (1.5 - 1) \left(\frac{2}{R} \right)$	$\frac{1}{2}$
$4 = \left(\frac{0.5 \times 2}{R} \right)$	
$R = \frac{1}{4} = 0.25m = 25 \text{ cm}$	$\frac{1}{2}$

23. The power of a thin lens is +5 D. When it is immersed in a liquid, it behaves like a concave lens of focal length 100 cm. Calculate the refractive index of the liquid. Given refractive index of glass = 1.5.

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23. The power of a thin lens is +5 D. When it is immersed in a liquid, it behaves like a concave lens of focal length 100 cm. Calculate the refractive index of the liquid. Given refractive index of glass = 1.5.

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2

$$P = +5 \text{ D}, \quad f_e = -100 \text{ cm}, \quad \mu_g = 1.5, \quad \mu_l = ?$$

$$f_a = \frac{1}{P} = \frac{1}{5} = 0.2 \text{ m} = 20 \text{ cm}$$

$$\frac{1}{f_a} = (\mu_{ga} - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\frac{1}{20} = (1.5 - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \text{-----(1)}$$

$$\frac{1}{f_l} = \left(\frac{\mu_g}{\mu_l} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\frac{1}{(-100)} = \left(\frac{1.5}{\mu_l} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \text{-----(2)}$$

From (1) and (2), on solving

$$\mu_l = \frac{5}{3} = 1.67$$

1/2

1/2

1/2

1/2

25. A point object in air is placed symmetrically at a distance of 60 cm in front of a concave spherical surface of refractive index 1.5. If the radius of curvature of the surface is 20 cm, find the position of the image formed. **2**

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25. A point object in air is placed symmetrically at a distance of 60 cm in front of a concave spherical surface of refractive index 1.5. If the radius of curvature of the surface is 20 cm, find the position of the image formed. **2**

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$$\mu = 1.5, u = -60 \text{ cm}, R = -20 \text{ cm}, v = ?$$

$$\frac{\mu}{v} - \frac{1}{u} = \frac{\mu - 1}{R}$$

$$\frac{1.5}{v} - \frac{1}{(-60)} = \frac{1.5 - 1}{(-20)}$$

$$v = -36 \text{ cm}$$

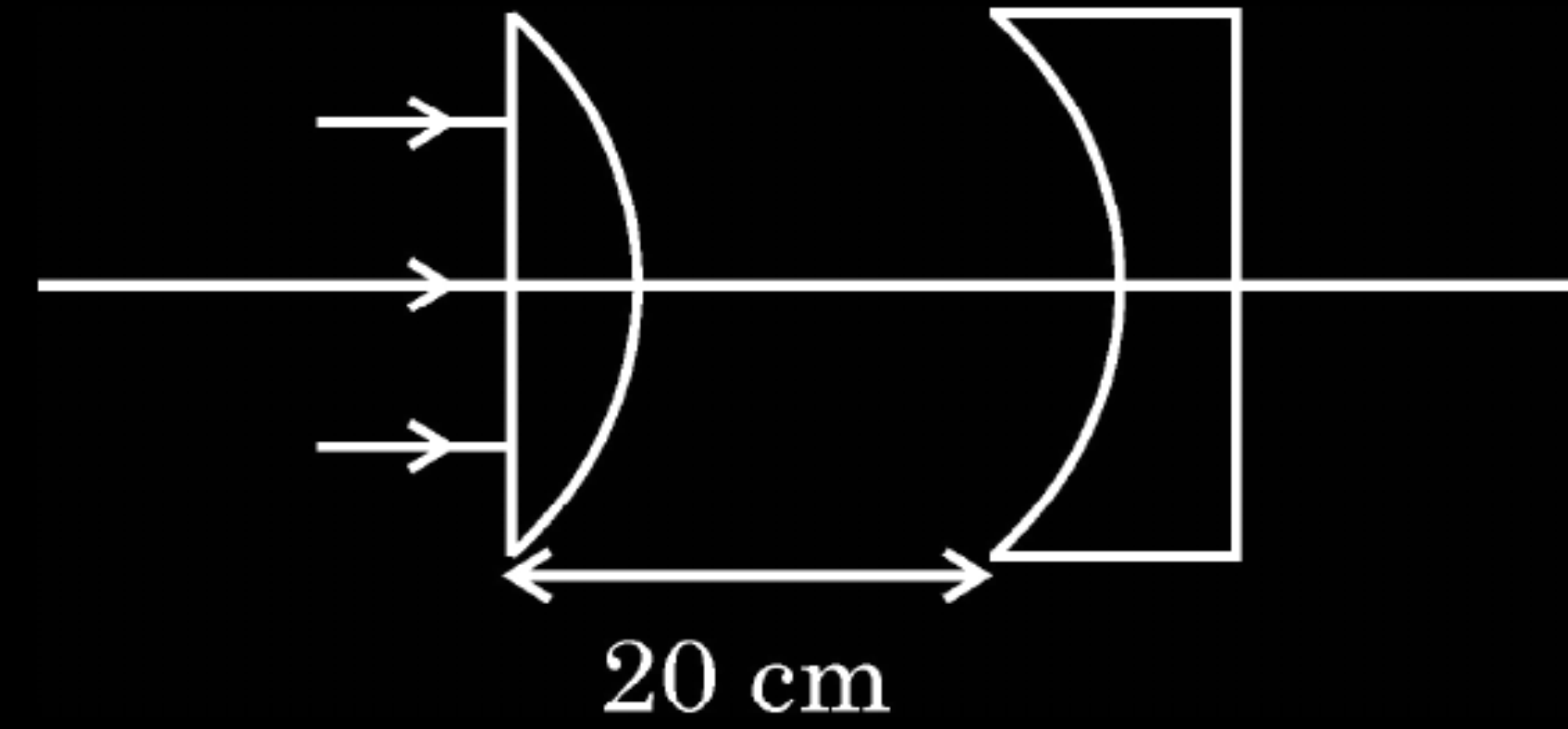
$$\frac{1}{2}$$

$$1$$

$$\frac{1}{2}$$

21. In the given figure the radius of curvature of curved face in the plano-convex and the plano-concave lens is 15 cm each. The refractive index of the material of the lenses is 1.5. Find the final position of the image formed.

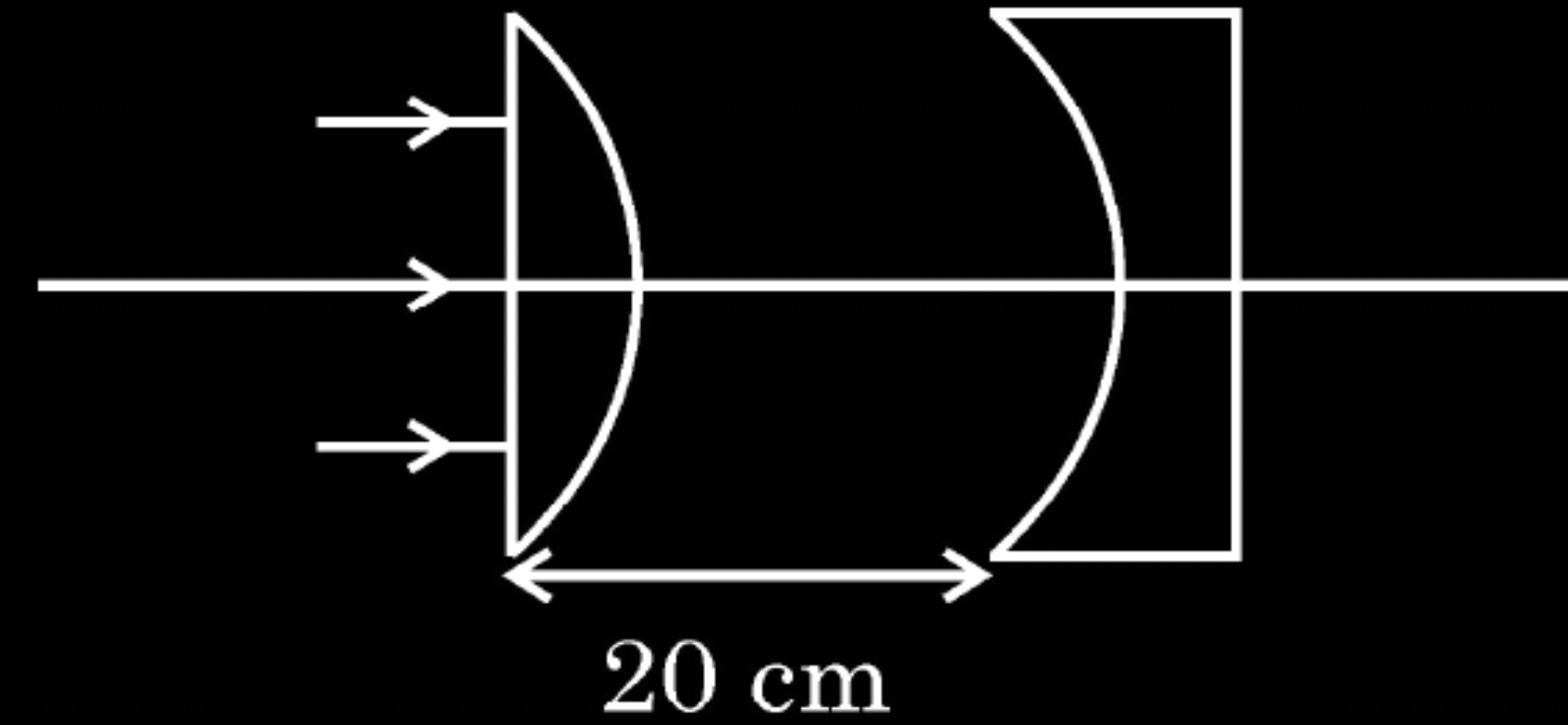
2



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21. In the given figure the radius of curvature of curved face in the plano-convex and the plano-concave lens is 15 cm each. The refractive index of the material of the lenses is 1.5. Find the final position of the image formed.

2



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Using the formula $\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$

- Focal length of plano-convex lens = +30 cm
- Focal length of plano-concave lens = -30 cm

- For plano-convex lens

As object is at ∞ , its real image will be formed at its focus i.e +30 cm

$$v_1 = +30 \text{ cm}$$

- For plano-concave lens $u = +(30-20) \text{ cm}$
= +10 cm

$$\frac{1}{f_2} = \left(\frac{1}{v_2} - \frac{1}{u_2} \right)$$

$$\frac{1}{-30} = \left(\frac{1}{v_2} - \frac{1}{10} \right)$$

$$\therefore v_2 = 15 \text{ cm}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

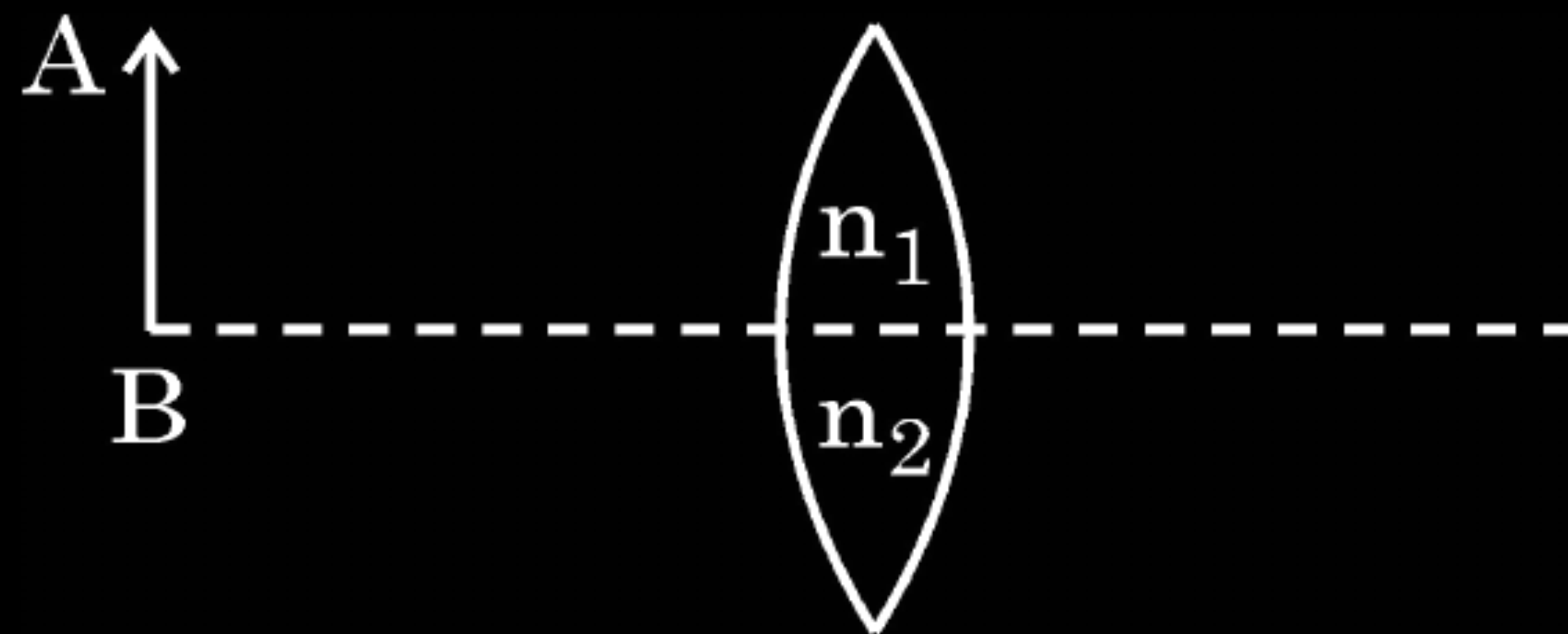
$\frac{1}{2}$

34. A lens is a transparent optical medium bounded by two surfaces; at least one of which should be spherical. Considering image formation by a single spherical surface successively at the two surfaces of a lens, lens maker's formula is obtained. It is useful to design lenses of desired focal length using surfaces of suitable radii of curvature. This formula helps us obtain a relation between u , v and f for a lens. Lenses form images of objects and they are used in a number of optical devices, for example microscopes and telescopes.

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- (i) An object AB is kept in front of a composite convex lens, as shown in figure. Will the lens produce one image ? If not, explain.



- (ii) A real image of an object formed by a convex lens is observed on a screen. If the screen is removed, will the image still be formed ? Explain.
- (iii) A double convex lens is made of glass of refractive index 1.55 with both faces of the same radius of curvature. Find the radius of curvature required if focal length is 20 cm.

OR

- (iii) Two convex lenses A and B of focal lengths 15 cm and 10 cm respectively are placed coaxially 'd' distance apart. A point object is kept at a distance of 30 cm in front of lens A. Find the value of 'd' so that the rays emerging from lens B are parallel to its principal axis.

(i) No, The lens is made up of two materials of different refractive indices. It has two focal lengths.	$\frac{1}{2}$ $\frac{1}{2}$	For lens A (f=15 cm)	
(ii) Yes Rays are still intersecting/ converging at the location of image.	$\frac{1}{2}$ $\frac{1}{2}$	$u = -30 \text{ cm}$ $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$	$\frac{1}{2}$
(iii) $\frac{1}{f} = (n-1) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$ $R_1 = +R, R_2 = -R$ $\frac{1}{f} = (n-1) \left[\frac{2}{R} \right]$ $R = 2(n-1)f$ $R = 2(1.55-1) \times 20 = 22 \text{ cm}$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	$\frac{1}{v_1} = \frac{1}{15} - \frac{1}{30} = \frac{1}{30}$ $v_1 = 30 \text{ cm}$ For lens B For rays to go parallel to principal axis out of lens B the image formed by lens A must lie at the focus of B So d= 30+10=40 cm	$\frac{1}{2}$ 1
		Alternatively: (Object is kept at 2f so image will also be formed at 2f on the other side of the lens i.e. at 30 cm. Now the final image is to be formed at infinity so the image formed must lie at the focus of the second lens (B). Thus separation is 30 +10 =40 cm)	

7. (a) Define the term 'Power of a lens' and mention its SI unit.
- (b) A diverging lens is made of a material of refractive index 1.6. The radii of curvature of its two surfaces are 10 cm and 30 cm. Calculate its power.

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7. (a) Define the term 'Power of a lens' and mention its SI unit.
- (b) A diverging lens is made of a material of refractive index 1.6. The radii of curvature of its two surfaces are 10 cm and 30 cm. Calculate its power.

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(a) Power of a lens is defined as the tangent of the angle by which it converges and diverges a beam of light falling at unit distance from the optical centre.

Alternatively

Power :- Ability to converge or diverge the beam of light incident on a lens.

Alternatively

Power of the lens is given by the reciprocal of its focal length.

S.I unit is diopetre or m^{-1} .

$$(b) P = \frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$= (1.6 - 1) \left(\frac{1}{-0.1} + \frac{1}{0.3} \right)$$

$$= (0.6) \left(\frac{-3+1}{0.3} \right)$$

$\frac{1}{2}$

$\frac{1}{2}$

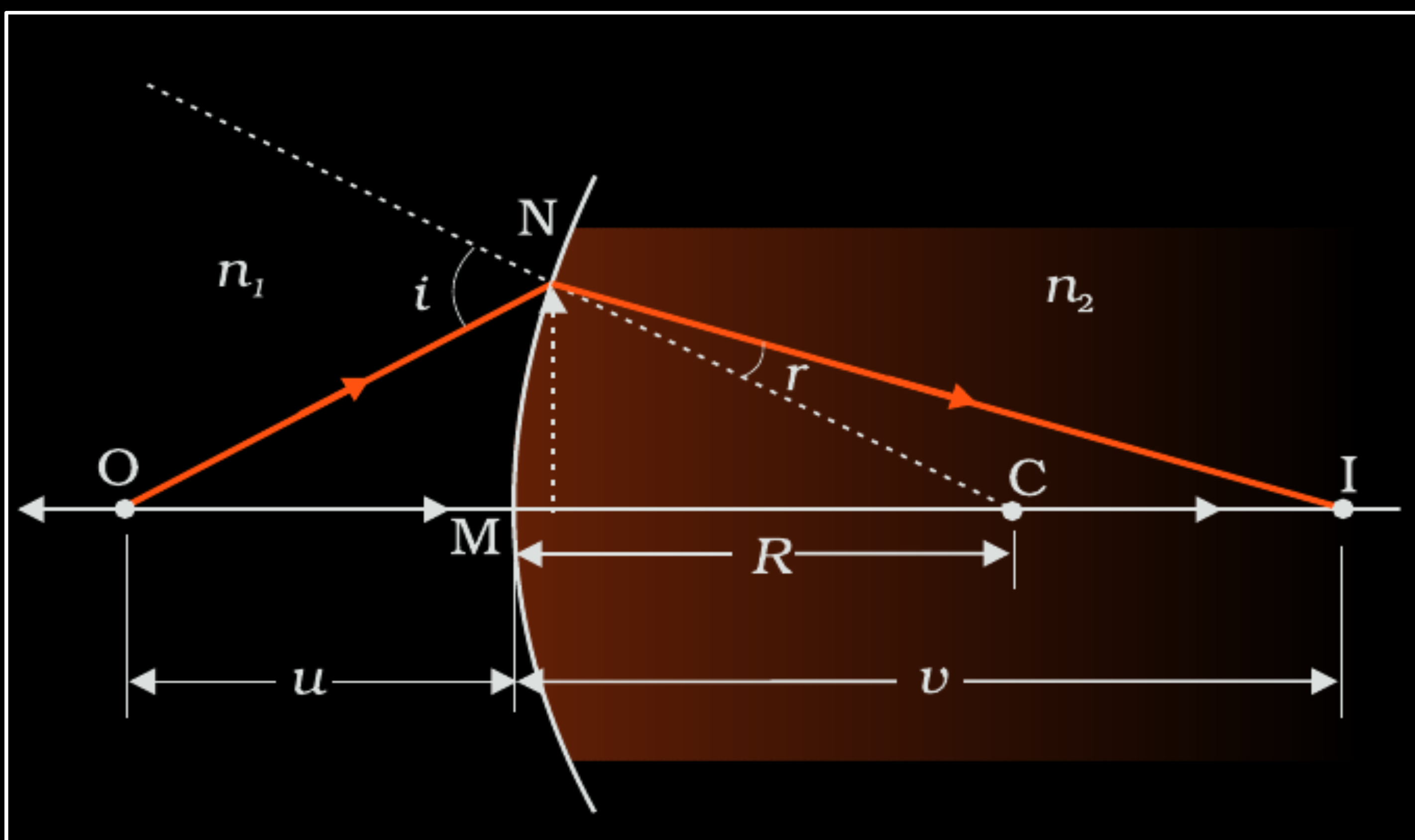
$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- (b) Consider a convex spherical surface of radius of curvature R , separating two media of refractive indices n_1 and n_2 ($> n_1$). A point object is placed at a distance u in front of the surface in medium of refractive index n_1 . Its real image is formed at a distance v . Obtain a relation between u and v in terms of n_1 , n_2 and R .

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for small angles,
 $\tan \angle NOM = \frac{MN}{OM}$

$$\tan \angle NCM = \frac{MN}{MC}$$

$$\tan \angle NIM = \frac{MN}{MI}$$

Now, for $\triangle NOC$, $\angle i$ is the exterior angle. Therefore, $\angle i = \angle NOM + \angle NCM$

$$i = \frac{MN}{OM} + \frac{MN}{MC}$$

Similarly,

$$\text{i.e. } r = \frac{MN}{MC} - \frac{MN}{MI}$$

now, by Snell's law

$$n_1 \sin i = n_2 \sin r$$

or for small angles, $\sin i \approx i$ and $\sin r \approx r$

$$n_1 i = n_2 r$$

Substituting i and r , we get

$$\frac{n_1}{OM} + \frac{n_2}{MI} = \frac{n_2}{MC} - \frac{n_1}{MI}$$

Here, OM , MI and MC represent magnitude of distances. Applying the Cartesian sign convention

$$OM = -u, \quad MI = +v, \quad MC = +R$$

Substituting these in equation, we get

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

12. A lens is a portion of a transparent medium bounded by two surfaces and one of these surfaces is essentially spherical. It is used to converge or diverge the light incident on it. Power of a lens is the measure of its ability to converge or diverge the light incident on it. Power of a lens depends on the refractive index of the material of lens with respect to the surrounding medium and the radii of curvature of its two surfaces.

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- (a) The power of a convex lens
- (i) increases when the lens is dipped in water.
 - (ii) increases when the wavelength of incident light decreases.
 - (iii) decreases when another convex lens is placed in contact with it.
 - (iv) increases when the lens is cut into two identical plano-convex lenses.

12. A lens is a portion of a transparent medium bounded by two surfaces and one of these surfaces is essentially spherical. It is used to converge or diverge the light incident on it. Power of a lens is the measure of its ability to converge or diverge the light incident on it. Power of a lens depends on the refractive index of the material of lens with respect to the surrounding medium and the radii of curvature of its two surfaces.

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- (a) The power of a convex lens
- (i) increases when the lens is dipped in water.
 - (ii) increases when the wavelength of incident light decreases.
 - (iii) decreases when another convex lens is placed in contact with it.
 - (iv) increases when the lens is cut into two identical plano-convex lenses.

(b) The focal length of a concave lens is 40 cm. The power of the lens is :

(i) 0.025 D

(ii) 2.5 D

(iii) – 0.025 D

(iv) – 2.5 D

(b) The focal length of a concave lens is 40 cm. The power of the lens is :

(i) 0.025 D

(ii) 2.5 D

(iii) – 0.025 D

(iv) – 2.5 D

(iv)

(c) The focal length of a concave lens ($\mu_g = 1.5$) in air is 20 cm. What should be the refractive index of the surrounding medium so that the lens behaves as a converging lens of focal length 60 cm ?

(i) 1.4

(ii) 1.8

(iii) 1.7

(iv) 1.2

(c) The focal length of a concave lens ($\mu_g = 1.5$) in air is 20 cm. What should be the refractive index of the surrounding medium so that the lens behaves as a converging lens of focal length 60 cm ?

(i) 1.4

(ii) 1.8

(iii) 1.7

(iv) 1.2

(d) Beams of red light, blue light, yellow light and violet light are incident on a convex lens, one-by-one. Which one of them converges nearest to the lens ?

- (i) Blue light
- (ii) Violet light
- (iii) Red light
- (iv) Yellow light

(d) Beams of red light, blue light, yellow light and violet light are incident on a convex lens, one-by-one. Which one of them converges nearest to the lens ?

- (i) Blue light
- (ii) Violet light
- (iii) Red light
- (iv) Yellow light

(e) A beam of light coming parallel to the principal axis of a convex lens L_1 of focal length 15 cm is incident on it. Another convex lens L_2 of focal length 25 cm is placed coaxially at a distance d from L_1 . For the final image to be formed at infinity, the value of d should be :

- (i) 10 cm
- (ii) 15 cm
- (iii) 25 cm
- (iv) 40 cm

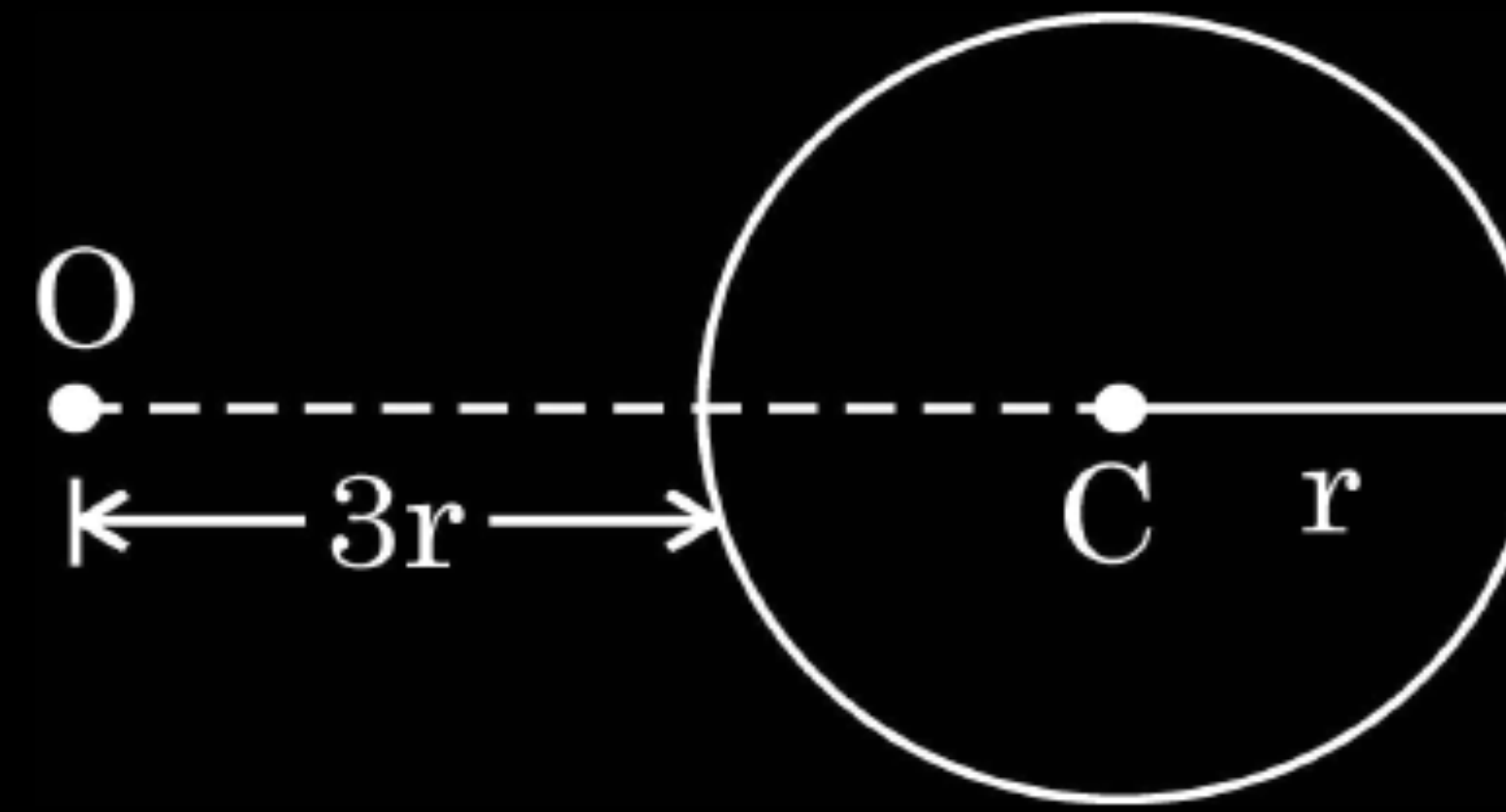
(e) A beam of light coming parallel to the principal axis of a convex lens L_1 of focal length 15 cm is incident on it. Another convex lens L_2 of focal length 25 cm is placed coaxially at a distance d from L_1 . For the final image to be formed at infinity, the value of d should be :

- (i) 10 cm
- (ii) 15 cm
- (iii) 25 cm
- (iv) 40 cm

4. (a) (i) Define SI unit of power of a lens.
- (ii) A plano convex lens is made of glass of refractive index 1.5.
The radius of curvature of the convex surface is 25 cm.
- (ii.i) Calculate the focal length of the lens.
- (ii.ii) If an object is placed 50 cm in front of the lens, find the nature and position of the image formed.

(i) Diopetre is the reciprocal of focal length of lens in metre. <u>Alternatively:</u> $\text{Diopetre} = \frac{1}{\text{focal length (m)}}$ <u>Alternatively:</u> One diopetre is the power of a lens of focal length of one metre	1
ii.i) $\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$ $R_1 = \infty, \quad R_2 = -25 \text{ cm}, \quad \mu = 1.5$ $\frac{1}{f} = (\mu - 1) \left(\frac{1}{\infty} + \frac{1}{25} \right)$ $\text{or } \frac{1}{f} = 0.5 \times \frac{1}{25} \text{ or } f = 50 \text{ cm}$ ii.ii) $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$ $\frac{1}{50} = \frac{1}{v} - \frac{1}{-50}$ $\frac{1}{v} = 0$ $\therefore v = \infty,$ Thus the image will be real and inverted (Note: Award only ½ mark if a student draws ray diagram showing correct position of the image instead of doing calculations)	½ ½ ½ ½

- (a) An object is placed in front of a converging lens. Obtain the conditions under which the magnification produced by the lens is (i) negative and (ii) positive.
- (b) A point object is placed at O in front of a glass sphere as shown in figure.



Show the formation of image by the sphere.

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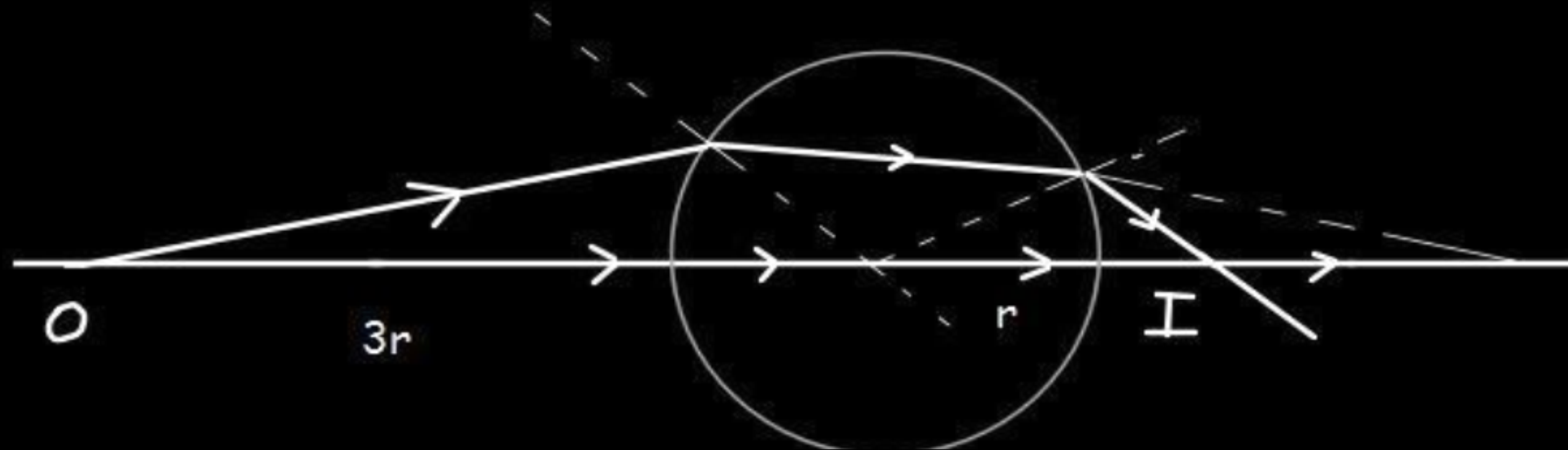
(a) Two conditions $m = \frac{h'}{h} = \frac{v}{u}$

(i) For real images / when object is placed beyond F , As u is negative and v is positive..

$$\frac{1}{2} + \frac{1}{2}$$

(ii) for virtual image / when object is kept between F and the optical centre of the lens. As u and v both are negative.

$$\frac{1}{2} + \frac{1}{2}$$



5. Monochromatic light of wavelength 600 nm is incident from air on a water surface. The refractive index of water is 1.33. Find the (i) wavelength, (ii) frequency and (iii) speed, of reflected and refracted light.

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(a) Reflected light

$$\lambda' = \lambda = 600 \text{ nm.}$$

No change in wavelength

$$c = 3 \times 10^8 \text{ m/s in air}$$

$$\nu = \frac{c}{\lambda} = \frac{3 \times 10^8}{600 \times 10^{-9}} \text{ Hz}$$

Refracted light

$$\lambda' = \frac{\lambda}{\mu} = \frac{600}{1.33} = 451.1 \text{ nm}$$

$$\simeq 450 \text{ nm}$$

$$\nu = \frac{c}{\mu} = \frac{3 \times 10^8}{1.33}$$

$$= 2.26 \times 10^8 \text{ m/s}$$

$$= 5 \times 10^{14} \text{ Hz}$$

1/2

1/2

1/2

1/2

1/2

1/2

(b) A thin equiconvex lens of radius of curvature R made of material of refractive index μ_1 is kept coaxially, in contact with an equiconcave lens of the same radius of curvature and refractive index $\mu_2 (> \mu_1)$. Find :

- (i) the ratio of their powers, and
- (ii) the power of the combination and its nature.

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(i) From $P = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$	$\frac{1}{2}$
$P_1 = P_{convex} = (\mu - 1) \left(\frac{1}{R_1} - \left(-\frac{1}{R_2} \right) \right)$ $= (\mu - 1) \left(\frac{2}{R} \right)$	$\frac{1}{2}$
$P_2 = P_{concave} = (\mu - 1) \left(-\frac{1}{R_1} - \frac{1}{R_2} \right)$ $= -(\mu - 1) \left(\frac{2}{R} \right)$	
$\therefore \frac{P_1}{P_2} = \frac{(\mu_1 - 1)}{-(\mu_2 - 1)} = \frac{(\mu_1 - 1)}{(1 - \mu_2)}$	$\frac{1}{2}$
(ii) $P = P_1 + P_2$	$\frac{1}{2}$
$= (\mu_1 - 1) \left(\frac{2}{R} \right) + (-(\mu_2 - 1)) \left(\frac{2}{R} \right)$ $P = \frac{2(\mu_1 - \mu_2)}{R}$	$\frac{1}{2}$
As $\mu_2 > \mu_1$, P is negative	
$\therefore$ Nature is diverging.	$\frac{1}{2}$

10. Find the two possible positions of an object kept in front of a lens of $+ 5.0$ D, so that the image formed in both cases is four times magnified. CBSE 2022 **3**

10. Find the two possible positions of an object kept in front of a lens of + 5.0 D, so that the image formed in both cases is four times magnified. CBSE 2022 **3**

Power of lens = +5 D	
Focal length of lens = $\frac{100}{5} = 20$ cm	
Magnification ,m = ± 4	$\frac{1}{2}$
Case I ,m = $4 = \frac{v}{u} \therefore v = 4u$	$\frac{1}{2}$
$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$	
$\frac{1}{20} = \frac{1}{4u} - \frac{1}{u}$	
$u = -15cm$	1
Case II , m = $-4 = \frac{v}{u} \therefore v = -4u$	$\frac{1}{2}$
$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$	
$\frac{1}{20} = \frac{1}{-4u} - \frac{1}{u}$	
$u = -25cm$	$\frac{1}{2}$

5. A point source in air is kept 24 cm in front of a concave spherical glass surface ($\mu_g = 1.5$) and radius of curvature 60 cm. Find the nature of the image formed and its distance from the point source.

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5. A point source in air is kept 24 cm in front of a concave spherical glass surface ($\mu_g = 1.5$) and radius of curvature 60 cm. Find the nature of the image formed and its distance from the point source.

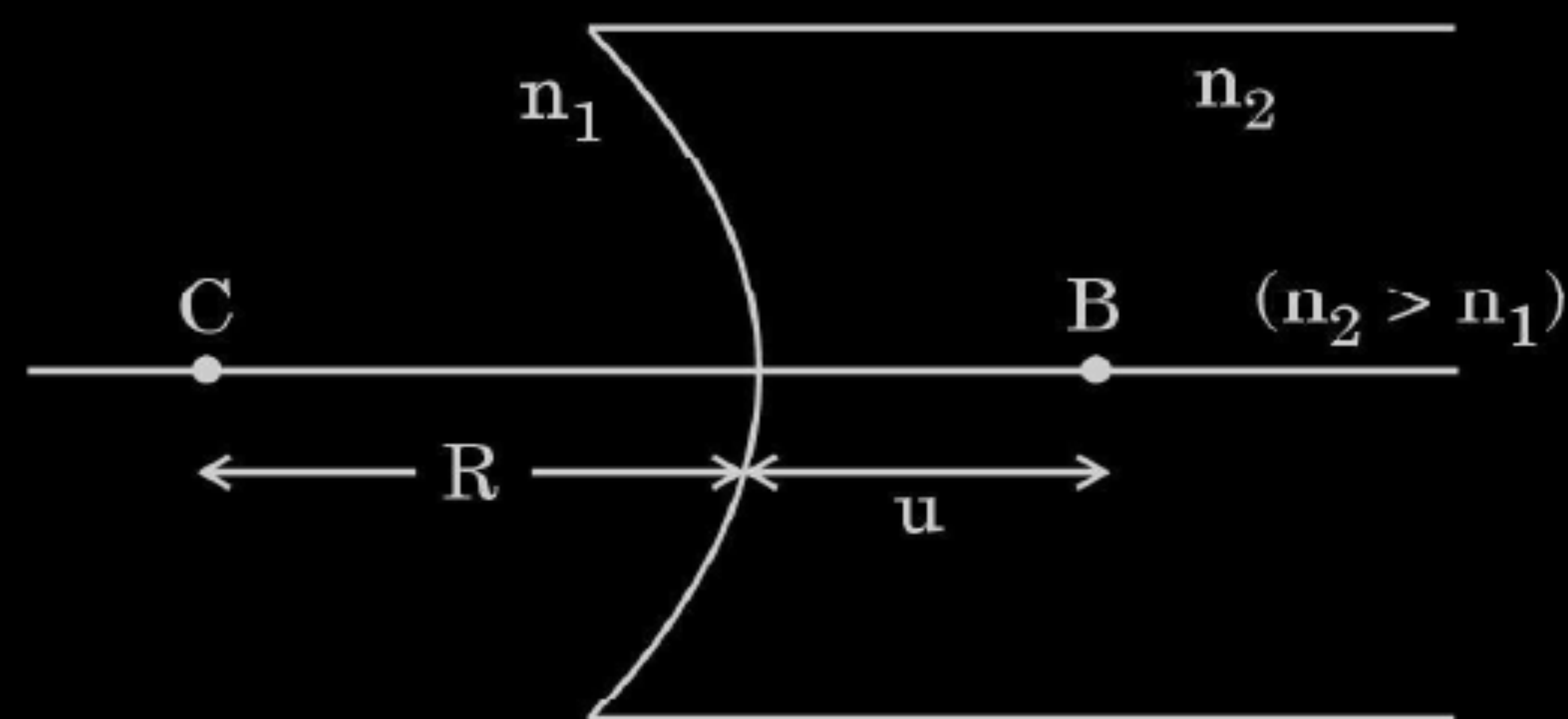
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Relation for concave spherical surface	
$\frac{\mu_1}{-u} + \frac{\mu_2}{v} = \frac{\mu_2 - \mu_1}{R}$	$\frac{1}{2}$
$\frac{1}{-u} + \frac{\mu}{v} = \frac{\mu - 1}{R}$	$\frac{1}{2}$
$\frac{1}{-(-24)} + \frac{1.5}{v} = \frac{1.5 - 1}{-60}$	$\frac{1}{2}$
$\therefore v = -30 \text{ cm}$	
Distance of image from point source = $-24 - (-30) = 6 \text{ cm}$	$\frac{1}{2}$
Nature of image = Virtual image	1

12. Two transparent media of refractive indices n_1 and n_2 are separated by a spherical transparent surface. The rays of light incident on the surface get refracted into the medium on the other side. The laws of refraction are valid at each point of the spherical surface. A lens is a transparent optical medium bounded by two surfaces, at least one of which should be spherical. The focal length of a lens is determined by the radii of curvature (R_1 and R_2) of its two surfaces and the refractive index (n) of the medium of the lens with respect to the surrounding medium. Depending on R_1 and R_2 , a lens behaves as a diverging or a converging lens. The ability of a lens to diverge or converge a beam of light incident on it defines its power.

- (a) An object is placed at the point B as shown in the figure. The object distance (u) and the image distance (v) are related as



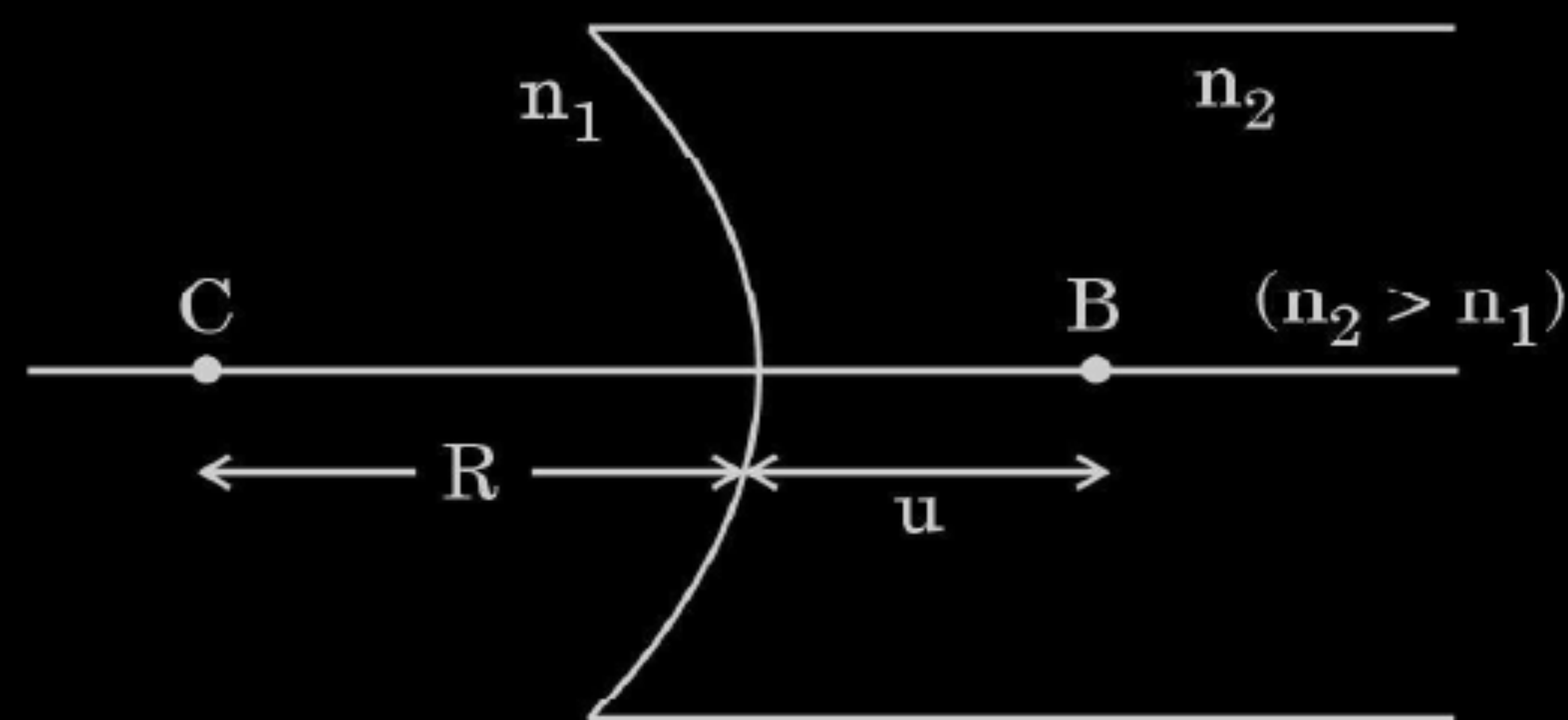
$$(i) \quad \frac{1}{v} - \frac{1}{u} = \left(\frac{n_2 - n_1}{n_1} \right) \frac{1}{R}$$

$$(ii) \quad \frac{1}{v} - \frac{1}{u} = \left(\frac{n_1 - n_2}{n_2} \right) \frac{1}{R}$$

$$(iii) \quad \frac{n_2}{v} - \frac{n_1}{u} = \frac{(n_2 - n_1)}{R}$$

$$(iv) \quad \frac{n_1}{v} - \frac{n_2}{u} = \frac{(n_1 - n_2)}{R}$$

- (a) An object is placed at the point B as shown in the figure. The object distance (u) and the image distance (v) are related as



$$(i) \quad \frac{1}{v} - \frac{1}{u} = \left(\frac{n_2 - n_1}{n_1} \right) \frac{1}{R}$$

$$(ii) \quad \frac{1}{v} - \frac{1}{u} = \left(\frac{n_1 - n_2}{n_2} \right) \frac{1}{R}$$

$$(iii) \quad \frac{n_2}{v} - \frac{n_1}{u} = \frac{(n_2 - n_1)}{R}$$

$$(iv) \quad \frac{n_1}{v} - \frac{n_2}{u} = \frac{(n_1 - n_2)}{R}$$

$$(iv) \quad \frac{n_1}{v} - \frac{n_2}{u} = \frac{(n_1 - n_2)}{R}$$

(b) A point object is placed in air at a distance ' R ' in front of a convex spherical refracting surface of radius of curvature R . If the medium on the other side of the surface is glass, then the image is :

- (i) real and formed in glass.
- (ii) real and formed in air.
- (iii) virtual and formed in glass.
- (iv) virtual and formed in air.

(b) A point object is placed in air at a distance ' R ' in front of a convex spherical refracting surface of radius of curvature R . If the medium on the other side of the surface is glass, then the image is :

- (i) real and formed in glass.
- (ii) real and formed in air.
- (iii) virtual and formed in glass.
- (iv) virtual and formed in air.

(iv) virtual and formed in air

(c) An object is kept at $2F$ in front of an equiconvex lens. The image formed is :

- (i) real and of the size of the object.
- (ii) virtual and of the size of the object.
- (iii) real and enlarged.
- (iv) virtual and diminished.

(c) An object is kept at $2F$ in front of an equiconvex lens. The image formed is :

- (i) real and of the size of the object.
- (ii) virtual and of the size of the object.
- (iii) real and enlarged.
- (iv) virtual and diminished.

(i) real and of the size of the object

(d) A thin converging lens of focal length 10 cm and a thin diverging lens of focal length 20 cm are placed coaxially in contact. The power of the combination is :

(i) -5 D

(ii) $+5 \text{ D}$

(iii) $+15 \text{ D}$

(iv) -15 D

(d) A thin converging lens of focal length 10 cm and a thin diverging lens of focal length 20 cm are placed coaxially in contact. The power of the combination is :

(i) -5 D

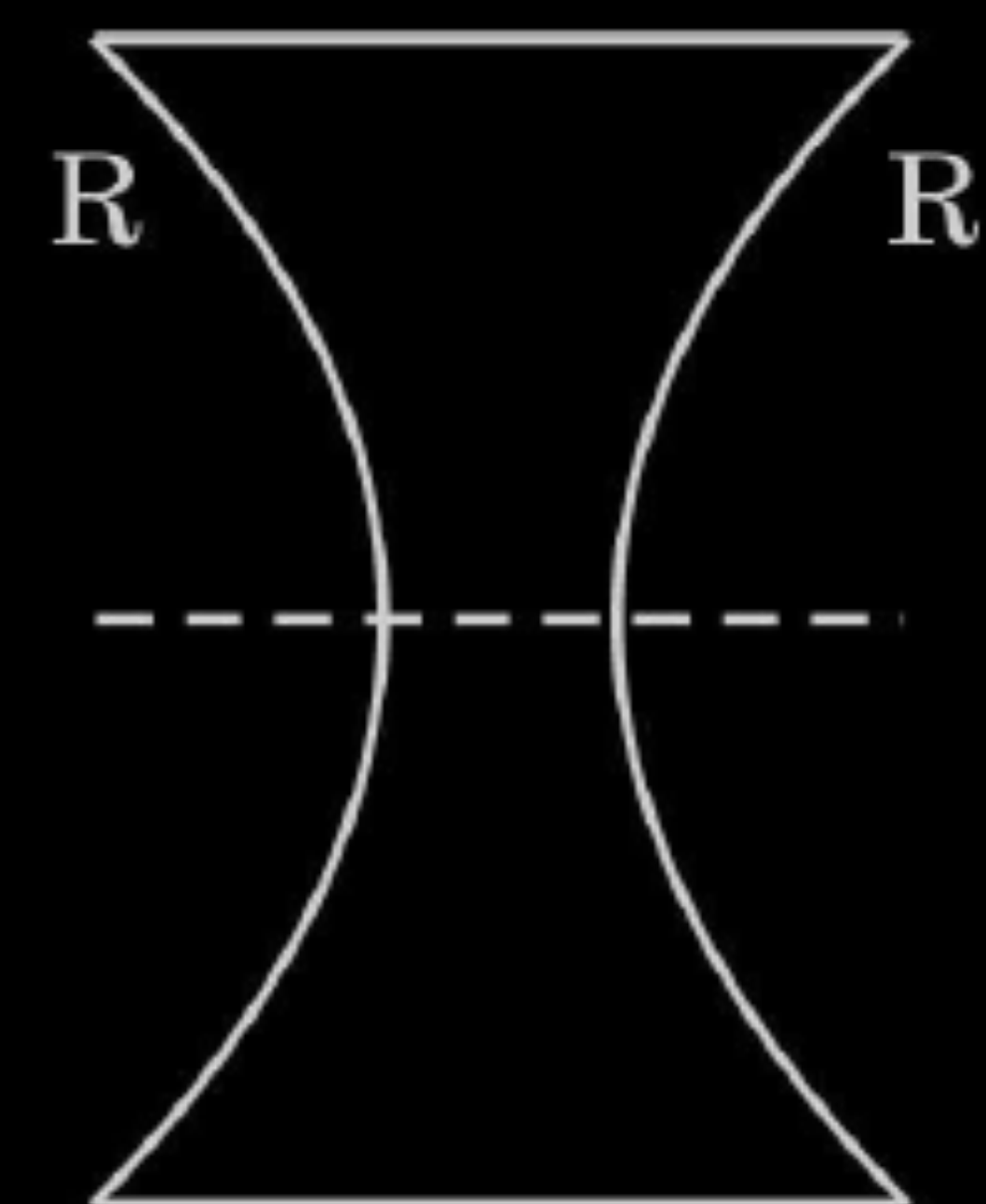
(ii) $+5\text{ D}$

(iii) $+15\text{ D}$

(iv) -15 D

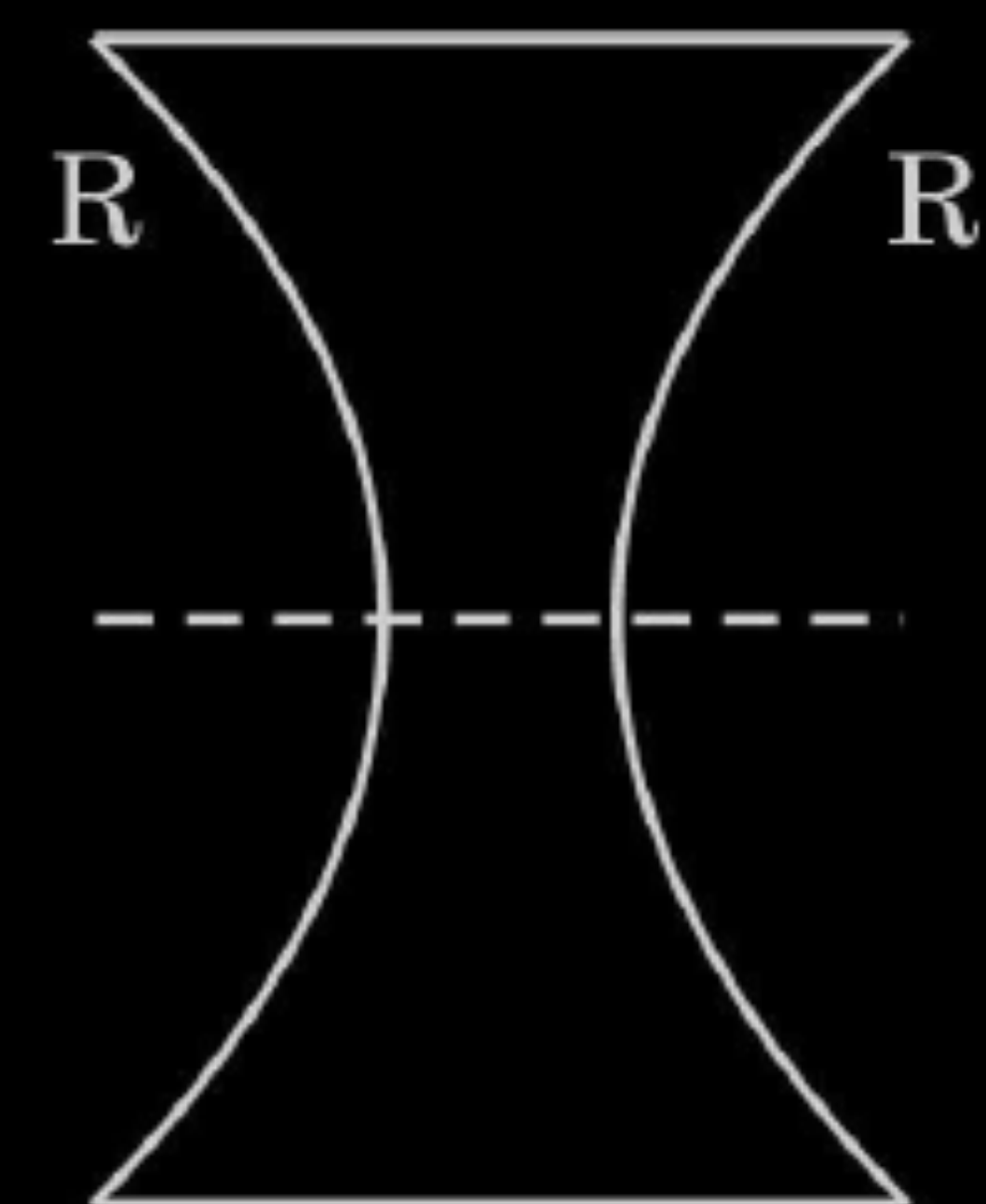
(ii) $+5\text{D}$

- (e) An equiconcave lens of focal length 'f' is cut into two identical parts along the dotted line as shown in the figure. The focal length of each part will be :



- (i) $\frac{f}{4}$
- (ii) $\frac{f}{2}$
- (iii) f
- (iv) $2f$

- (e) An equiconcave lens of focal length 'f' is cut into two identical parts along the dotted line as shown in the figure. The focal length of each part will be :



(i) $\frac{f}{4}$

(ii) $\frac{f}{2}$

(iii) f

(iv) $2f$

(iii) f

6. An equiconvex lens forms a two times enlarged real image when an object is kept 16 cm from it. The lens is cut into two identical plano-convex lenses. If the object is again kept 16 cm in front of one of these lenses, then find the nature and position of the image formed.

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6. An equiconvex lens forms a two times enlarged real image when an object is kept 16 cm from it. The lens is cut into two identical plano-convex lenses. If the object is again kept 16 cm in front of one of these lenses, then find the nature and position of the image formed.

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$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$\frac{1}{2}$

$$\frac{1}{f} = \frac{1}{32} + \frac{1}{16} = \frac{1+2}{32} = \frac{3}{32}$$

$\frac{1}{2}$

$$f = \frac{32}{3} \text{ cm}$$

$\frac{1}{2}$

When lens is cut into two equal halves the new focal length = $f' = 2f = \frac{64}{3} \text{ cm}$

$$\frac{1}{f'} = \frac{1}{v} - \frac{1}{u} \Rightarrow \frac{3}{64} = \frac{1}{v} + \frac{1}{16}$$

$\frac{1}{2}$

$$\frac{1}{v} = \frac{3}{64} - \frac{1}{16} = \frac{3-4}{64} = \frac{-1}{64}$$

$$v = -64 \text{ cm}$$

$\frac{1}{2}$

Image formed is virtual & erect.

$\frac{1}{2}$

7. A converging lens made of glass ($\mu = 1.5$) has its spherical faces of radii of curvature 10 cm and 20 cm. Find its focal length

(a) in air, and

(b) when it is immersed in a liquid of refractive index 1.25.

3

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7. A converging lens made of glass ($\mu = 1.5$) has its spherical faces of radii of curvature 10 cm and 20 cm. Find its focal length

(a) in air, and

(b) when it is immersed in a liquid of refractive index 1.25.

3

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(a)

$$\frac{1}{f_a} = (\mu_a^g - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$R_1 = 10 \text{ cm} \text{ \& } R_2 = -20 \text{ cm}, \mu_a^g = 1.5$$

$$\frac{1}{f_a} = (1.5 - 1) \left(\frac{1}{10} - \frac{1}{-20} \right)$$

$$f_a = \frac{40}{3} = 13.3 \text{ cm}$$

b)

$$\frac{1}{f_l} = (\mu_l^g - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$= \left(\frac{1.5}{1.25} - 1 \right) \left(\frac{1}{10} - \frac{1}{-20} \right)$$

$$f_l = \frac{100}{3} = 33.3 \text{ cm}$$

1/2

1/2

1/2

1/2

1/2

1/2

10. A biconcave lens of power P vertically splits into two identical plano concave parts. The power of each part will be

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(a) $2P$

(b) $P/2$

(c) P

(d) $P/\sqrt{2}$

1

10. A biconcave lens of power P vertically splits into two identical plano concave parts. The power of each part will be

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(a) $2P$

(b) $P/2$

(c) P

(d) $P/\sqrt{2}$

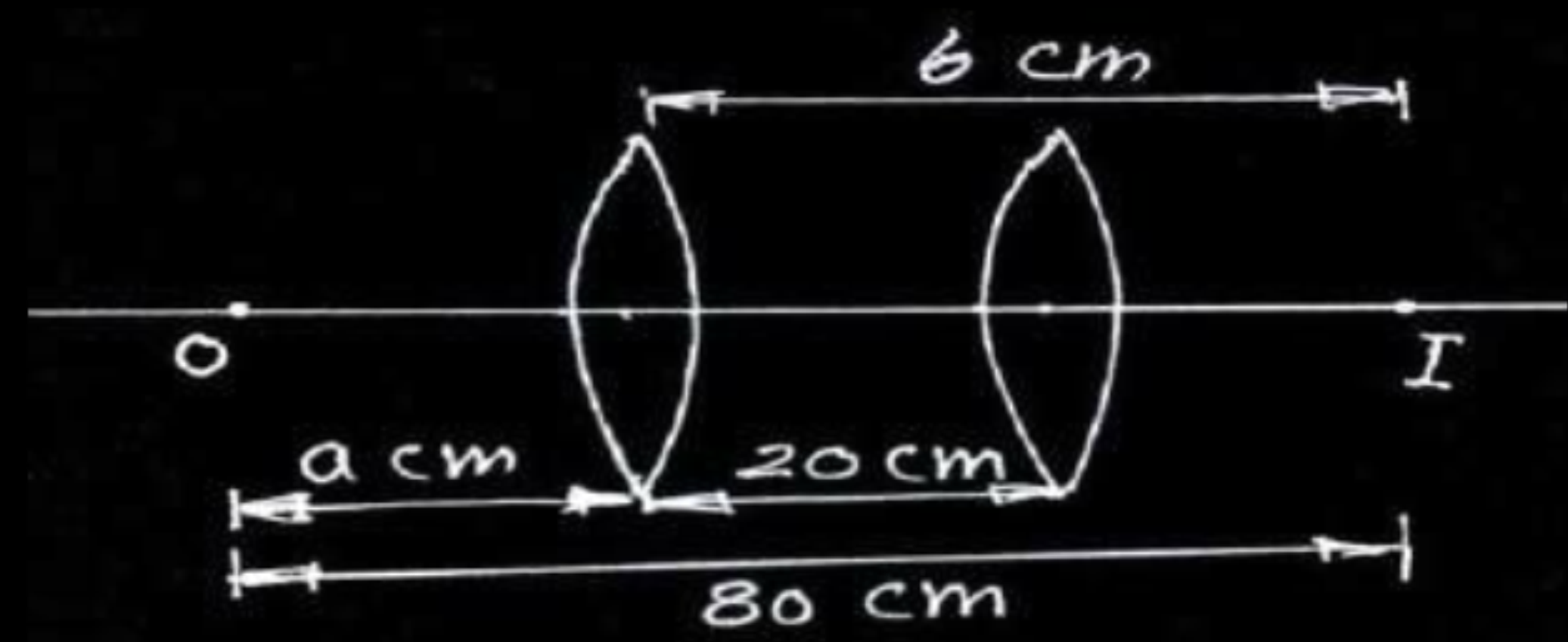
1

(b) $P/2$

- (b) A screen is placed 80 cm from an object. The image of the object on the screen is formed by a convex lens placed between them at two different locations separated by a distance 20 cm. Determine the focal length of the lens.

CBSE 2020

The two positions of the lens are as shown



for position 1

$$u = -a \text{ cm}$$

$$v = b = +(80 - a) \text{ cm}$$

for position 2

$$u = -(a + 20) \text{ cm}$$

$$v = +(b - 20) = +(60 - a) \text{ cm}$$

By lens formula

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\therefore \frac{1}{(80 - a)} + \frac{1}{a} = \frac{1}{(60 - a)} + \frac{1}{(20 + a)}$$

This gives

$$a(80 - a) = (a + 20)(60 - a)$$

$$\text{or } 80a - a^2 = 40a + 1200 - a^2$$

$$\text{or } a = 30 \text{ cm}$$

$$\therefore \frac{1}{f} = \frac{1}{50} + \frac{1}{30} = \frac{3 + 5}{150} = \frac{8}{150}$$

$$\therefore f = \frac{150}{8} \text{ cm} = 18.75 \text{ cm}$$

Alternatively

$$F = \left(\frac{D^2 - x^2}{4D} \right) = \frac{80^2 - 20^2}{4 \times 80} = 18.75 \text{ cm}$$

Alternatively

The values of

$a = |u|$ and $b = |v|$ simply get interchanged in the two positions.

$$b + a = 80 \text{ cm}$$

$$b - a = 20 \text{ cm}$$

This gives $b = 50 \text{ cm}$ and $a = 30 \text{ cm}$

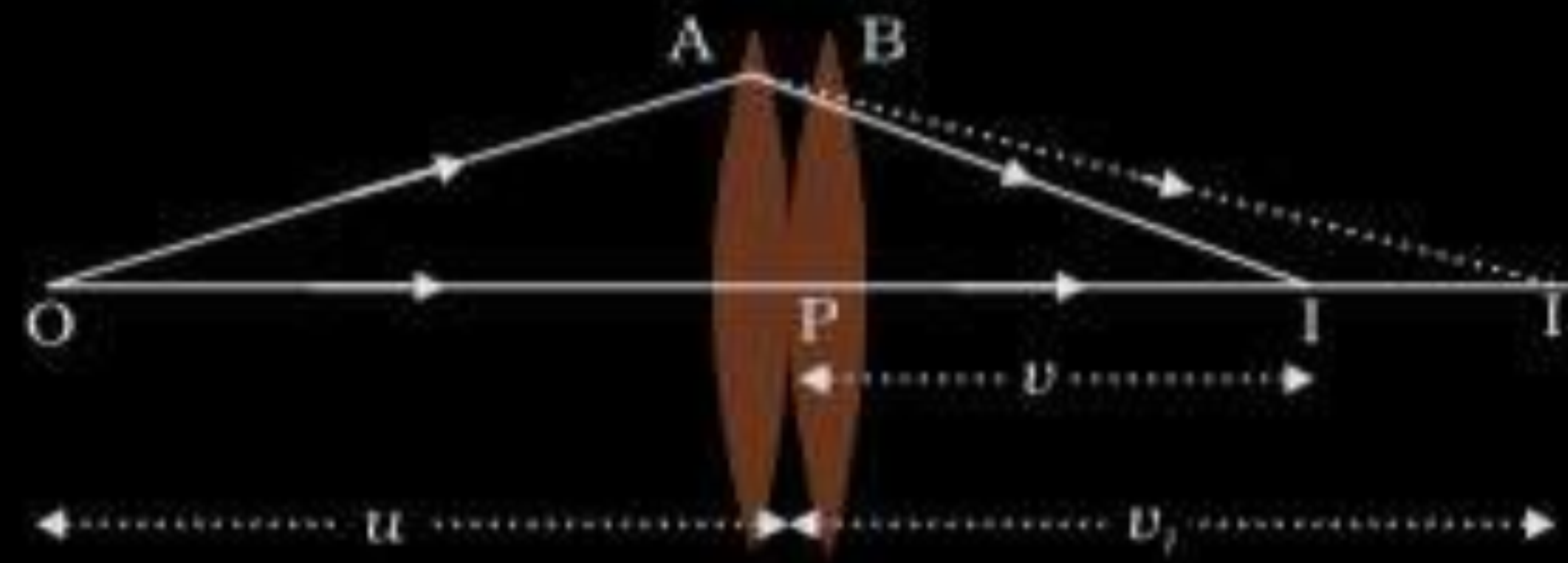
$$\therefore \frac{1}{f} = \frac{1}{50} - \left(\frac{1}{-30} \right) = \frac{8}{150}$$

$$f = 18.75 \text{ cm}$$

- (a) Two thin lenses are placed coaxially in contact. Obtain the expression for the focal length of this combination in terms of the focal lengths of the two lenses.
- (b) A converging lens of refractive index 1.5 has a power of 10 D. When it is completely immersed in a liquid, it behaves as a diverging lens of focal length 50 cm. Find the refractive index of the liquid.

5

(a)



For lens A

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f_1}$$

(i)

For lens B :

virtual image I_1 formed by A acting as object

$$\frac{1}{v} - \frac{1}{v_1} = \frac{1}{f_2}$$

(ii)

Adding equations (i) and (ii)

$$\frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{v} - \frac{1}{u}$$

$$\text{Hence } \frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\text{Therefore } \frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{F}$$

(b)

$$\text{In air } P_1 = \frac{1}{f_1} = (a^{\mu_g} - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

(i)

For Liquid

$$P_2 = \frac{1}{f_2} = (l^{\mu_g} - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

(ii)

From (i) and (ii)

$$\frac{P_1}{P_2} = \frac{(a^{\mu_g} - 1)}{(l^{\mu_g} - 1)}$$

$$\frac{10}{-2} = \frac{(1.5 - 1)}{\left(\frac{1.5}{\mu_l} - 1 \right)}$$

$$\mu_l = \frac{5}{3}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- 12.** The refractive index of the material of a converging lens is 1.5. If air is replaced by a medium of refractive index 1.6, then the lens will now behave as a _____ lens.

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12. The refractive index of the material of a converging lens is 1.5. If air is replaced by a medium of refractive index 1.6, then the lens will now behave as a _____ lens.

CBSE 2020

1

Divergent lens/ Concave lens

- 25.** A converging lens of focal length f_1 is placed coaxially in contact with a diverging lens of focal length f_2 ($f_1 > f_2$). Determine the power and nature of the combination in terms of f_1 and f_2 .

CBSE 2020

- 25.** A converging lens of focal length f_1 is placed coaxially in contact with a diverging lens of focal length f_2 ($f_1 > f_2$). Determine the power and nature of the combination in terms of f_1 and f_2 .

CBSE 2020

2

$$\frac{1}{f} = \frac{1}{f_1} - \frac{1}{f_2}$$

$$\frac{1}{f} = \frac{f_2 - f_1}{f_1 f_2}$$

$$\therefore P = \frac{f_2 - f_1}{f_1 f_2}$$

Because $f_2 < f_1 \therefore P$ is negative

$\therefore$ nature is diverging lens

$\frac{1}{2}$

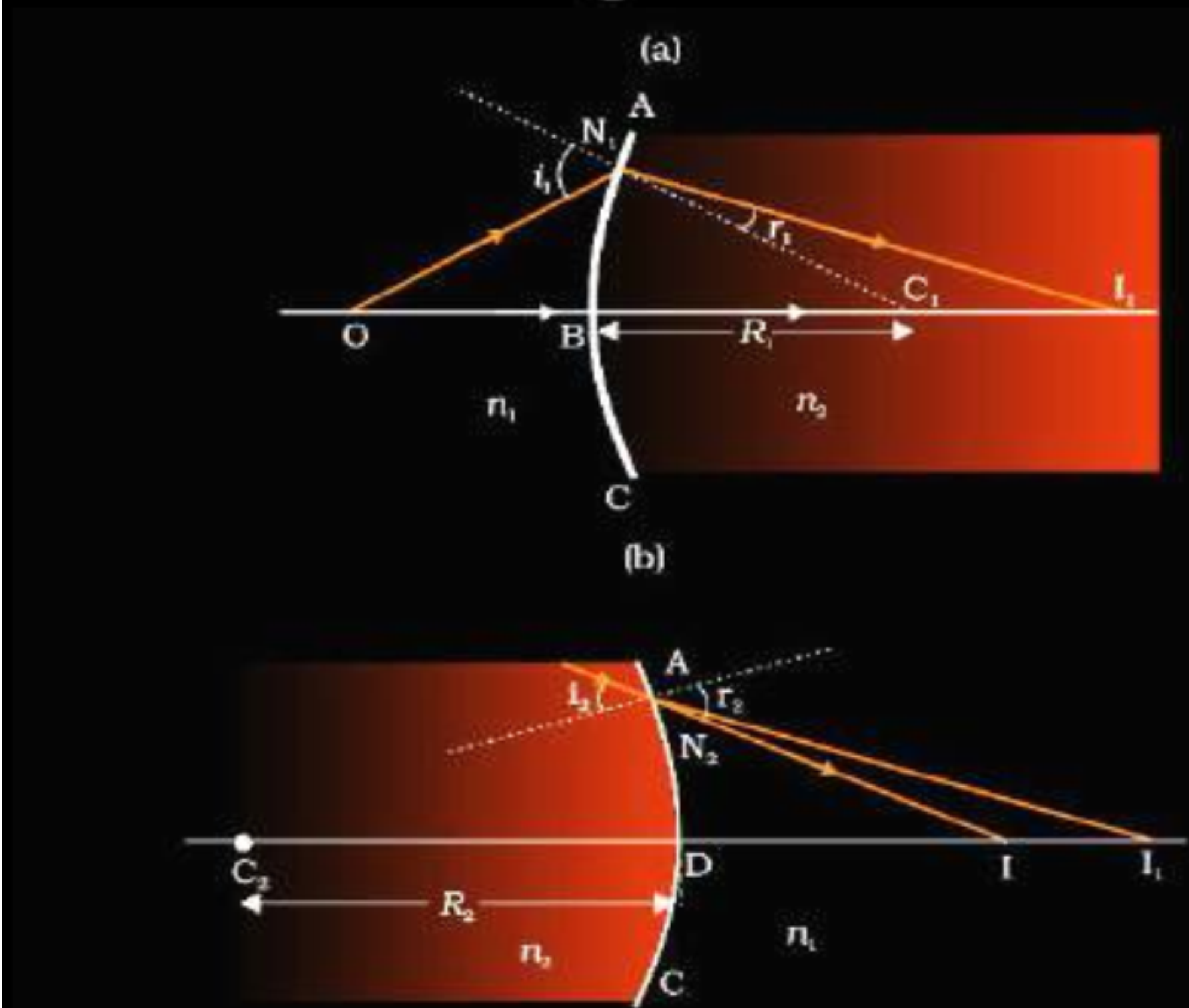
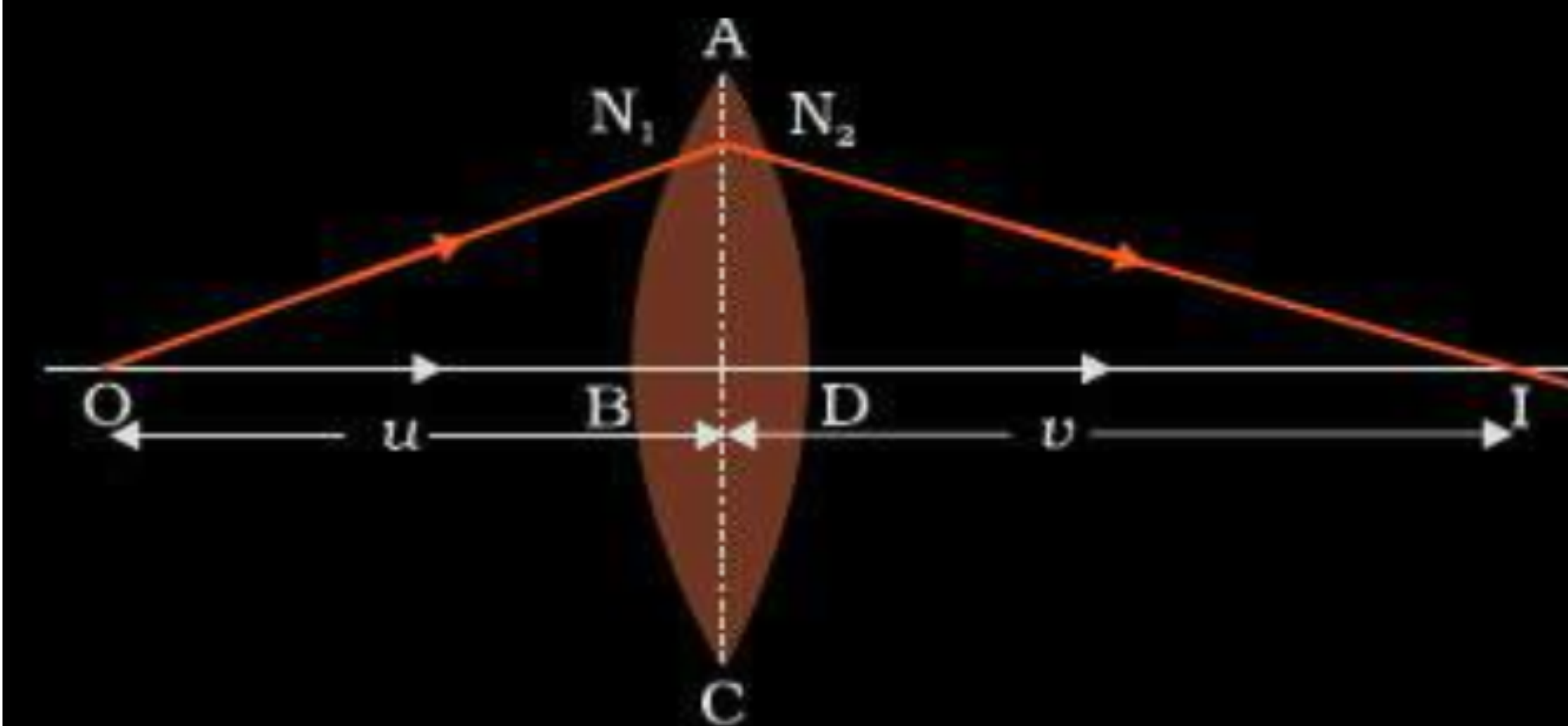
$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- 37.** (a) Derive lens maker's formula for a biconvex lens.
- (b) A point object is placed at a distance of 12 cm on the principal axis of a convex lens of focal length 10 cm. A convex mirror is placed coaxially on the other side of the lens at a distance of 10 cm. If the final image coincides with the object, sketch the ray diagram and find the focal length of the convex mirror.

CBSE 2020



1/2

1/2

For first refracting surface

$$\frac{\mu_2}{v_1} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R_1} \text{-----1}$$

For second refracting surface ADC

$$\frac{\mu_1}{v} - \frac{\mu_2}{v_1} = \frac{\mu_1 - \mu_2}{R_2} \text{-----2}$$

Adding equations 1 and 2, we get

$$\frac{\mu_1}{v} - \frac{\mu_1}{u} = (\mu_2 - \mu_1) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

$$\Rightarrow \frac{1}{v} - \frac{1}{u} = \left(\frac{\mu_2}{\mu_1} - 1 \right) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

$$\therefore \frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\therefore \frac{1}{f} = \left(\frac{\mu_2}{\mu_1} - 1 \right) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

also

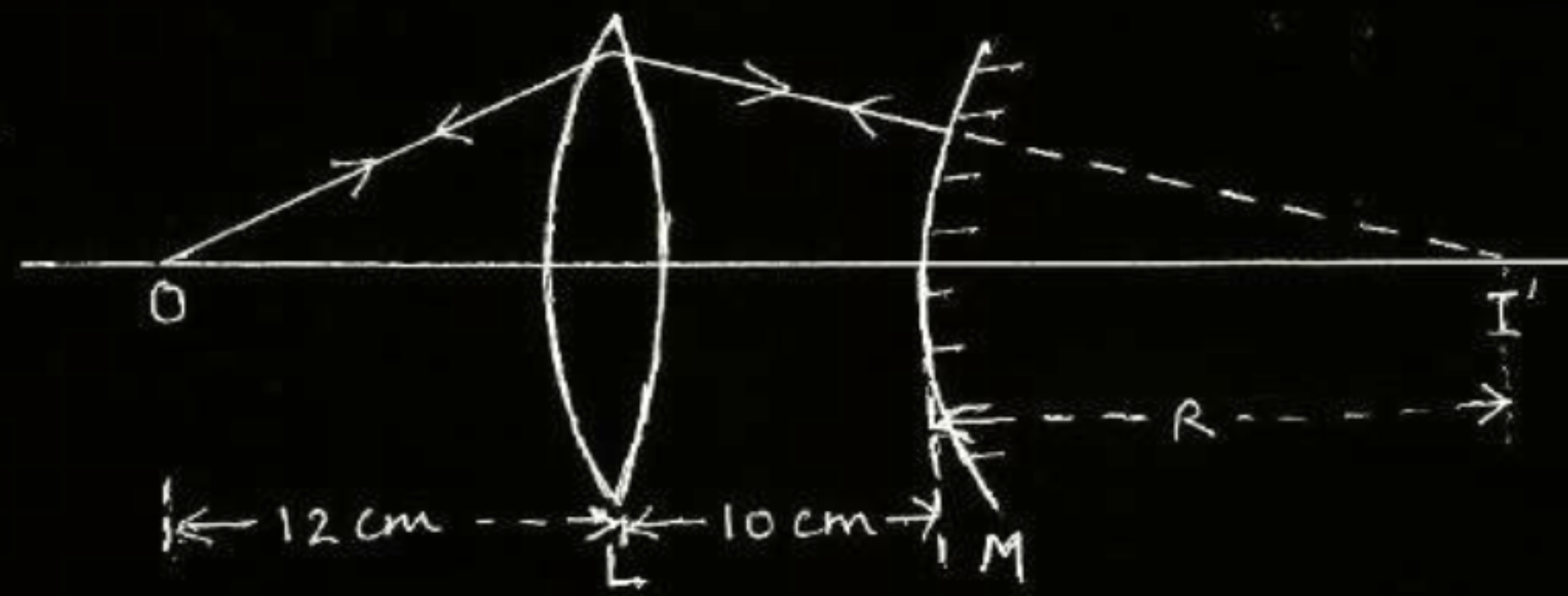
$$\frac{\mu_2}{\mu_1} = \mu$$

$$\therefore \frac{1}{f} = (\mu - 1) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

1/2

1/2

1/2



$$f_1 = 10 \text{ cm}$$

$$u = -12 \text{ cm}$$

Applying lens formula

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{10} = \frac{1}{v} - \frac{1}{-12}$$

$$\Rightarrow v = 60 \text{ cm}$$

$\therefore$ radius of curvature of the mirror

$$R = 60 \text{ cm} - 10 \text{ cm} = 50 \text{ cm}$$

$$\text{hence focal length of the mirror } f_m = \frac{R}{2} = 25 \text{ cm}$$

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

7. A biconvex lens of glass having refractive index 1.47 is immersed in a liquid. It becomes invisible and behaves as a plane glass plate. The refractive index of the liquid is

CBSE 2020

- (A) 1.47
- (B) 1.62
- (C) 1.33
- (D) 1.51

7. A biconvex lens of glass having refractive index 1.47 is immersed in a liquid. It becomes invisible and behaves as a plane glass plate. The refractive index of the liquid is

CBSE 2020

1

- (A) 1.47
- (B) 1.62
- (C) 1.33
- (D) 1.51

(A)

1.47

Using lens maker's formula, derive the thin lens formula $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$ for a biconvex lens.

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Using lens maker's formula, derive the thin lens formula $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$ for a biconvex lens.

2

CBSE 2020

$$\therefore \frac{1}{v} - \frac{1}{u} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \text{-----1}$$

When $u = \infty$ and $v = f$

$$\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \text{-----2}$$

$$\left[n = \frac{n_2}{n_1} \right]$$

From Eq 1 and 2

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \text{ then lens formula}$$

[Even if the student derives $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$ for biconvex lens, award 1 ½ marks]

1

1/2

1/2

- (b) An object is placed 30 cm in front of a plano-convex lens with its spherical surface of radius of curvature 20 cm. If the refractive index of the material of the lens is 1.5, find the position and nature of the image formed.

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(b) According to lens maker's formula

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

for plano convex lens $R_1 \rightarrow R$ and $R_2 \rightarrow \infty$

$$\frac{1}{f} = \frac{(\mu - 1)}{R} = \frac{1.5 - 1}{20}$$

$$\therefore f = 40 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$
$$\frac{1}{40} = \frac{1}{v} - \frac{1}{-30}$$

$$v = -12 \text{ cm}$$

Nature: virtual

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

A screen is placed 90 cm from an object. The image of the object on the screen is formed by a convex lens at two different positions separated by 20 cm. Calculate the focal length of the lens.

CBSE 2019

A screen is placed 90 cm from an object. The image of the object on the screen is formed by a convex lens at two different positions separated by 20 cm. Calculate the focal length of the lens.

CBSE 2019

$v+u = 90$	$\frac{1}{2}$
$v-u = 20$	$\frac{1}{2}$
$v = 55 \text{ cm}$	$\frac{1}{2}$
$u = 35 \text{ cm}$	$\frac{1}{2}$
$f = \frac{55 \times 35}{55 + 35} = 21.4 \text{ cm}$	1

A convex lens of focal length 20 cm and a concave lens of focal length 15 cm are kept 30 cm apart with their principal axes coincident. When an object is placed 30 cm in front of the convex lens, calculate the position of the final image formed by the combination. Would this result change if the object were placed 30 cm in front of the concave lens ? Give reason.

CBSE 2019

For Image formed by convex lens:

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{20} = \frac{1}{v} + \frac{1}{30}$$

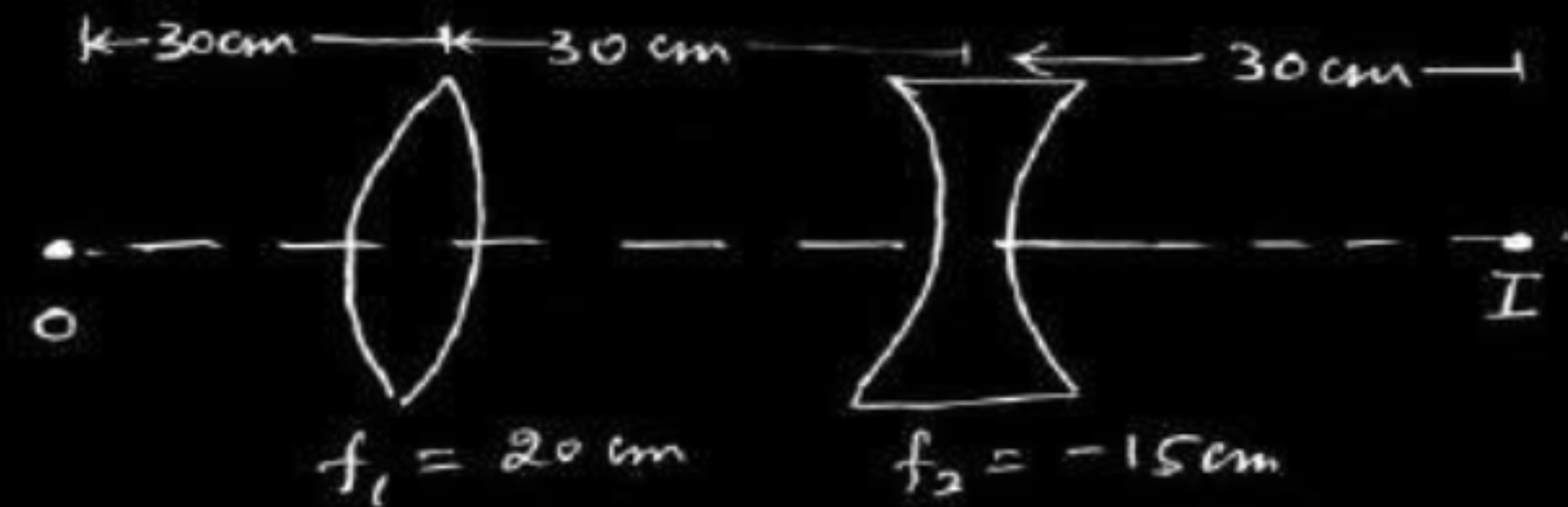
$$v = \frac{20 \times 30}{30 - 20} = 60 \text{ cm}$$

u for concave lens = + 30 cm

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{-15} = \frac{1}{v} - \frac{1}{30}$$

$$v = \frac{15 \times 30}{15 - 30} = -30 \text{ cm}$$



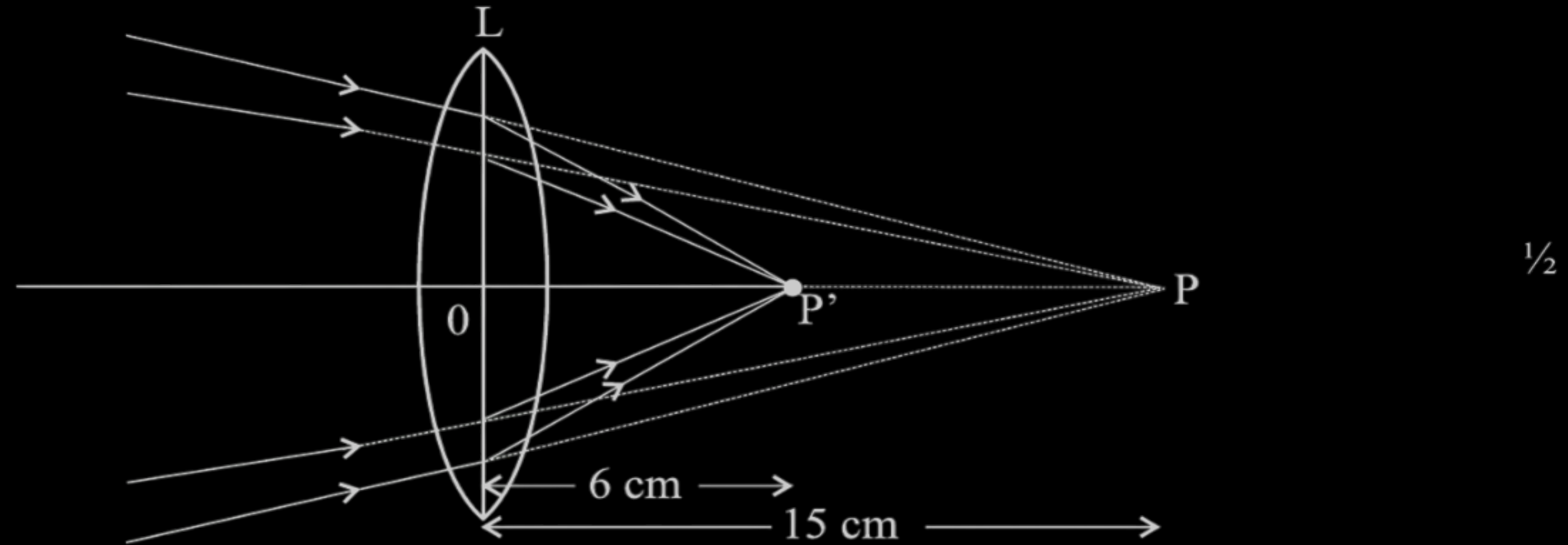
No, the result will not change from principle of reversibility

A beam of light converges at a point P. Now a convex lens is placed in the path of the convergent beam at 15 cm from P. At what point does a beam converge if the convex lens has a focal length 10 cm ?

CBSE 2019

A beam of light converges at a point P. Now a convex lens is placed in the path of the convergent beam at 15 cm from P. At what point does a beam converge if the convex lens has a focal length 10 cm ?

CBSE 2019



$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \quad (\text{lens formula})$$

Here $u = +15\text{cm} ; f = +10\text{ cm}$

$$\therefore \frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{1}{10} + \frac{1}{15}$$

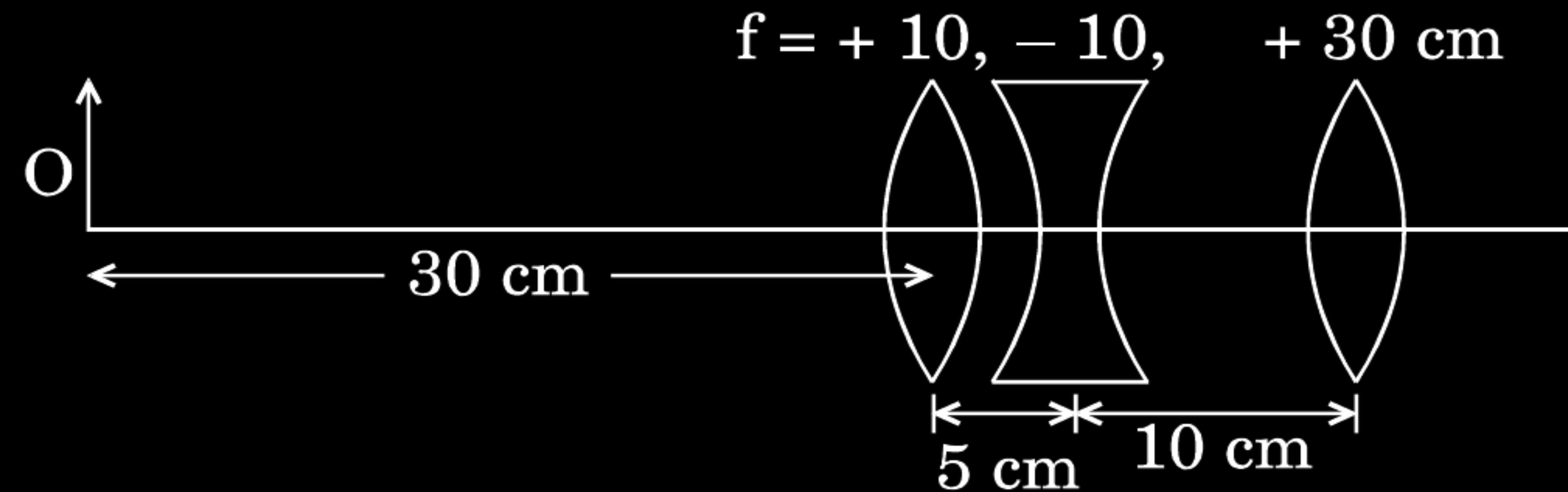
$$\Rightarrow v = 6\text{ cm}$$

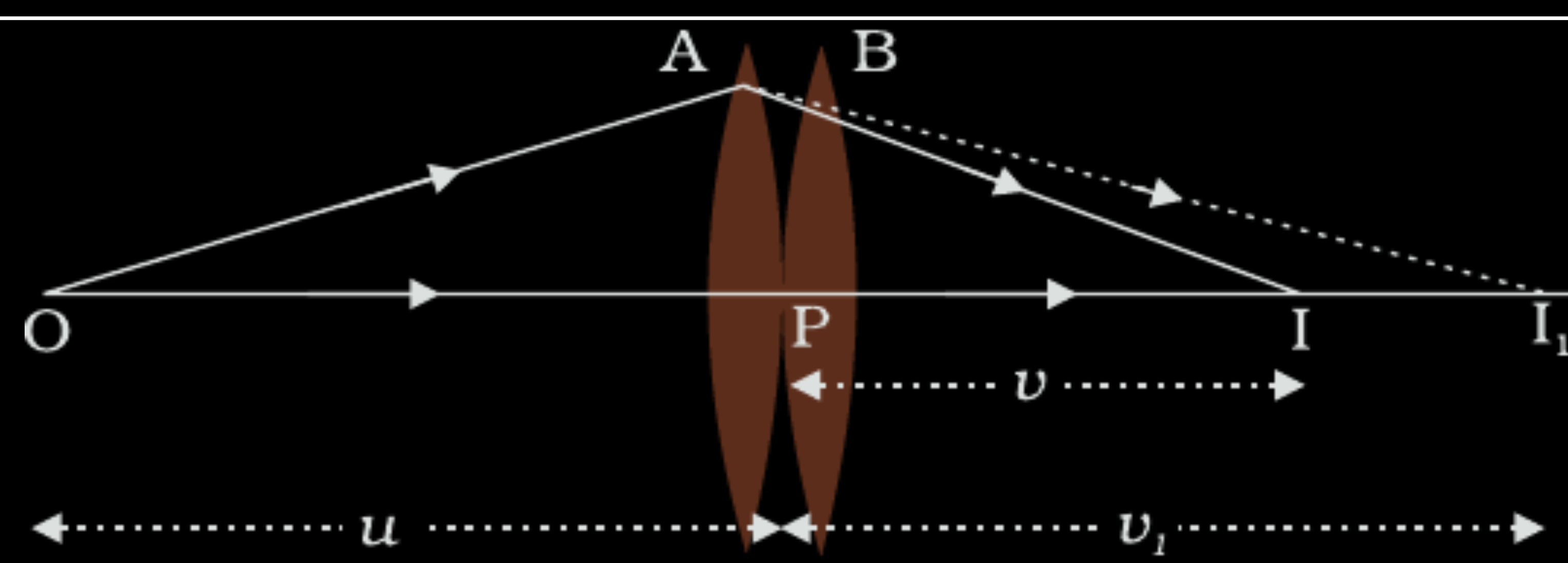
- (a) Using the ray diagram for a system of two lenses of focal lengths f_1 and f_2 in contact with each other, show that the two lens system can be regarded as equivalent to a single lens of focal length f , where

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}.$$

Also write the relation for the equivalent power of the lens combination.

- (b) Determine the position of the image formed by the lens combination given in the figure.





For lens A

$$\frac{1}{f_1} = \frac{1}{v_1} - \frac{1}{u}$$

eg. (I)

$\frac{1}{2}$

For lens B

$$\frac{1}{f_2} = \frac{1}{v} - \frac{1}{v_1}$$

eg. (ii)

$\frac{1}{2}$

Adding eqn. (i) & eqn. (ii)

$\frac{1}{2}$

$$\frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{v_1} - \frac{1}{u} + \frac{1}{v} - \frac{1}{v_1}$$

$$\frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{F}$$

$\frac{1}{2}$

$\therefore$ equivalent power

$$P = P_1 + P_2$$

$\frac{1}{2}$

b) Image formed by lense of $f = +10$ cm

$$\frac{1}{v_1} - \frac{1}{u_1} = \frac{1}{f_1}$$

$$\frac{1}{v_1} - \frac{1}{30} = \frac{1}{10}$$

$$\therefore v_1 = 15 \text{ cm}$$

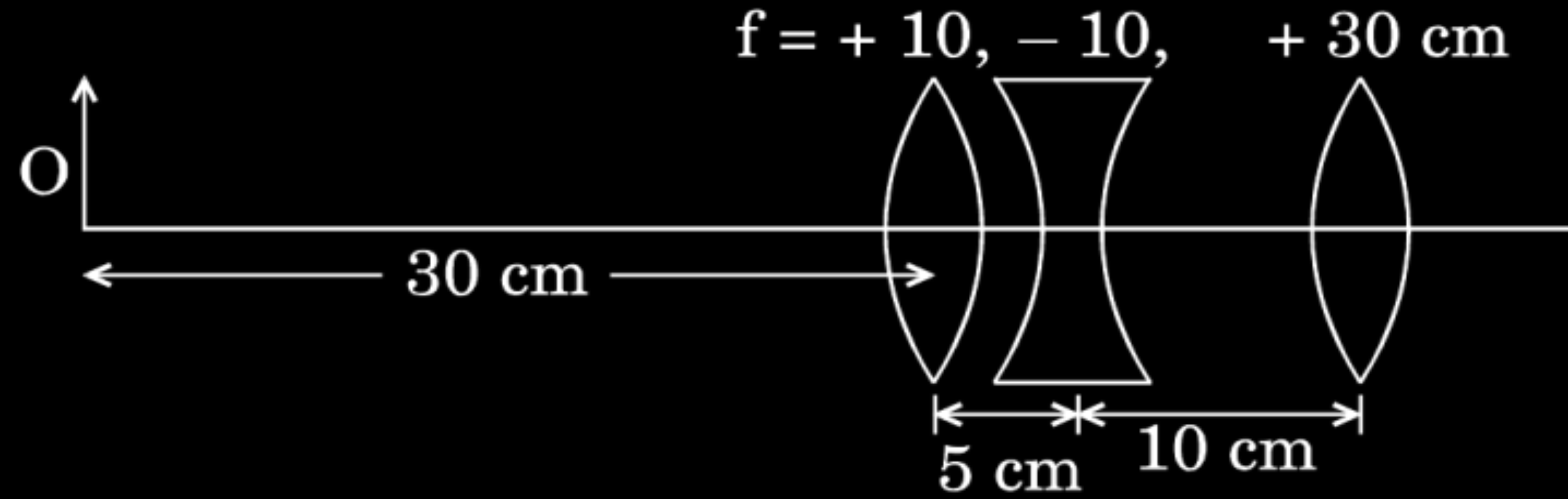
This image formed by the lens act as object from concave lens

$$\therefore u_2 = 15 - 5 = 10 \text{ cm}$$

$$\frac{1}{f_2} + \frac{1}{v_2} = \frac{1}{u_2}$$

$$\frac{1}{-10} = \frac{1}{v} - \frac{1}{10}$$

$$v = \infty$$



Therefore virtual image forms at right of concave lens at $v = \infty$ and act as convex lens. ($f = +30$ cm)

$$\therefore u_2 = 15 - 5 = 10 \text{ cm}$$

$$\frac{1}{v_3} = \frac{1}{4} - \frac{1}{f_3}$$

$$\frac{1}{v_3} = \frac{1}{\infty} = \frac{1}{30}$$

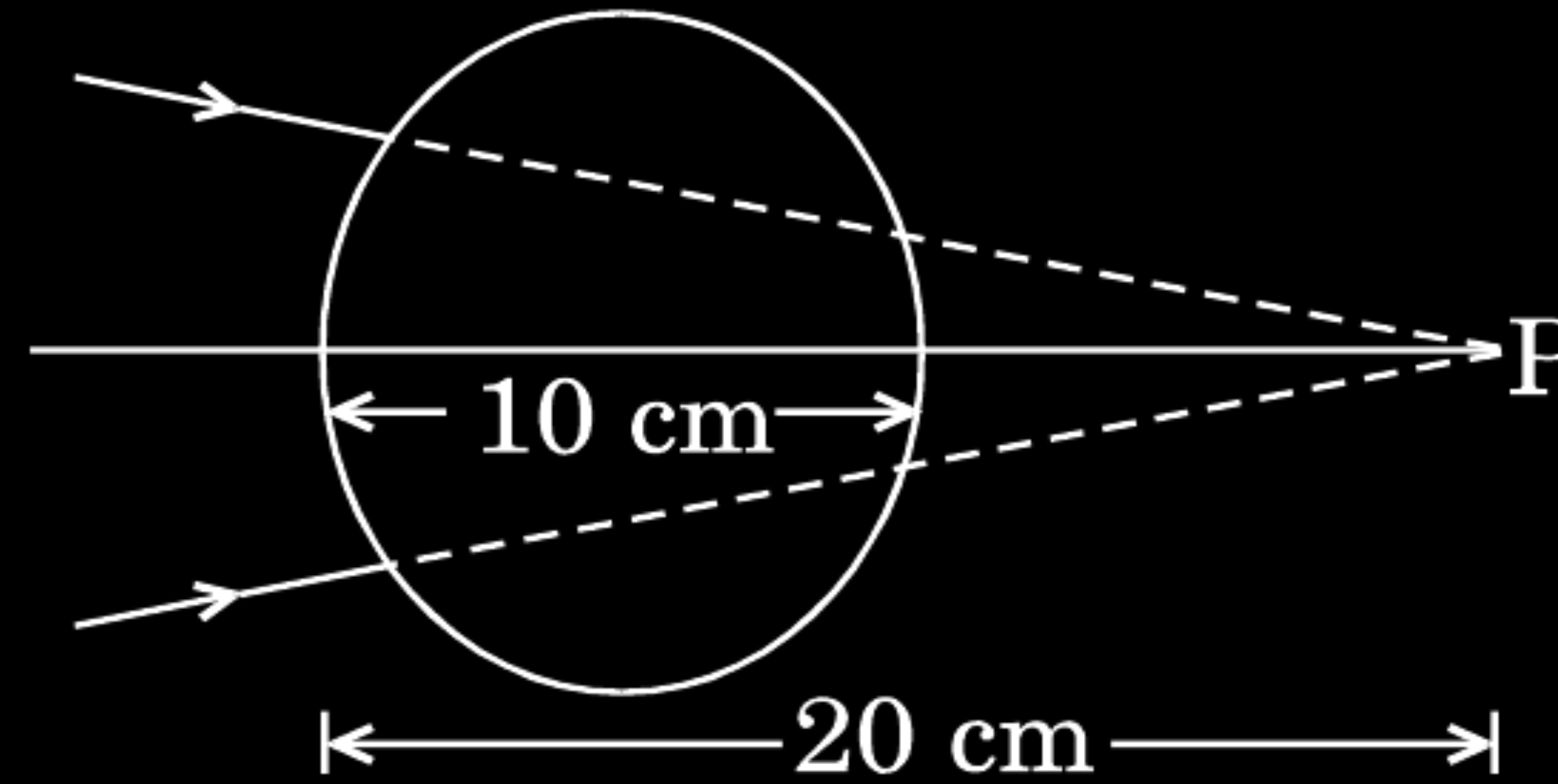
$$v_3 = 30 \text{ cm}$$

(If student calculate upto $v = \infty$, give full marks)

$\frac{1}{2}$

$\frac{1}{2}$

A converging beam of light travelling in air converges at a point P as shown in the figure. When a glass sphere of refractive index 1.5 is introduced in between the path of the beam, calculate the new position of the image. Also draw the ray diagram for the image formed.



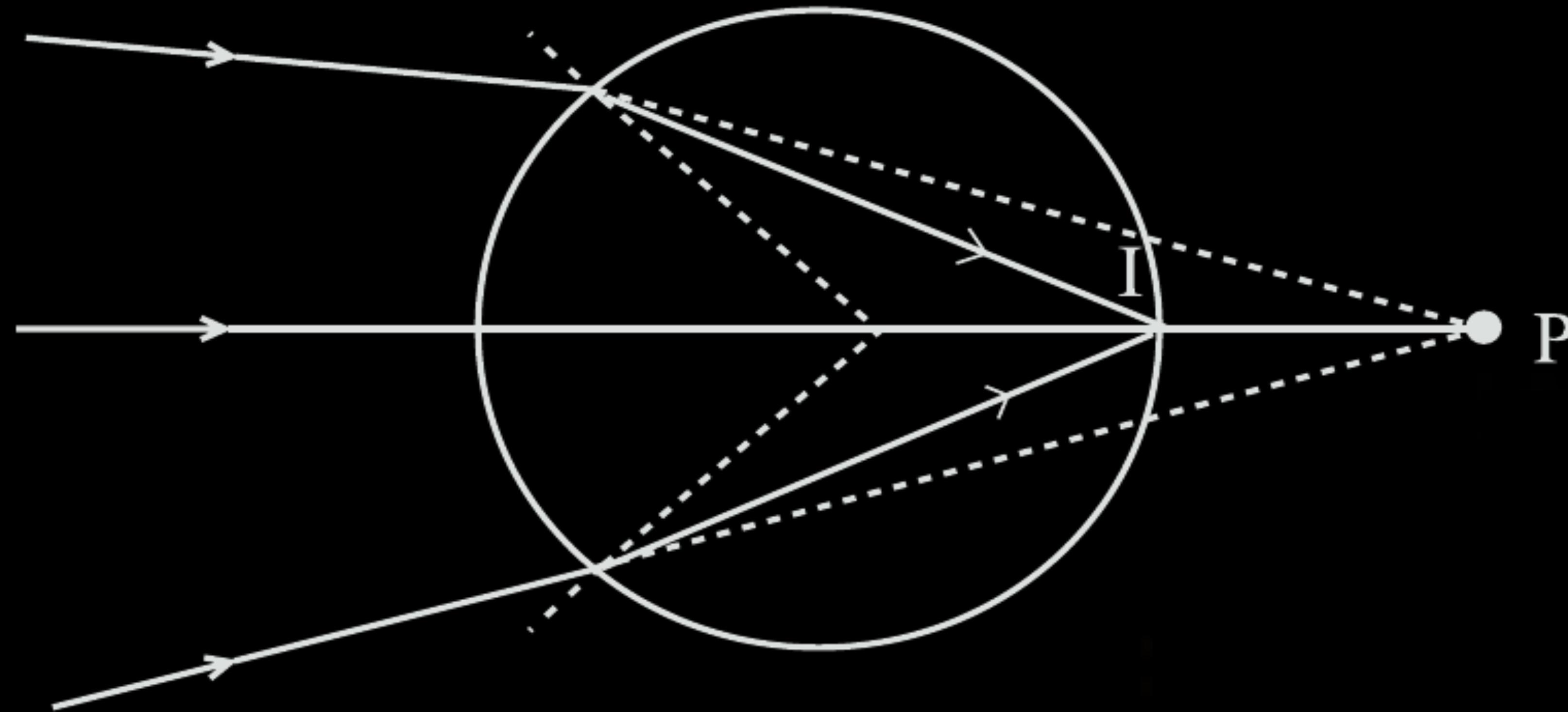
CBSE 2019

a) $u = 20\text{cm}$, $n_2 = 1.5$, $n_1 = 1$, $R = 5\text{cm}$

Using $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$

$$\frac{1.5}{v} - \frac{1}{20} = \frac{1.5 - 1}{5}$$

$$v = 10\text{ cm}$$



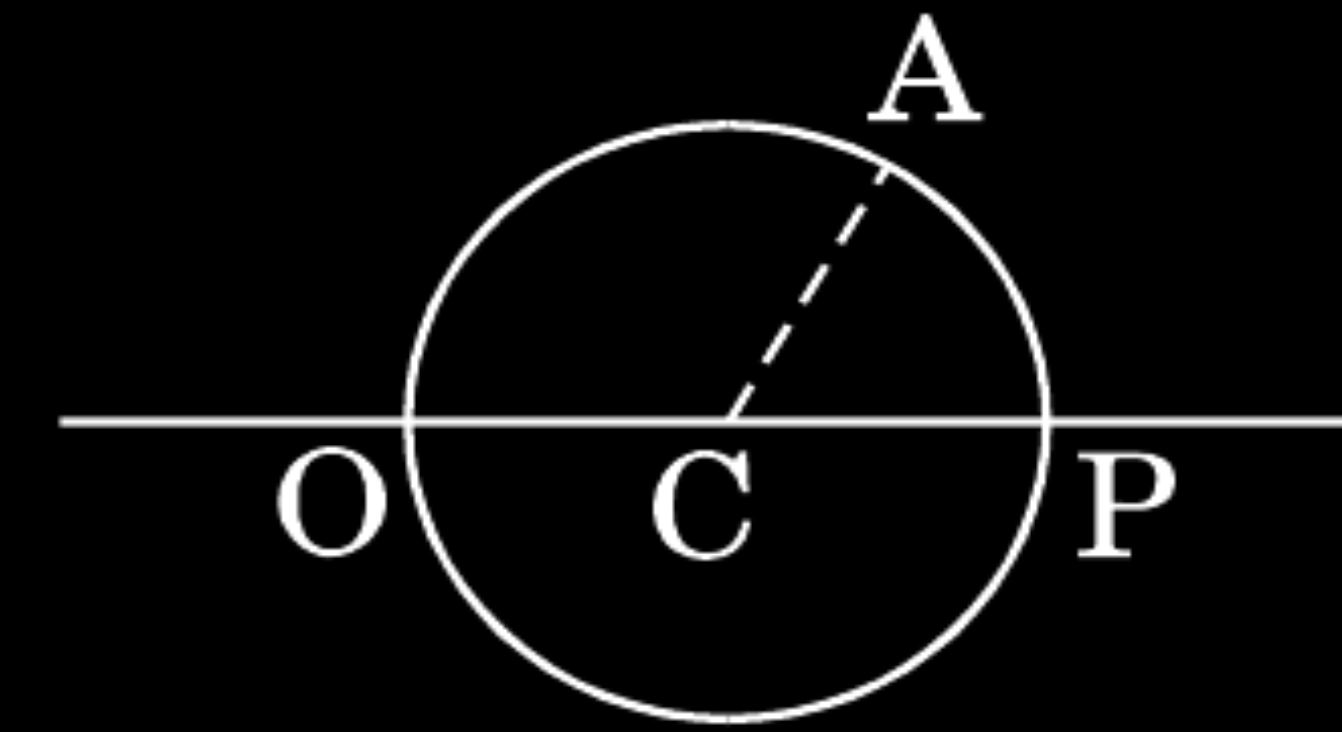
$\frac{1}{2}$

1

$\frac{1}{2}$

1

A point 'O' marked on the surface of a glass sphere of diameter 20 cm is viewed through glass from the position directly opposite to the point O. If the refractive index of the glass is 1.5, find the position of the image formed. Also, draw the ray diagram for the formation of the image.



CBSE 2019

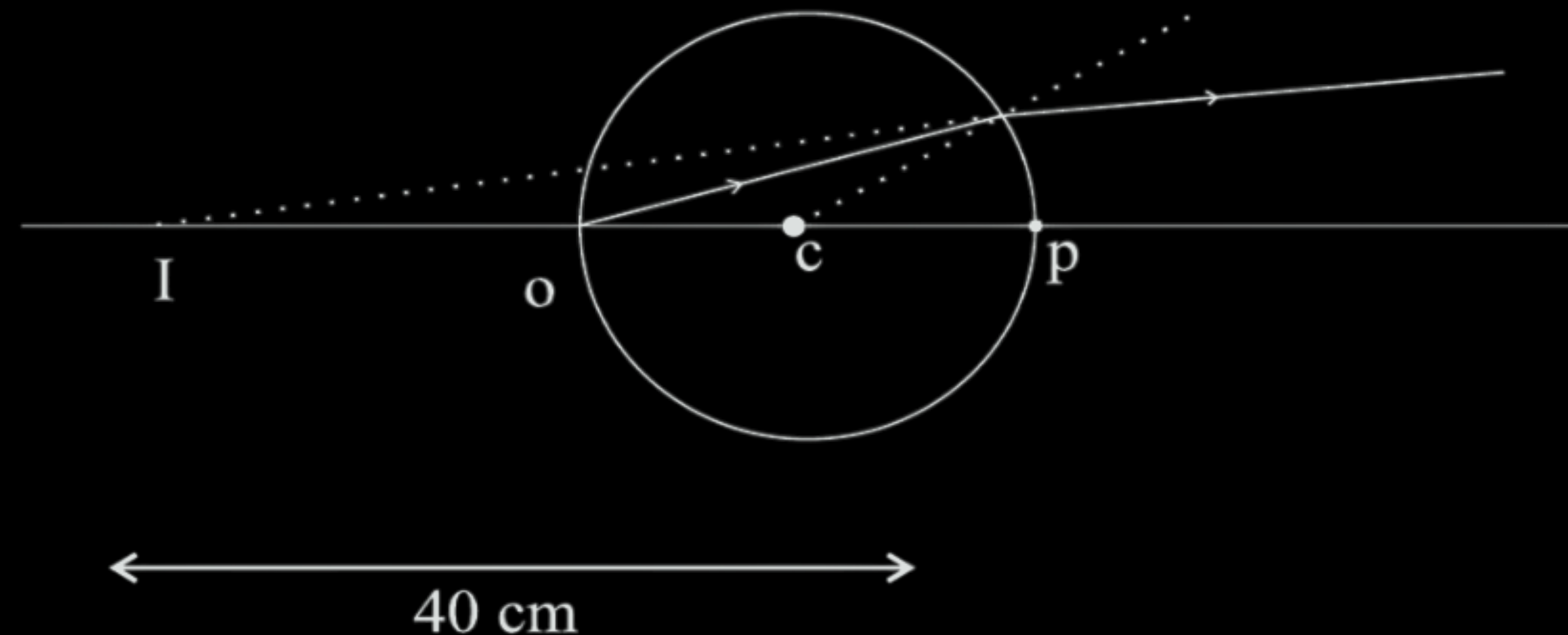
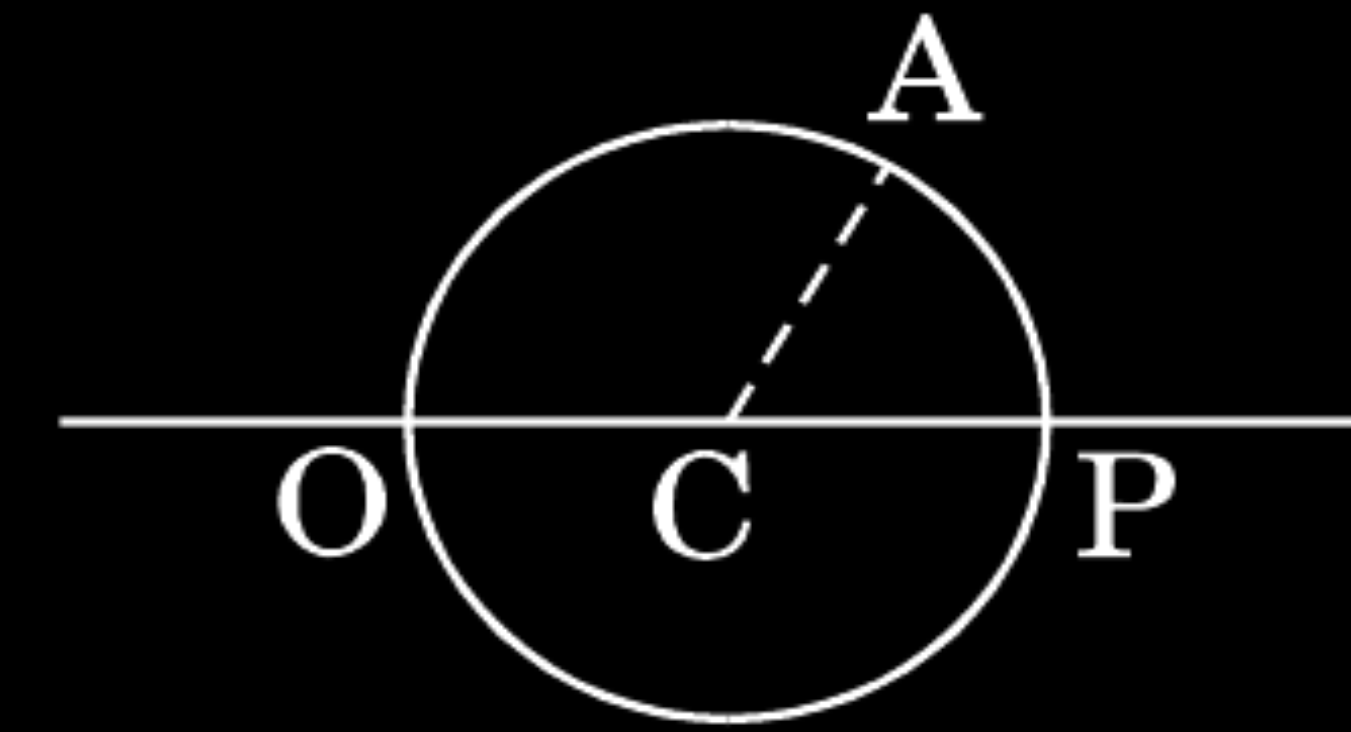
A point 'O' marked on the surface of a glass sphere of diameter 20 cm is viewed through glass from the position directly opposite to the point O. If the refractive index of the glass is 1.5, find the position of the image formed. Also, draw the ray diagram for the formation of the image.

CBSE 2019

$$\frac{n_1}{-u} + \frac{n_2}{v} = \frac{n_2 - n_1}{R}$$

$$\frac{1.5}{20} + \frac{1}{v} = \frac{1 - 1.5}{-10}$$

$$v = -40 \text{ cm}$$



1

$\frac{1}{2}$

$\frac{1}{2}$

1

Calculate the radius of curvature of an equi-concave lens of refractive index 1.5, when it is kept in a medium of refractive index 1.4, to have a power of -5D ?

CBSE 2019

Calculate the radius of curvature of an equi-concave lens of refractive index 1.5, when it is kept in a medium of refractive index 1.4, to have a power of -5D ?

CBSE 2019

$$f = \frac{1}{P} = \frac{1}{-5} \text{ m} = -\frac{100}{5} \text{ cm} = -20 \text{ cm}$$

$\frac{1}{2}$

$$\frac{1}{f} = \left(\frac{\mu_2}{\mu_1} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$\frac{1}{2}$

$$\mu_2 = 1.5, \quad \mu_1 = 1.4, \quad R_1 = -R$$

$$R_2 = R$$

$$\frac{1}{-20} = \left(\frac{1.5}{1.4} - 1 \right) \left(-\frac{1}{R} - \frac{1}{R} \right)$$

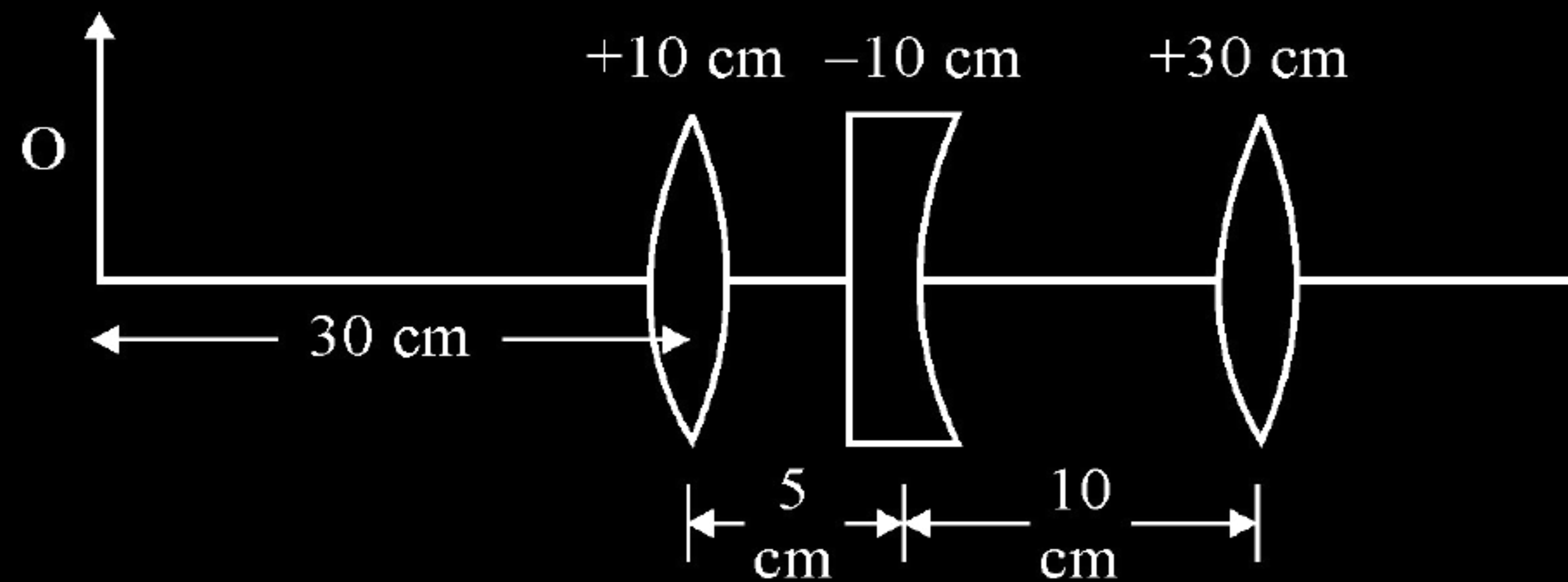
$\frac{1}{2}$

$$\frac{1}{-20} = \left(\frac{0.1}{1.4} \right) \left(-\frac{2}{R} \right)$$

$\frac{1}{2}$

$$R = \frac{20}{7} \text{ cm} (= 2.86 \text{ cm})$$

- (a) Under what conditions is the phenomenon of total internal reflection of light observed ? Obtain the relation between the critical angle of incidence and the refractive index of the medium.
- (b) Three lenses of focal lengths $+10\text{ cm}$, -10 cm and $+30\text{ cm}$ are arranged coaxially as in the figure given below. Find the position of the final image formed by the combination.



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- (a) (i) Light travels from denser to rarer medium.
(ii) Angle of incidence is more than the critical angle

For the Grazing incidence

$$\mu \sin i_c = 1 \sin 90^\circ$$

$$\mu = \frac{1}{\sin i_c}$$

- (b) For convex lens of focal Length 10 cm

$$\frac{1}{f_1} = \frac{1}{v_1} - \frac{1}{u_1}$$

$$\frac{1}{10} = \frac{1}{v_1} - \frac{1}{-30} \Rightarrow v_1 = 15 \text{ cm}$$

Object distance for concave lens $u_2 = 15 - 5 = 10 \text{ cm}$

$$\frac{1}{f_2} = \frac{1}{v_2} - \frac{1}{u_2}$$

$$\frac{1}{-10} = \frac{1}{v_2} - \frac{1}{10}$$

$$v_2 = \infty$$

For third lens

$$\frac{1}{f_3} = \frac{1}{v_3} - \frac{1}{u_3}$$

$$\frac{1}{30} = \frac{1}{v_3} - \frac{1}{\infty} \Rightarrow v_3 = 30 \text{ cm}$$

1

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- (b) A double convex lens is made of a glass of refractive index 1.55, with both faces of the same radius of curvature. Find the radius of curvature required, if the focal length is 20 cm.

CBSE 2017

- (b) A double convex lens is made of a glass of refractive index 1.55, with both faces of the same radius of curvature. Find the radius of curvature required, if the focal length is 20 cm.

CBSE 2017

$$\text{b) } \frac{1}{f} = \left[\frac{\mu_2}{\mu_1} - 1 \right] \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

$$\therefore \frac{1}{20} = \left[\frac{1.55}{1} - 1 \right] \frac{2}{R}$$

$$\therefore R = (20 \times 1.10) \text{ cm} = 22 \text{ cm}$$

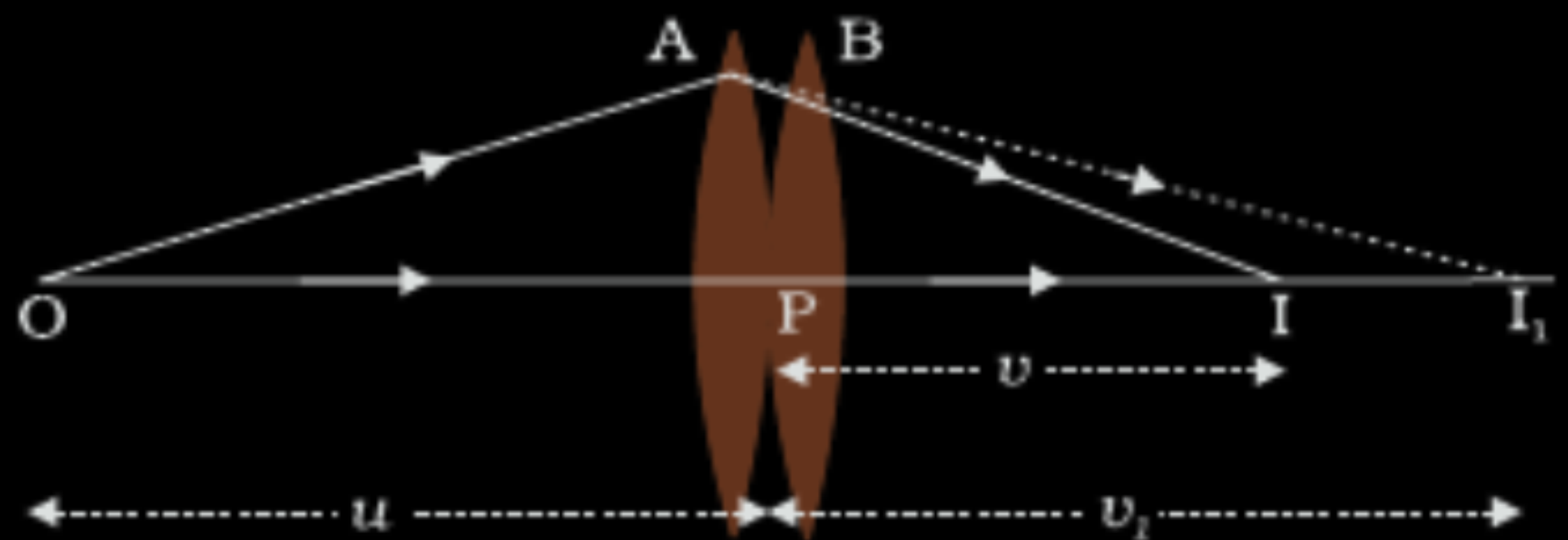
1/2

1/2

1/2

- (a) Draw a ray diagram to show the image formation by a combination of two thin convex lenses in contact. Obtain the expression for the power of this combination in terms of the focal lengths of the lenses.
- (b) A ray of light passing from air through an equilateral glass prism undergoes minimum deviation when the angle of incidence is $\frac{3}{4}$ th of the angle of prism. Calculate the speed of light in the prism.

a)



Two thin lenses, of focal length f_1 and f_2 are kept in contact. Let O be the position of object and let u be the object distance. The distance of the image (which is at I_1), for the first lens is v_1 .

This image serves as object for the second lens.

Let the final image be at I. We then have

$$\frac{1}{f_1} = \frac{1}{v_1} - \frac{1}{u}$$

$$\frac{1}{f_2} = \frac{1}{v} - \frac{1}{v_1}$$

Adding , we get

$$\frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\therefore \frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$\therefore P = P_1 + P_2$$

b) At minimum deviation

$$r = A/2 = 30^\circ$$

We are given that

$$i = \frac{3}{4}A = 45^\circ$$

$$\therefore \mu = \frac{\sin 45^\circ}{\sin 30^\circ} = \sqrt{2}$$

$$\therefore \text{Speed of light in the prism} = \frac{c}{\sqrt{2}}$$

$$(\cong 2.1 \times 10^8 \text{ ms}^{-1})$$

[Award $\frac{1}{2}$ mark if the student writes the formula:

$$\mu = \frac{\sin(A + D_m)/2}{\sin(A/2)}$$

but does not do any calculations.]

1

 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

- (i) Draw a ray diagram showing the geometry of formation of image of a point object situated on the principal axis and on the convex side of a spherical surface of radius of curvature R . Taking the rays as incident from a rarer medium of refractive index n_1 to a denser medium of refractive index n_2 , derive the relation.

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}, \text{ where symbols have their usual meaning.}$$

- (ii) Explain briefly how the focal length of a convex lens changes with increase in wavelength of incident light.
- (iii) What happens to the focal length of convex lens when it is immersed in water ? Refractive index of the material of lens is greater than that of water.

$$\tan \alpha = \frac{AN}{ON} \approx \alpha$$

$$\tan \beta = \frac{AN}{ON} \approx \beta$$

$$\tan \gamma = \frac{AN}{ON} \approx \gamma$$

$$\alpha + \gamma = i; r = \gamma - \beta$$

$$\frac{AN}{ON} + \frac{AN}{CN} = i; r = \frac{AN}{CN} - \frac{AN}{NI}$$

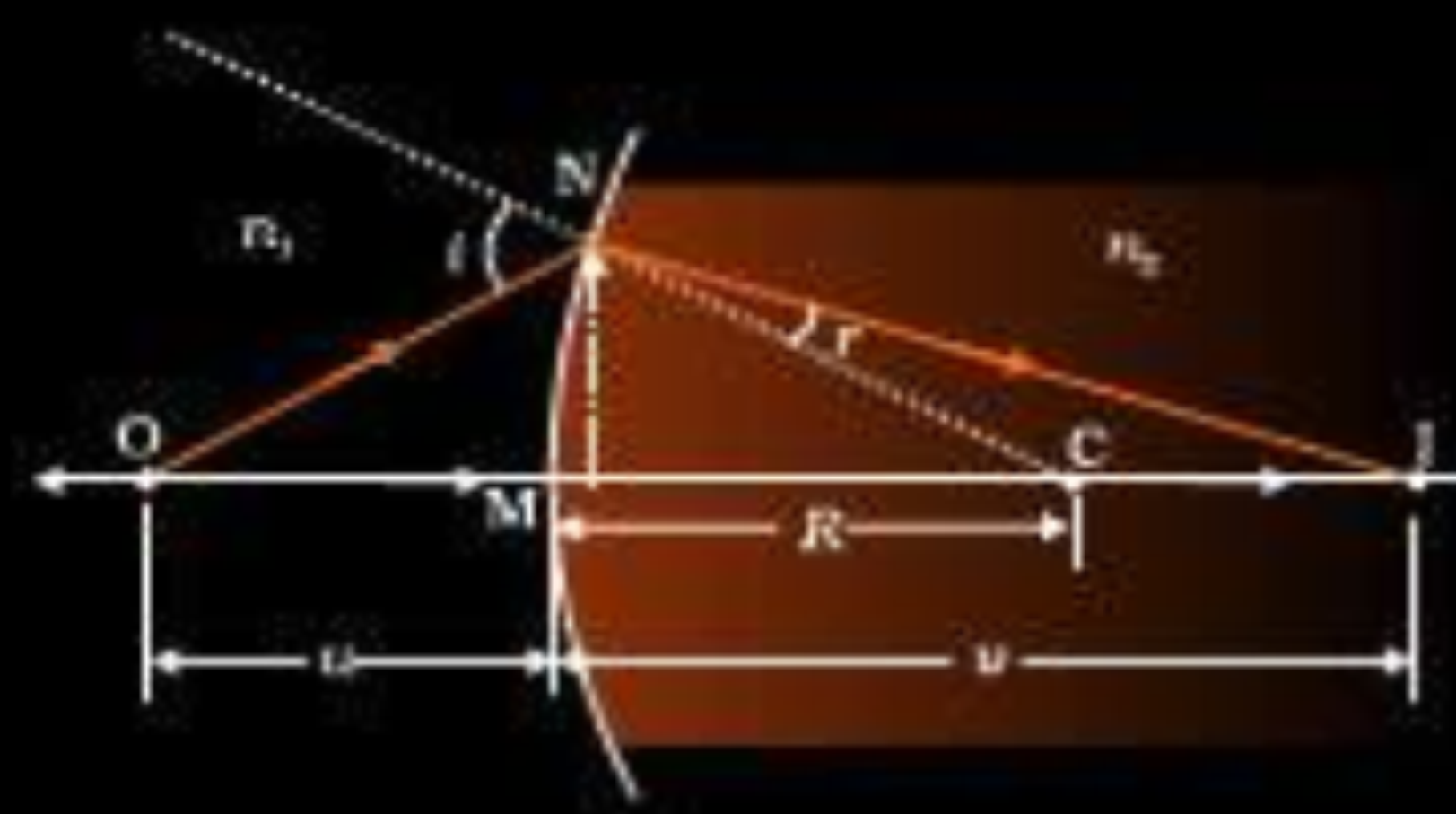
$$n_{21} = \frac{\sin i}{\sin r} \approx \frac{i}{r}$$

$$\frac{n_2}{n_1} = \frac{\frac{AN}{ON} + \frac{AN}{CN}}{\frac{AN}{CN} - \frac{AN}{NI}}$$

$$n_2 \left(\frac{AN}{CN} - \frac{AN}{NI} \right) = n_1 \left(\frac{AN}{ON} + \frac{AN}{CN} \right)$$

$$CN = R; NI = V; ON = -u$$

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$



(ii) focal length increases with increase of wavelength

$\frac{1}{f} = \left(\frac{\mu_2}{\mu_1} - 1 \right) \frac{2}{R}$ as wavelength increases μ_2/μ_1 decreases hence focal length increases

(iii) As μ_1 increases focal length increases

$$\frac{1}{f} = \left(\frac{\mu_2}{\mu_1} - 1 \right) \frac{2}{R}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

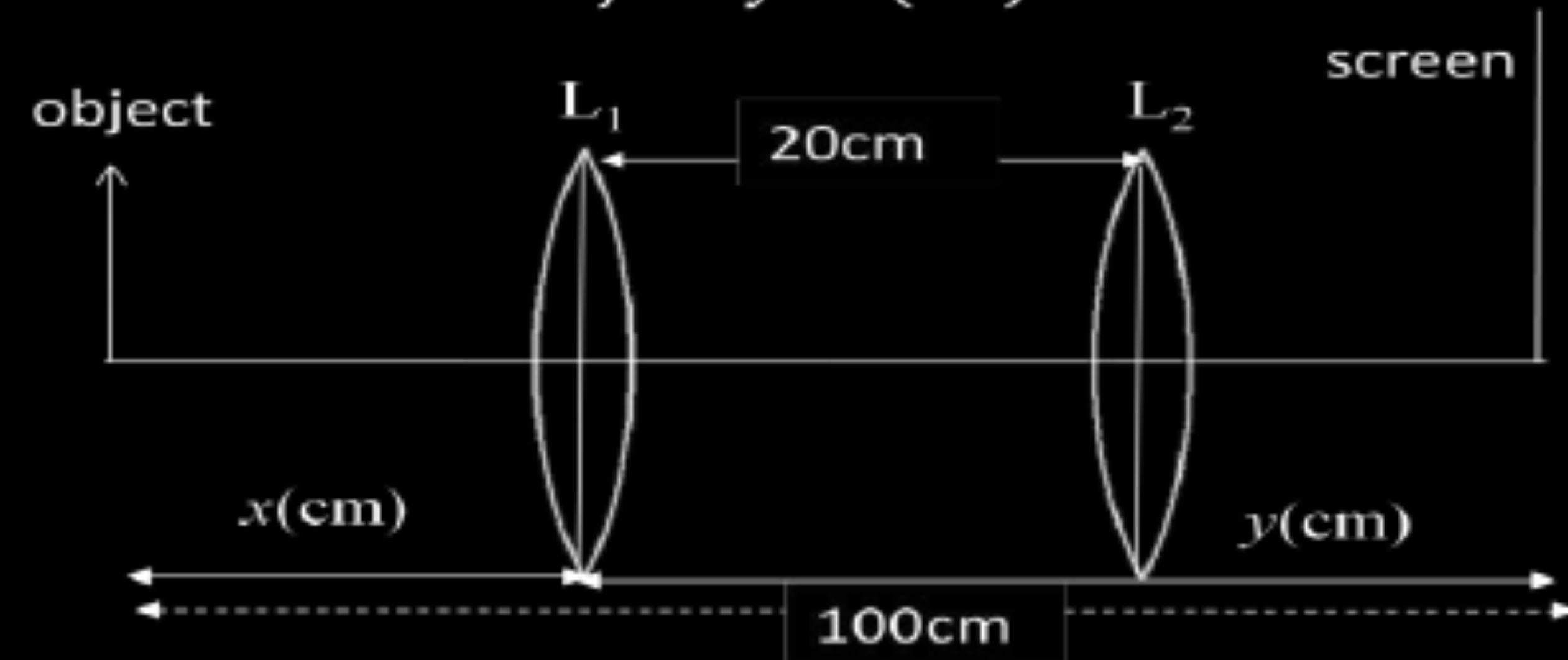
$\frac{1}{2}$

- (i) A screen is placed at a distance of 100 cm from an object. The image of the object is formed on the screen by a convex lens for two different locations of the lens separated by 20 cm. Calculate the focal length of the lens used.
- (ii) A converging lens is kept coaxially in contact with a diverging lens - both the lenses being of equal focal length. What is the focal length of the combination ?

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(i) For first position of the lens , we have

$$\frac{1}{f} = \frac{1}{y} - \frac{1}{(-x)}$$



For second position of the lens , we have

$$\frac{1}{f} = \frac{1}{y-20} - \frac{1}{-(x+20)}$$

$$\frac{1}{y} + \frac{1}{x} = \frac{1}{(y-20)} + \frac{1}{(x+20)}$$

$$\begin{aligned} \frac{x+y}{xy} &= \frac{(x+20)+(y-20)}{(y-20)(x+20)} \\ \therefore xy &= (y-20)(x+20) \\ &= xy - 20x + 20y - 400 \end{aligned}$$

$$\therefore x - y = -20$$

Also , $x + y = 100$

$\therefore x = 40$ cm

and $y = 60$ cm

$$\therefore \frac{1}{f} = \frac{1}{60} - \frac{1}{-40} = \frac{2+3}{120} = \frac{5}{120}$$

$$\therefore f = 24 \text{ cm}$$

Alternatively ,

We have

$$f = \frac{D^2 - d^2}{4D}$$

$$= \frac{100^2 - 20^2}{4 \times 100}$$

$$= \frac{120 \times 80}{400}$$

$$= 24 \text{ cm}$$

Alternatively,

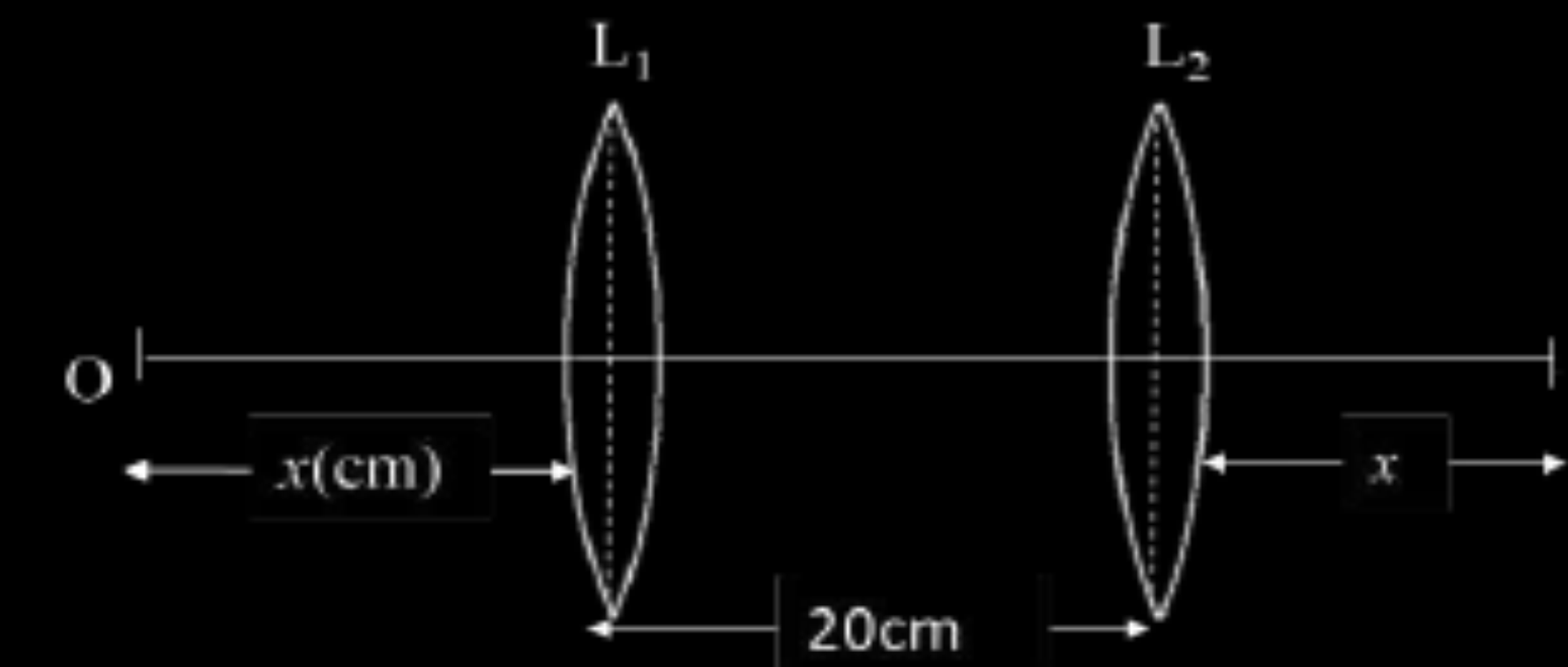
For the two positions of the lens , the values of the magnitudes of u and v , get interchanged.

Hence , $|u + v| = 100$

$|u - v| = 20$, This gives $|u| = 60$ $|v| = 40$

$\therefore f = 24 \text{ cm}$

Alternatively ,



$$2x + 20 = 100$$

$$\therefore x = 40 \text{ cm}$$

For lens at position L_1 ; $u = -x = -40 \text{ cm}$

$$v = 20 + 40 = 60 \text{ cm}$$

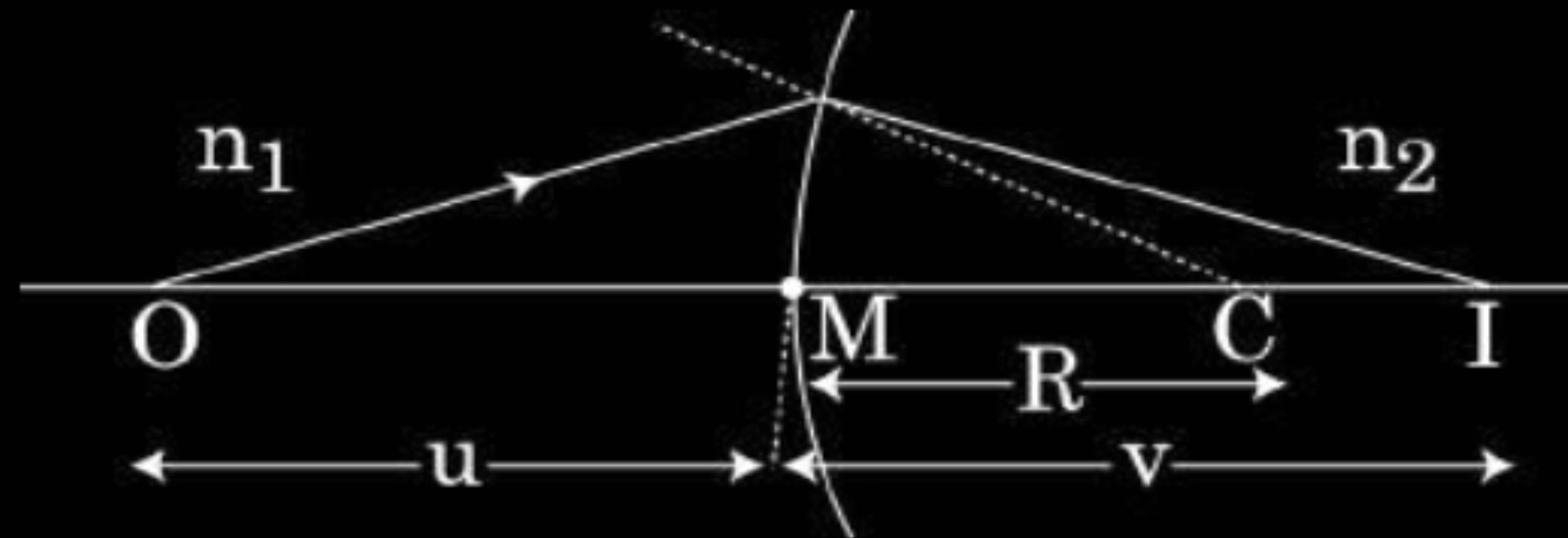
This gives $f = 24 \text{ cm}$

(i) For combination of two lenses in contact .	
Net Power of combination ,	
$P = P_1 + P_2$	$\frac{1}{2}$
$P_1 = +P$, $P_2 = -P$	
So $P = 0$ and $F = \text{infinite}$	$\frac{1}{2}$
Alternatively , $\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$	$\frac{1}{2}$
$= \frac{1}{f} + \left(\frac{-1}{f} \right) = \mathbf{0}$	
$F = \text{infinite}$	$\frac{1}{2}$

- (a) A point object O on the principle axis of a spherical surface of radius of curvature R separating two media of refractive indices n_1 and n_2 forms an image 'I' as shown in the figure.

Prove that

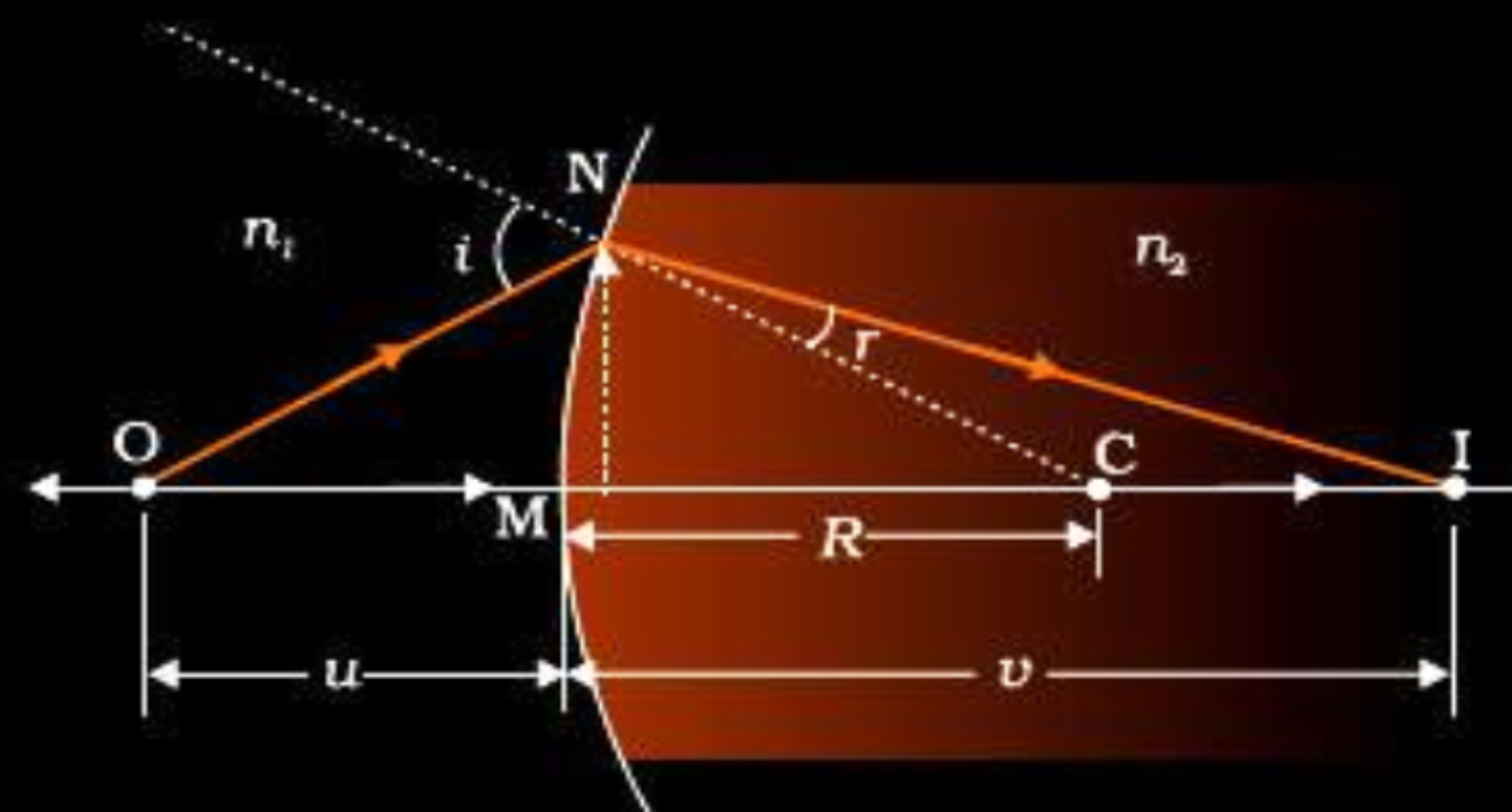
$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$



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- (b) Use this expression to derive lens maker's formula. Draw the necessary diagram.
- (c) A convex lens is placed over a plane mirror. A pin is now positioned so that there is no parallax between the pin and its image formed by this lens-mirror combination. How will you use this observation to find focal length of the lens ? Explain briefly.

(a)



$$i = \frac{MN}{OM} + \frac{MN}{MC}$$

$$r = \frac{MN}{MC} - \frac{MN}{MI}$$

$$n_1 i = n_2 r$$

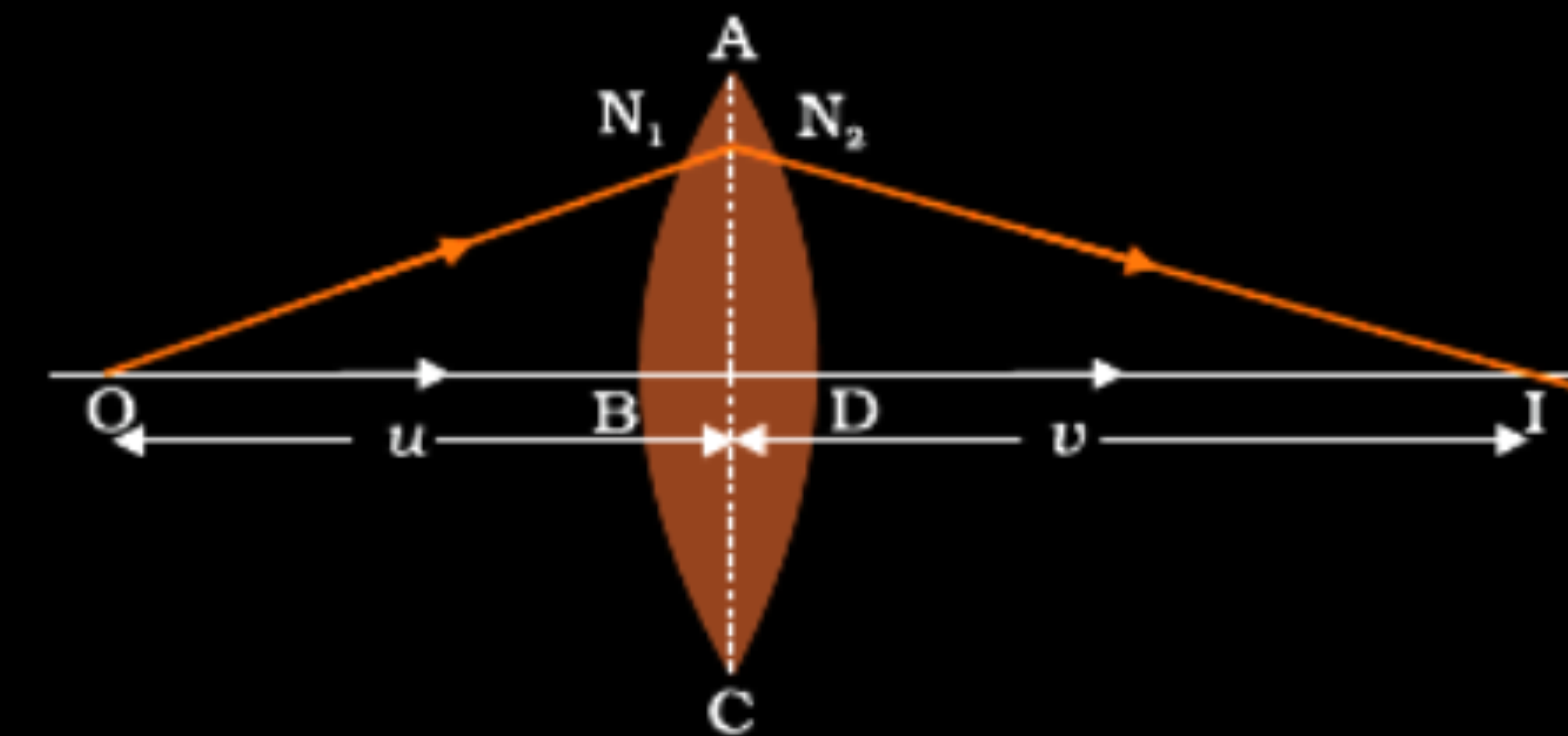
$$\frac{n_1}{OM} + \frac{n_2}{MI} = \frac{n_2 - n_1}{MC}$$

$$OM = -u, MI = +v, MC = +R$$

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

(b) Applying above relation to refraction of light through a convex lens ABCD

 $\frac{1}{2}$

$$\text{For interface ABC } \frac{n_2}{v_1} - \frac{n_1}{u} = \frac{n_2 - n_1}{R_1}$$

$$\text{For interface ADC } \frac{n_1}{v} - \frac{n_1}{v_1} = \frac{n_1 - n_2}{R_2}$$

 $\frac{1}{2}$ $\frac{1}{2}$

$$\therefore \frac{n_1}{v} - \frac{n_1}{u} = (n_2 - n_1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

 $\frac{1}{2}$

$$\text{or } \frac{1}{f} = (n_{21} - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

 $\frac{1}{2}$

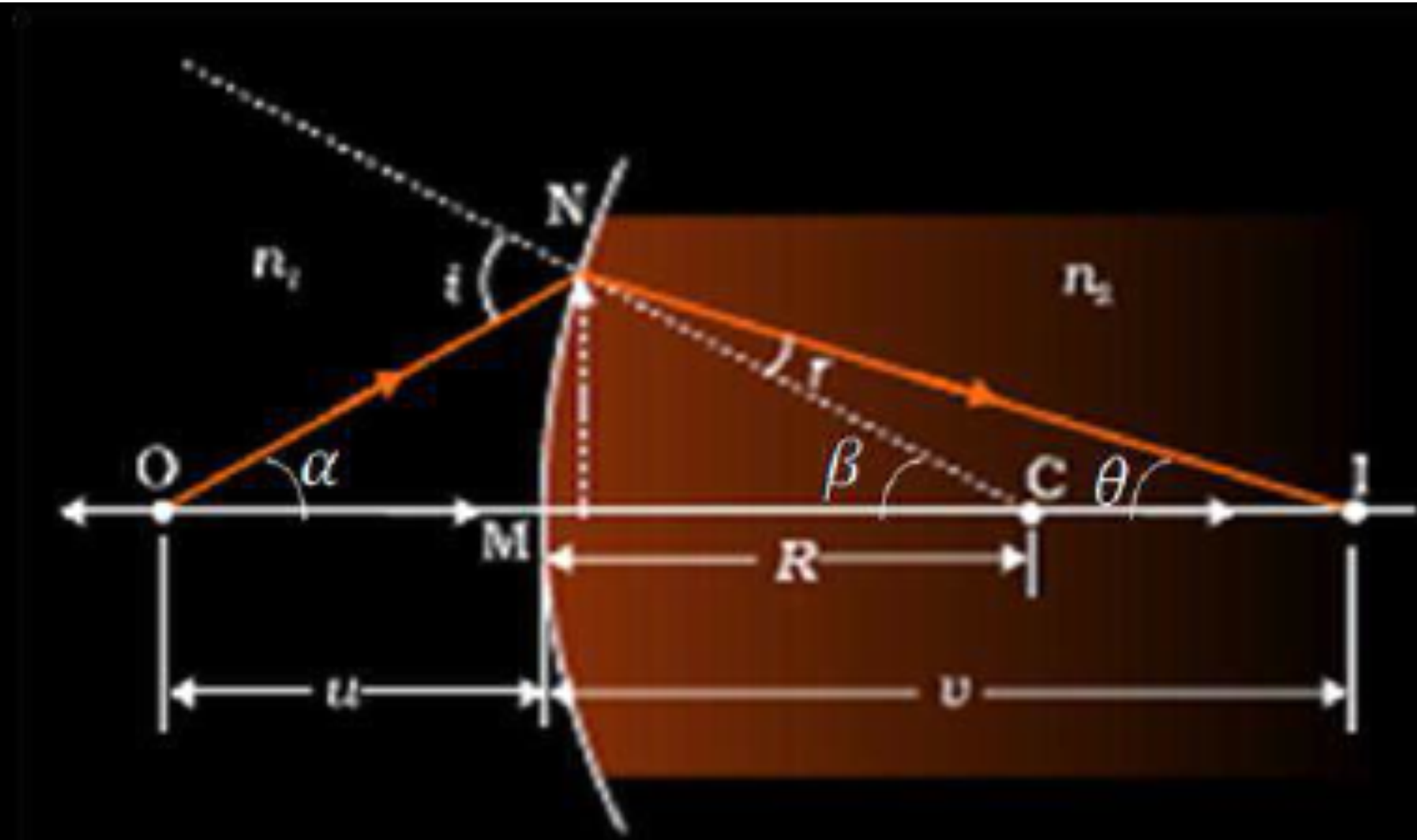
(c) Focal length = distance of the pin from the mirror.

The rays from the object after refraction from lens should fall normally on the Plane mirror. So they retrace their path. Hence, rays must be originating from focus and thus distance of the pin from the plane mirror gives focal length of the lens.

 $\frac{1}{2}$

- (i) Derive the mathematical relation between refractive indices n_1 and n_2 of two media and radius of curvature R for refraction at a convex spherical surface. Consider the object to be a point lying on the principal axis in rarer medium of refractive index n_1 and a real image formed in the denser medium of refractive index n_2 . Hence, derive lens maker's formula.
- (ii) Light from a point source in air falls on a convex spherical glass surface of refractive index 1.5 and radius of curvature 20 cm. The distance of light source from the glass surface is 100 cm. At what position is the image formed ?

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Ray diagram showing real image formation as per prescription

$$\theta_1 = \alpha + \beta$$

$$\theta_2 = \beta - \gamma \quad \therefore \gamma = \beta - \theta$$

For paraxial rays θ_1 and θ_2 are small

Therefore, $n_2 \sin \theta_2 = n_1 \sin \theta_1$ (Snells law)

Reduces to

$$\text{At N } \frac{\sin i}{\sin r} \sim \frac{i}{r} = \frac{n_2}{n_1}$$

$$\therefore n_1 = r \sin i$$

$$(\alpha + \beta)n_1 = (\beta - \theta)n_2$$

$$n_1 \left(\frac{NM}{OM} + \frac{NM}{MC} \right) = \left(\frac{NM}{MC} - \frac{NM}{MI} \right) n_2$$

$$n_1 \left(\frac{1}{-u} + \frac{1}{+R} \right) = \left(\frac{1}{+R} - \frac{1}{u} \right) n_2$$

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{(n_2 - n_1)}{R_1}$$

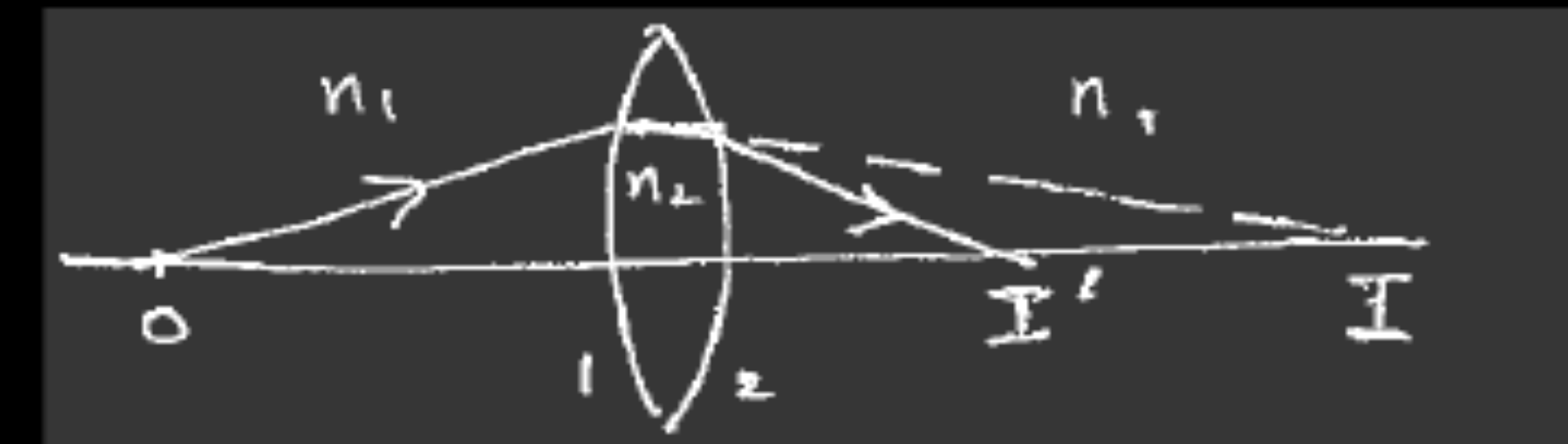
Applying above relations to refraction through a lens

1/2

1/2

1/2

1



For surface 1

$$\frac{n_2 - n_1}{R_1} = \frac{n_2}{v'} - \frac{n_1}{u} \quad \dots(i)$$

For surface 2

$$\frac{n_1 - n_2}{R_2} = \frac{n_1}{v} - \frac{n_2}{v'} \quad \dots(ii)$$

Adding eqn. (i) and (ii)

$$(n_2 - n_1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) = n_1 \left(\frac{1}{v} - \frac{1}{u} \right)$$

For $u = \infty$ $v = f$

$$\therefore \frac{n_1}{f} = (n_2 - n_1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\Rightarrow \frac{1}{f} = \left(\frac{n_2}{n_1} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

(iii) $R = 20 \text{ cm}$ $n_2 = 1.5$ $n_1 = 1$ $u = -100 \text{ cm}$

$$\frac{n_2}{v} = \frac{(n_2 - n_1)}{R} + \frac{n_1}{u}$$

$$= \frac{0.5}{20 \text{ cm}} - \frac{1}{100 \text{ cm}}$$

$$= \frac{1.5}{100} \text{ cm}$$

$\Rightarrow V = 100 \text{ cm}$ a real image on the other side, 100 cm away from the surface.

1/2

1/2

1/2

1/2

1/2